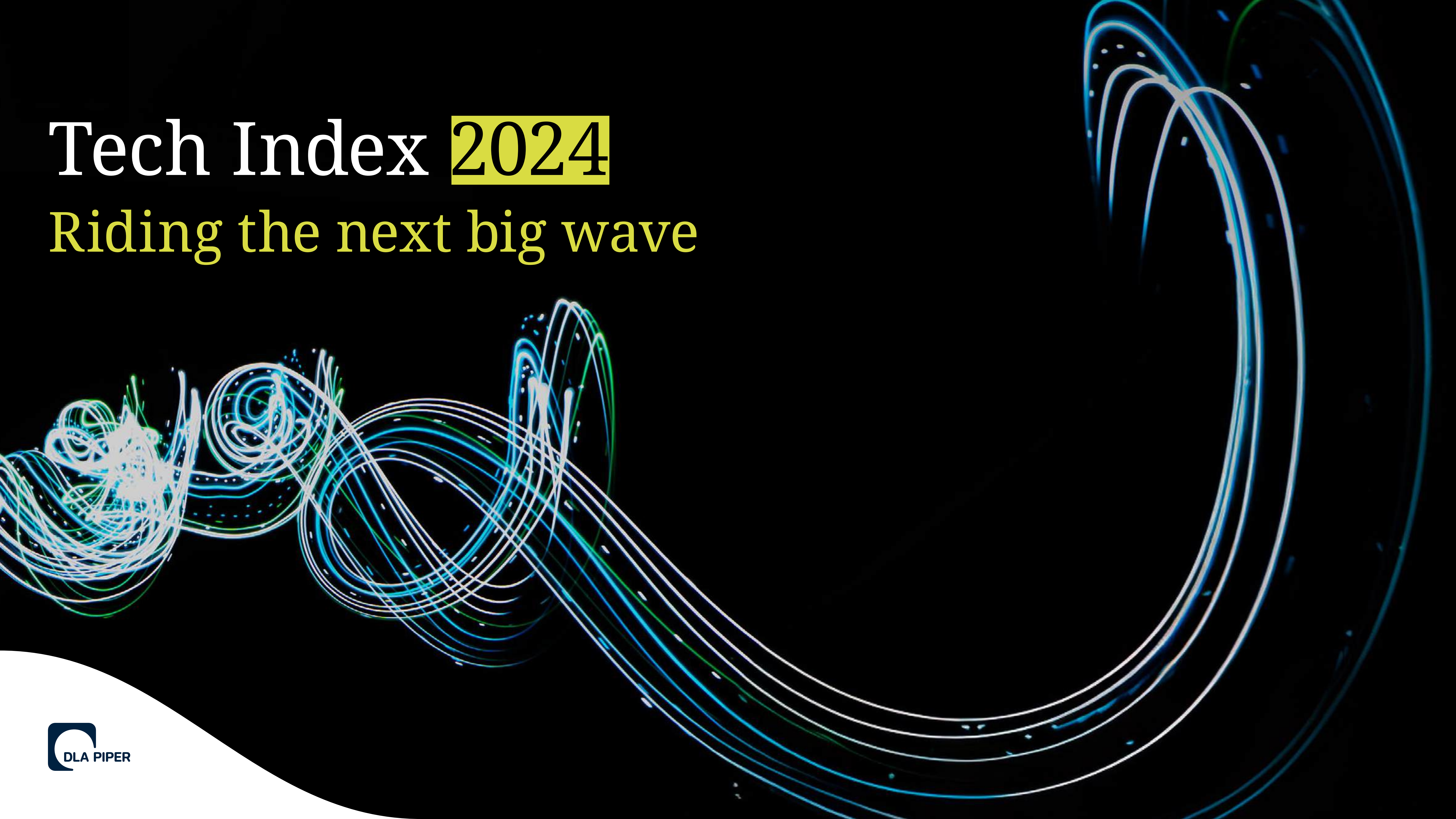
Tech Index 2024

Riding the next big wave



DLA PIPER

Welcome

Welcome to the 2024 Tech Index report. For the first time, this report offers a truly global perspective on the tech sector, drawing insights from industry leaders and policymakers worldwide.

Unlike previous editions that focused solely on Europe, this report captures the views of individuals from every region. It provides their insights into the sector's growth prospects, the challenges they anticipate, and the opportunities they aim to seize.

We hope you find the report both useful and insightful. We would love to hear your thoughts on our findings. If you'd like to discuss any of the issues raised in the report, please feel free to reach out to one of the many DLA Piper contributors involved with the Tech Index.

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Alongside the data and insights, you'll find some captivating images taken by DLA Piper colleagues using light photography. You can learn more about this process at the end of the report.

Riding the next big wave

In a deeply unstable geopolitical environment and with economic recovery proving extremely sluggish in many key markets, the tech industry is nevertheless in a very confident mood as it looks towards opportunities for future growth.

That is the clear conclusion of DLA Piper's biennial Tech Index which, unlike the previous six editions that focused exclusively on Europe, for the first time takes a truly global snapshot of opinion from across the tech sector.

We've widened our sample, interviewing 1,200 industry executives, investors and government and regulatory officials right across North America, Latin America, Europe, the Middle East, Africa and the Asia Pacific Region.

Our survey takes in companies of all sizes from small enterprises employing less than 100 people right up to major businesses with more than 10,000 employees and with revenues ranging from USD10 million to USD10 billion.

> Given the expanded scope of the latest Index, there are some surprising findings, not least a strikingly high level of confidence among respondents the world over.

In buoyant mood

Given the expanded scope of the latest Index and the diversity of markets it covers, there are some surprising findings not least – given that we are still living in troubled times – a strikingly high level of confidence among respondents the world over.

That buoyant mood is reflected in the overall tech score for 2024 which stands at a record-breaking 71. This is 3 points higher than the score we recorded in our last Tech Index of 68, which was itself the previous highest overall score on record.

It's also significant that sense of optimism is pretty consistent in all the regions where we took soundings, with only slight regional variations. It's highest in North America, Latin America and Africa, and only slightly lower in the Middle East, Europe and Asia Pacific.

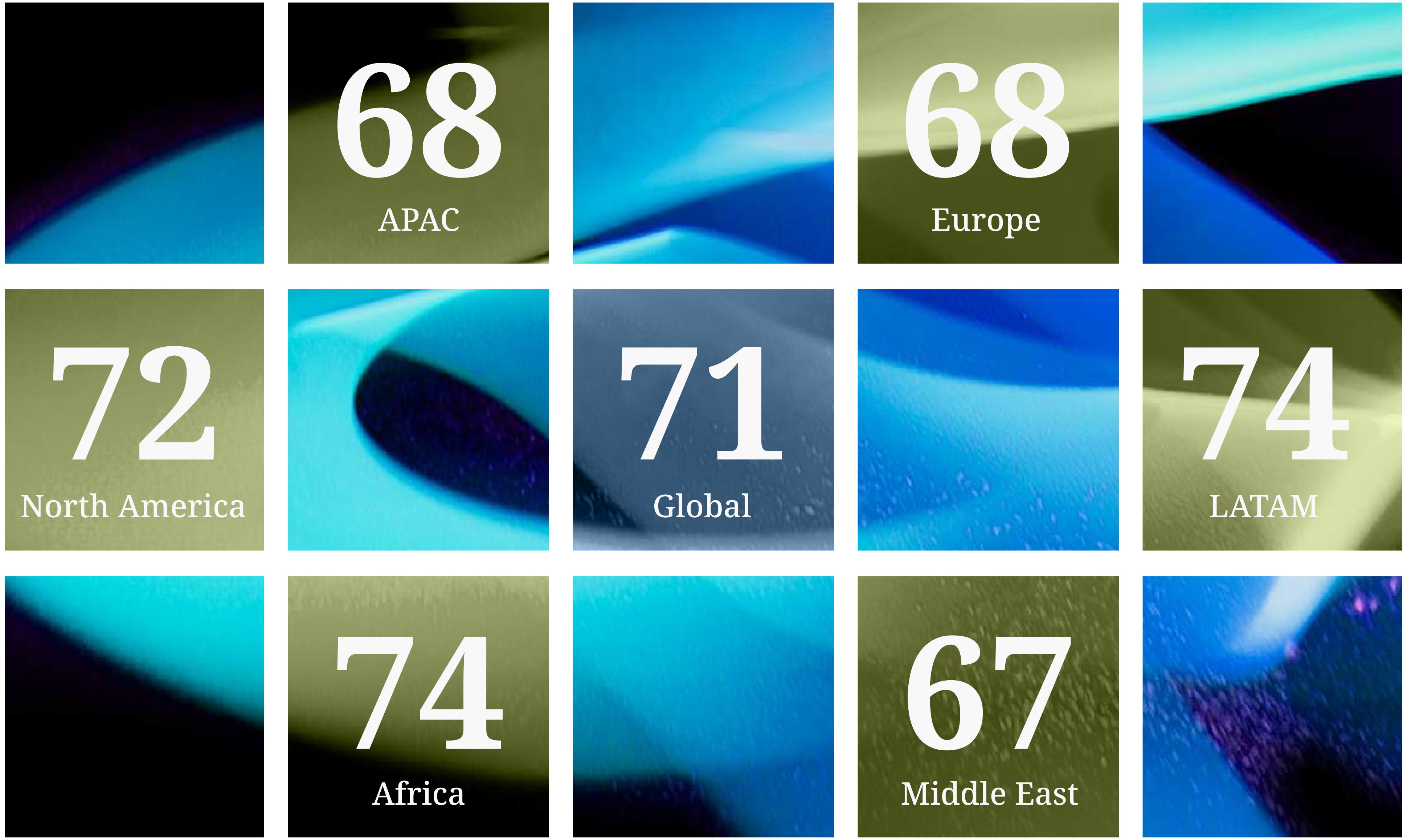
> The tech industry is in a very confident mood as it looks towards opportunities for future growth despite a deeply unstable geo-political environment.

Our respondents express confidence about the current economic environment, capital venture markets, the availability of talent and about the overall regulatory environment, although they are less comfortable with the tax regimes in certain markets.

Tech Index Methodology

The results of our survey have been collated and weighted to produce DLA Piper's Tech Index score. This score is based on a diffusion index, which evaluates the percentage of respondents' answers that are positive, negative, or neutral. The results are presented as a scorecard, showing the tech score for each business area monitored, along with an "overall" tech score.

The overall tech scores: globally and regionally



The global overall tech score for 2024 is 71. The overall tech score is broadly similar across all regions – but is slightly higher in LATAM and Africa and slightly lower in the Middle East, Europe and APAC.

Impact of the current economic environment on growth in your sector

70 Global

66 Europe | 73 North America | 68 APAC

74 LATAM | 74 Africa | 70 Middle East

Impact of the current venture/ capital markets model on growth in your sector

69 Global

67 Europe | 72 North America | 70 APAC

71 LATAM | 71 Africa | 69 Middle East

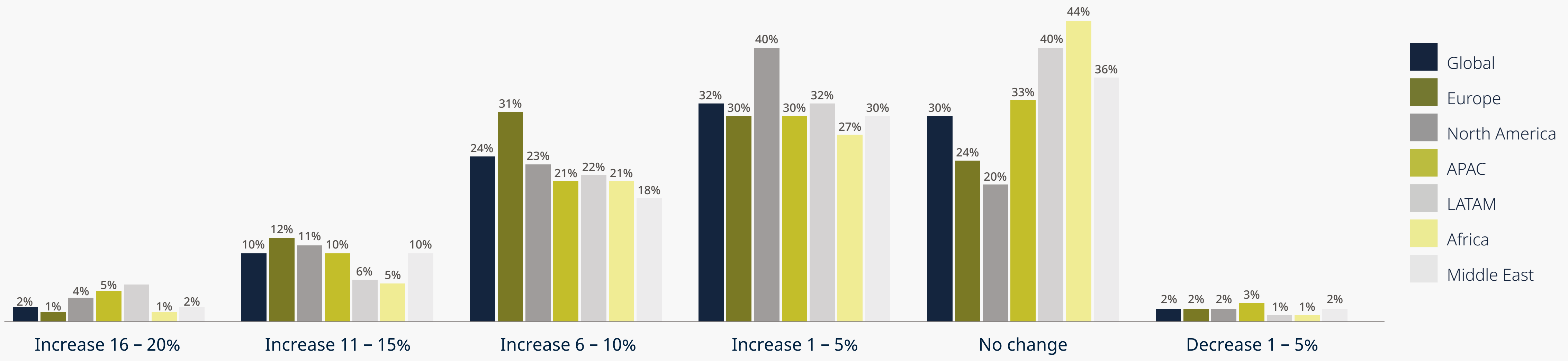
We see generally positive tech scores from respondents regarding the effects of the current economic environment and venture capital market model on growth in their sector.

Against that background two-thirds of our respondents are expecting to see their revenues increase in the coming 12 months with 36% saying they expect turnover to grow by more than 6% in that period.

Revenue expectations are highest in North America and Europe, even though the former is experiencing much faster economic growth right now and the latter is on a much slower recovery path.

> The overall tech score for 2024 stands at a record-breaking 71, reflecting a buoyant mood in the industry.

Expected change in revenues in the next 12 months



2/3 of organizations expect to see at least some increase in revenues over the next 12 months and almost 4 in 10 (36%) expect an increase of more than 6%. Those in North America and Europe are most likely to expect their revenues to increase. Very few expect to see a decrease in revenues in the next 12 months.

Impact of the current geopolitical situation on growth in your sector

53 Global		
52 Europe	54 North America	48 APAC
57 LATAM	56 Africa	55 Middle East

When companies were asked about the impact of the geopolitical environment on their growth outlook, the score for this particular question was only 53, reflecting the dampening effect of this factor on their overall growth outlook. The geopolitical situation score is not included in the calculation of the "overall" tech score.

Geopolitics remains a big concern

Only one, admittedly very large, issue is dampening this overriding sense of optimism – the increasingly volatile geopolitical environment in which companies are operating.

The continuing war in Ukraine, the conflict in Gaza and rising tensions between China and Taiwan are clearly having an impact on growth expectations and constraining activity in key markets.

We've observed that investors are, for instance, wary about doing transactions in Central and Eastern Europe because of the region's proximity to the Ukraine conflict, while Taiwan continues to put an added strain on already strained US/China diplomatic and trade relations.

But other headwinds are also continuing to have an impact. 2024 is the busiest election year in history with more than 50 countries and around half the world's population going to the polls and some of the elections we have seen so far have had unpredictable outcomes.

In certain regions, notably Europe, concerns center on developments in national politics, particularly around the rise of populist political movements with highly protectionist economic policies and socially divisive agendas.

And one poll looms particularly large for investors around the world – November's US election with the clear possibility that it will see the return of Donald Trump to the White House with the US once again pursuing an isolationist agenda.

Against the backdrop of this uncertainty, it's not surprising that geopolitics remains a significant concern for our respondents.

Across all regions, tech scores are much lower, indicating a much lower level of positivity, for the impact of the current geopolitical situation on growth, falling to just 53.

A changed tech landscape

Those worries aside it's perhaps easy to see why the overall mood of our respondents is high.

Since our last survey, the tech landscape has evolved at a dizzying pace, with two words increasingly on everybody's lips – Artificial Intelligence (AI).

In 2022, large language and generative AI models such as Open AI's ChatGPT were on the horizon but only just. The developments in this area have been quite extraordinary in so short a space of time, while the move to digitalizing the economy has been accelerated by the pandemic with services like zoom and Teams and online banking and shopping becoming everyday.

Such developments have put renewed impetus behind the digital transformation of organizations we are working with, providing a reminder that, when tech is advancing so fast, they must be prepared to embrace continuous technological change to truly transform rather than just fine-tune their operations and business models. It should be a case of digital evolution rather than one-off optimization.

The economic environment has also improved, if not quite as much as many would have hoped.

Inflation has come down rapidly and interest rates in key economies, while remaining relatively high compared with the fall in prices, are at least lower although still subject to some volatility. It's not a return to that long period of cheap debt that fuelled a booming transaction market for so many years, but at least the peaks we saw in 2022 and 2023 have been smoothed.

Given the shocks that organizations have had to deal with in recent years, including Covid, economic and political instability, supply chain disruptions, the impact of global trade tensions, to mention just a few, there's a sense among our clients that companies have become more resilient in the face of perpetual uncertainty.

They are prepared to keep looking for new and alternative pathways to growth, hunting for opportunities irrespective of the short-term challenges confronting them.

> Two-thirds of organizations expect to see at least some increase in revenues over the next 12 months, with almost 4 in 10 anticipating growth of more than 6%.

AI and the last mile

When asked to identify the key opportunities for growth there is no doubting what tops the league – AI.

Overall, 63% of respondents see this as the key area for potential growth, with that score rising to 72% in Europe.

As we explore in greater detail in our dedicated AI chapter later in this report, a significant proportion of companies are expecting the costs of investing in AI to be significant. But many are still willing to make that investment because of the huge potential they see in this technology.

But it's important to put that initial enthusiasm in context.

AI remains at relatively early stage of development. And it has a last mile problem, to date not able to fully demonstrate what it's ultimate potential might be.

Rival solutions are vying for the attention of businesses, big and small, but while the benefits remain unproven, it is very difficult for organizations to assess where and how much to invest.

Opposing forces are at play here – the fear of being left behind battling against a concern that companies will take a bet on a solution that ultimately turns out to be the Kodak or Betamax of the AI world.

And it's important to remember that other areas of technology, several years ahead, are still in the process of proving themselves. That's very true, for example, of Internet of Things solutions (still seen by over half of our respondents as a key growth area) and cloud computing. AI is now on a similar journey, even if the pace has quickened.

Another factor causing pause for thought is risk. This technology has the potential to be a force for great good, but, in the wrong hands or poorly managed, is also a potent force for evil-doing, with complex ethical and regulatory questions to be answered.

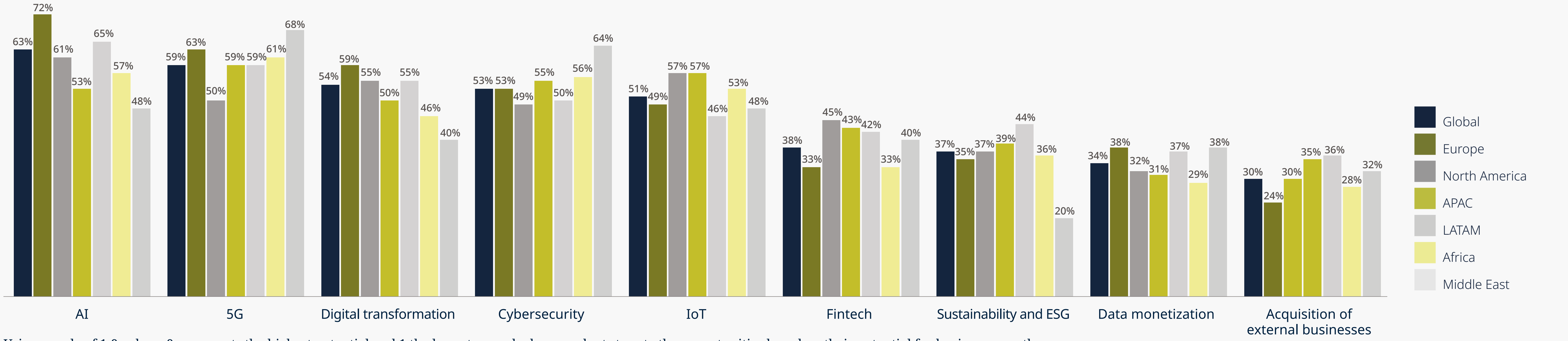
In these relatively early days for AI, many of our clients are, therefore, much more focused on the continued digital transformation or evolution of their operations, keenly aware, however, that AI will supercharge the transformative impact of digitalization.

In that sense they are preparing themselves to ride the next massive wave in tech development, in the full knowledge that they are on the brink of what will be the next industrial and, indeed, social revolution.

And AI will be a force for change with regards to almost all the other areas where respondents see potential for growth, whether that's digital transformation, cybersecurity, IoT, fintech, data monetization and even sustainability and ESG.

> 63% of respondents see AI as the key area for potential growth, with the score rising to 72% in Europe.

Potential opportunities for future business growth



Using a scale of 1-9, where 9 represents the highest potential and 1 the lowest, we asked respondents to rate the opportunities based on their potential for business growth. The percentages indicate the proportion of respondents who rated a particular opportunity as 7, 8, or 9. AI, 5G, digital transformation and cybersecurity were ranked highest.

M&A cautiously returns

It's no big surprise that while respondents include the acquisition of external businesses in the mix of potential growth areas, this ranks at the bottom of their list.

This is clearly a reflection of the sharp downturn in M&A activity that we've seen in the last 18 months after an exceedingly long bull market for transactions which peaked in early 2022 as interest rates rose sharply closing a prolonged period of cheap debt finance.

The boom years saw a proliferation of deals at very high – and often unsustainable – multiples, some of which were rushed and ill-thought through and subsequently proved to be duds.

With transactions coming to an almost complete standstill, many early-stage companies came under pressure to rein in spending and cut headcount and last year saw a raft of redundancies as well as even previously successful new businesses forced to delay follow up financing or to compelled to do down rounds.

Some of the talented people who were let go are now returning with ideas of their own and new start-up ventures seeking finance.

We're seeing the M&A market return in 2024, but the pattern of recovery is not even. North America is leading the turn around mostly led by strategic rather than financial buyers thanks to relatively high and volatile interest rates. Other markets, notably Australia, however, are benefitting from a resurgence in PE activity, while Europe is lagging behind, with venture funds still playing a safe hand.

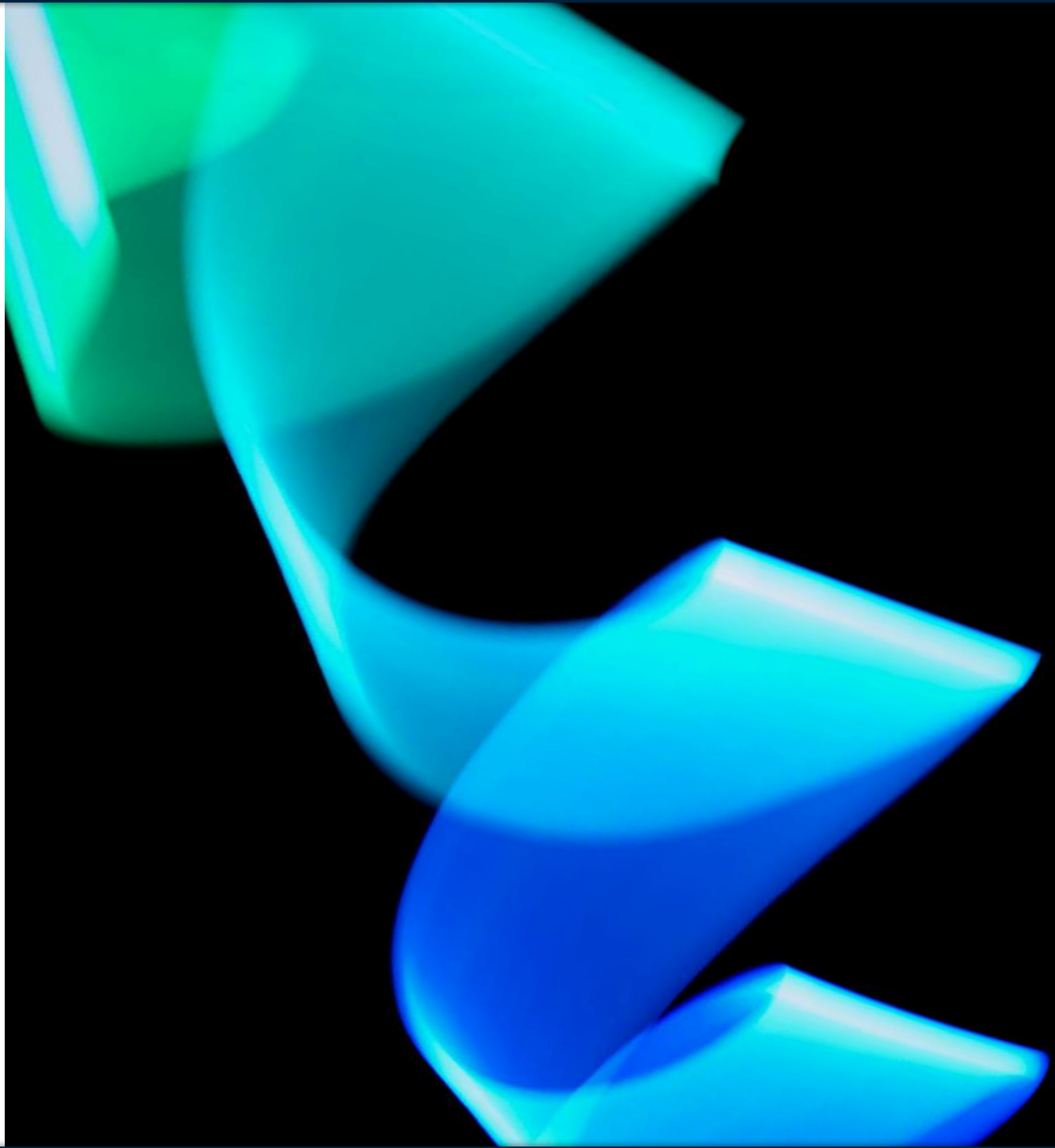
Our survey reflects this mixed picture, with North America, APAC, Latin America and the Middle East all expecting transactions to provide greater growth potential than Europe.

What's different this time is the degree of caution being taken by acquirers in all markets.

Careful assessments are being made of the technology being bought or of the capabilities of the team behind those technologies, if the acquihiring is the purpose behind the deal. Due diligence processes are far more disciplined, while multiples have moderated at much more sustainable levels.

Having said that we expect the market to pick up quite sharply, not least as companies begin to post better financial results in the months ahead – something, as we've said, our respondents fully expect to see.

> Careful regulation, properly implemented, can act as a spur to innovation, giving investors the confidence to back developers and support wider economic development.



Impact of the regulatory environment in your country on growth in your sector

75 Global

75 Europe	74 North America	73 APAC
79 LATAM	77 Africa	71 Middle East

Impact of the current tax regime in your country on growth in your sector

64 Global

63 Europe	66 North America	57 APAC
70 LATAM	70 Africa	61 Middle East

When we asked respondents about the impact of the regulatory environment on growth in their sector, their responses were positive, reflected by a tech score of 75. However, respondents are less positive about their local tax regime.

Regulation – two distinct points of view

Perhaps one of the most surprising findings is the degree to which respondents believe regulation is promoting growth in the sector, reflected by a tech score of 75.

The sheer volume of often highly complex tech-related regulations–most of them driven by developments in Europe–has led many to warn that regulatory overreach could stifle innovation.

Following the introduction of the GDPR regime, which wrong-footed many organizations at first, we are seeing a raft of new or proposed regulation creating new frameworks around AI, cybersecurity and vital national infrastructure, both physical and financial, data governance and sustainability and ESG reporting.

In the US, state legislatures tend to be at the forefront of applying new laws while at the federal level things are moving far slower over issues as diverse as fintech regulation and measures to prevent greenwashing.

Given that background it's easy to understand why some question whether this will put a brake on invention and entrepreneurialism.

But there is an entirely contrary view and one which our survey may be pointing to. It's the idea that carefully thought-through regulation, properly implemented, can instead act as a spur to innovation giving investors the confidence to back developers and eventually helping to support wider economic development.

That's clearly the message being sent by Dubai with its new Virtual Asset Regulatory Authority, which we have played a key role in implementing, where creating a secure market for virtual assets is seen as having huge growth potential acting as a magnet for inbound investment as the region diversifies away from its reliance on fossil fuels.

And we are seeing a similar mindset in both Singapore and Hong Kong, in the latter case backed by government financial assistance for developers, where there is also a determined effort to build an industry around crypto and other virtual assets on a secure regulatory basis.

In key African countries, regulatory developments in Europe are increasingly being seen as a kind of gold standard for new laws in the region in areas as diverse as AI, privacy, ESG and the modernisation of IP laws on patents and trademarks. Nigeria's approach to data privacy, for instance, is now being advanced at a rapid rate. This is an important development, and more and more companies are becoming compliant with the legislation.

It is inevitable that regulatory complexity will be a fact of life for organizations in the sector as new rules are developed in different jurisdictions. In the short-term this could lead to a greater complexity for companies trying to stay within the law, but which might ultimately lead to greater standardization and commonality.

In the end the biggest threat to innovation may lie in having a vast patchwork of potentially conflicting rules that companies have to navigate. A move to greater standardization across jurisdictions might well have the opposite effect.

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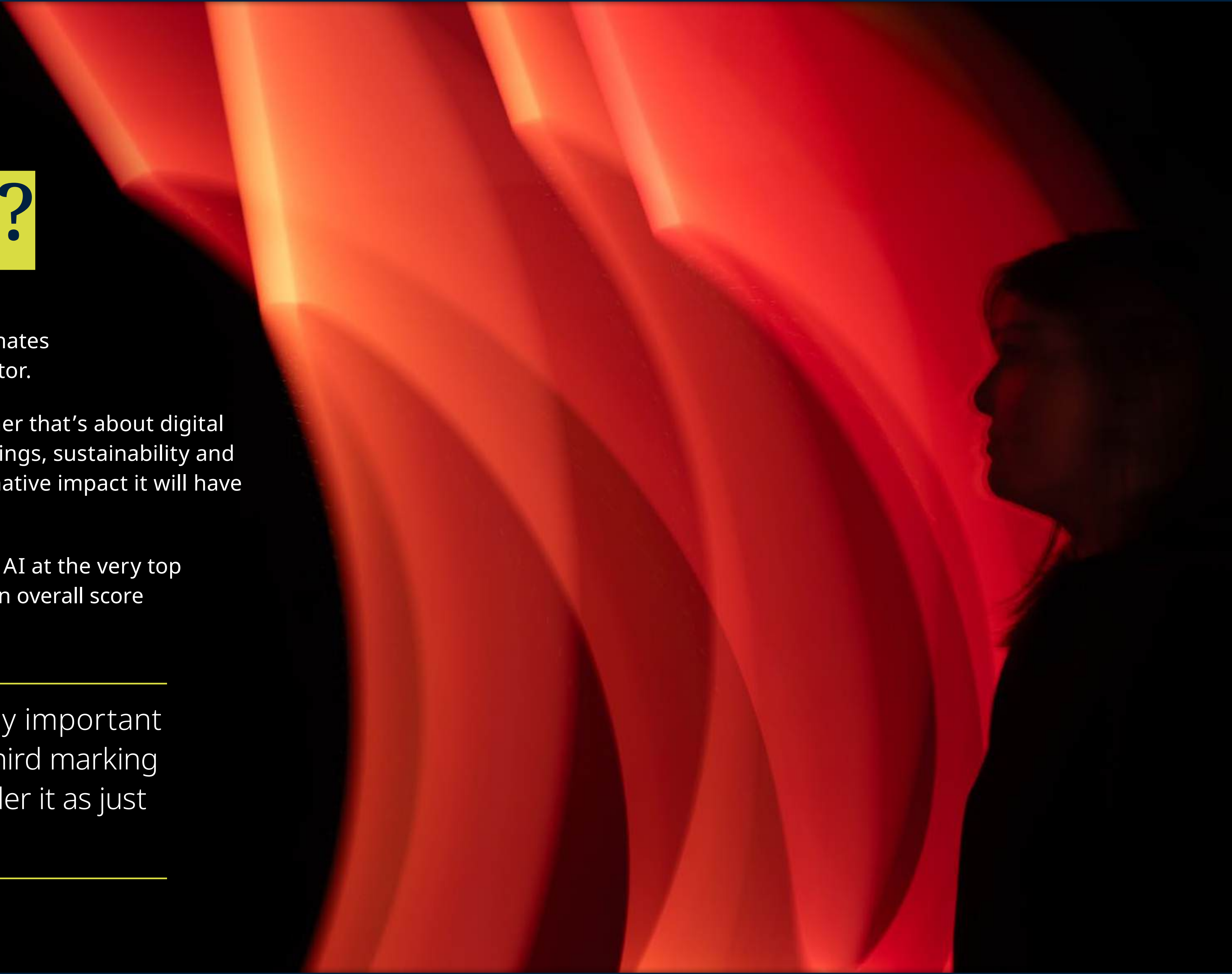
It’s all about AI – but how and why?

Without a doubt Artificial Intelligence is the one issue that dominates the agenda of most organizations, whether private or public sector.

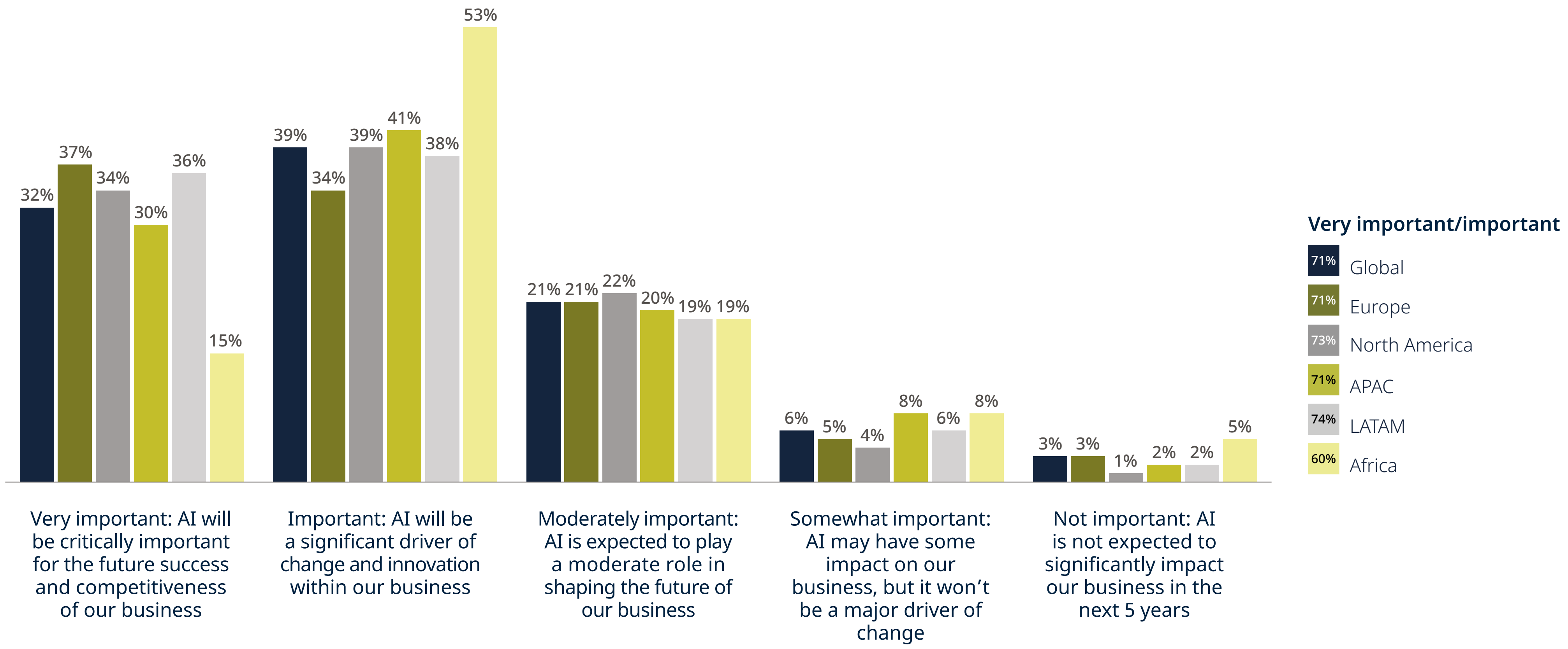
Almost any conversation about the future of technology – whether that’s about digital transformation, cybersecurity, data monetization, Internet of Things, sustainability and ESG – turns within a couple of sentences to AI and the transformative impact it will have on the way we work and live. Sometimes it comes up quicker!

Our global survey bears this out very clearly. Respondents place AI at the very top when asked to identify potential areas for business growth with an overall score of 63%, which rises to 72% in Europe.

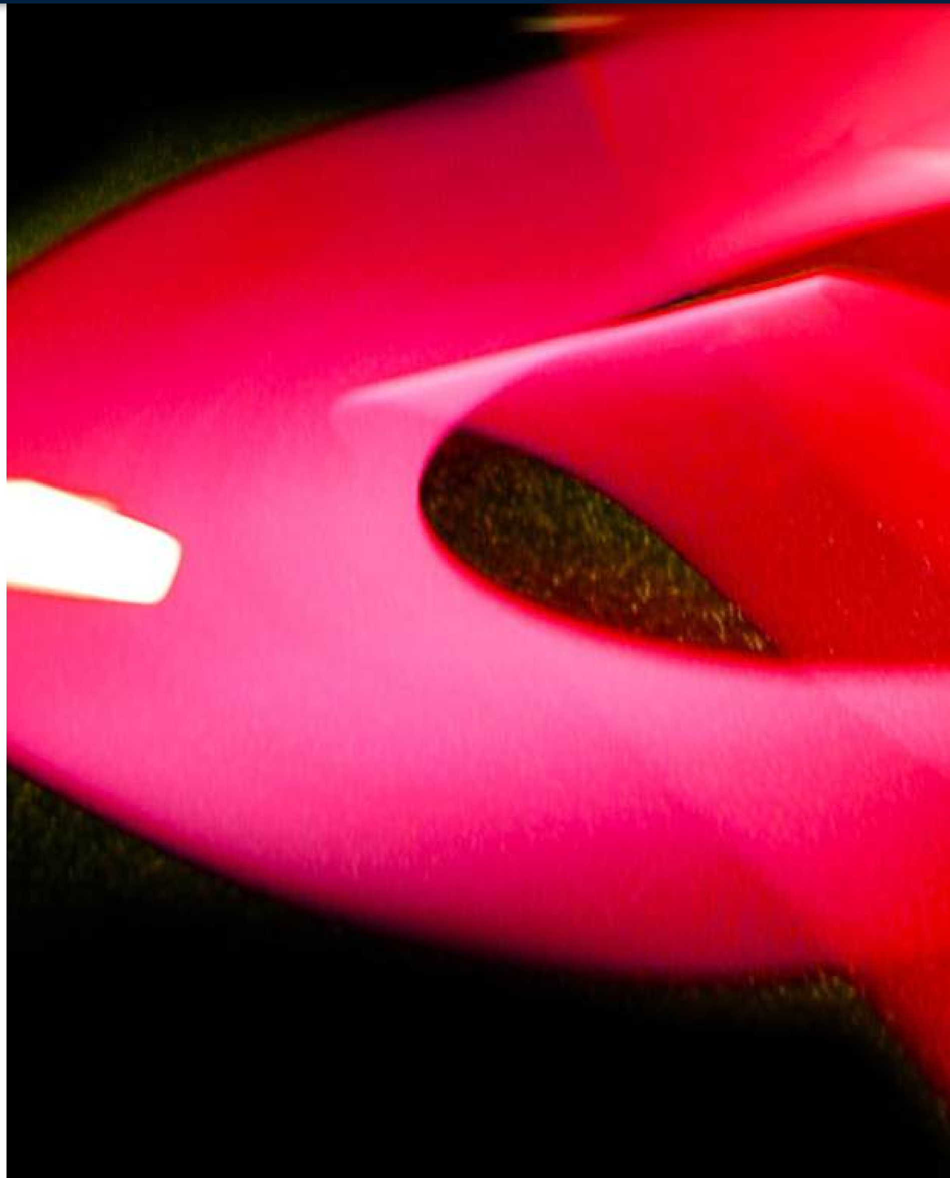
Around 7 in 10 rate AI as either important or very important for their business over the next five years, with a third marking it as “critical” to their future success. Just 9% consider it as just somewhat important or not important at all.



Importance of AI for business in the next five years



7 in 10 organizations see AI as an important development driving significant change and innovation over the next 5 years.
1 in 3 organizations see AI as critically important for their future competitiveness and success.



Since the emergence of ChatGPT and other generative AI models, it's clear that many organizations are experiencing a FOMO moment, convinced that everyone else is further ahead in adopting AI and desperate not to be left behind.

Placing bets

No doubting, then, how most organizations view AI and its likely impact. But there's a problem.

No one yet knows how that impact will be felt and most organizations sit in expectation of a transformed world without knowing quite what shape that transformation will take.

As we note in the introduction, AI has come a long way, but it has a last mile problem. While it is being developed at incredible speed it has yet to prove itself and it will continue to evolve rapidly.

That places organizations in something of a quandary, unsure where and how much to invest in AI. Since the emergence of ChatGPT and other generative AI models, it's clear that many organizations are experiencing a FOMO moment, convinced that everyone else is further ahead in adopting AI and desperate not to be left behind.

So, we are seeing investment rise rapidly. But, since few organizations have unlimited funds, most businesses are having to place bets on which applications will succeed and which will not. That means that very often AI is being implemented in a piecemeal way rather than holistically across all functions to harness its ability to fundamentally transform the organization.

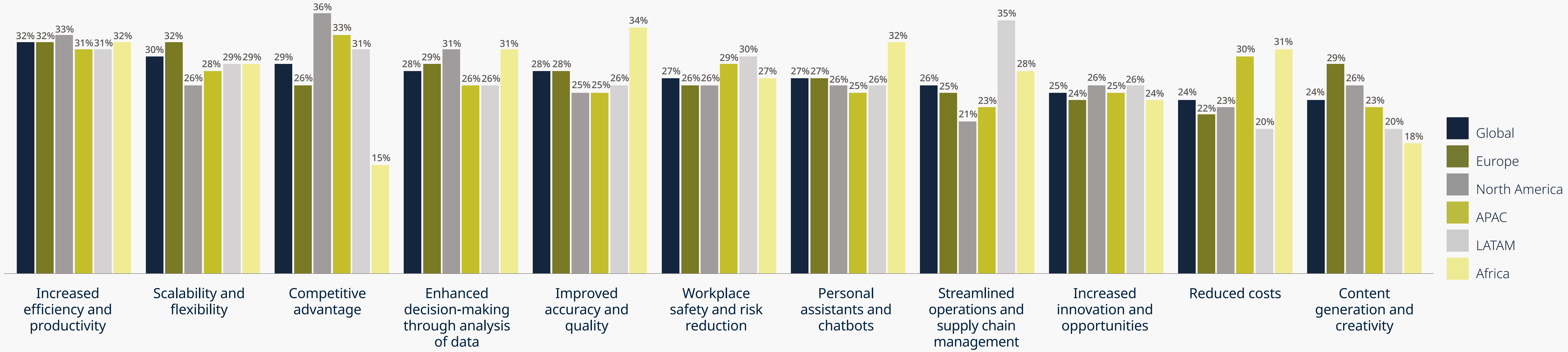
Our study once again clearly illustrates current thinking around the costs of AI investment.

Less than one in ten believe that the cost is so prohibitive that it is deterring them from investing in AI at all. An even smaller proportion think the cost of implementation will be minimal.

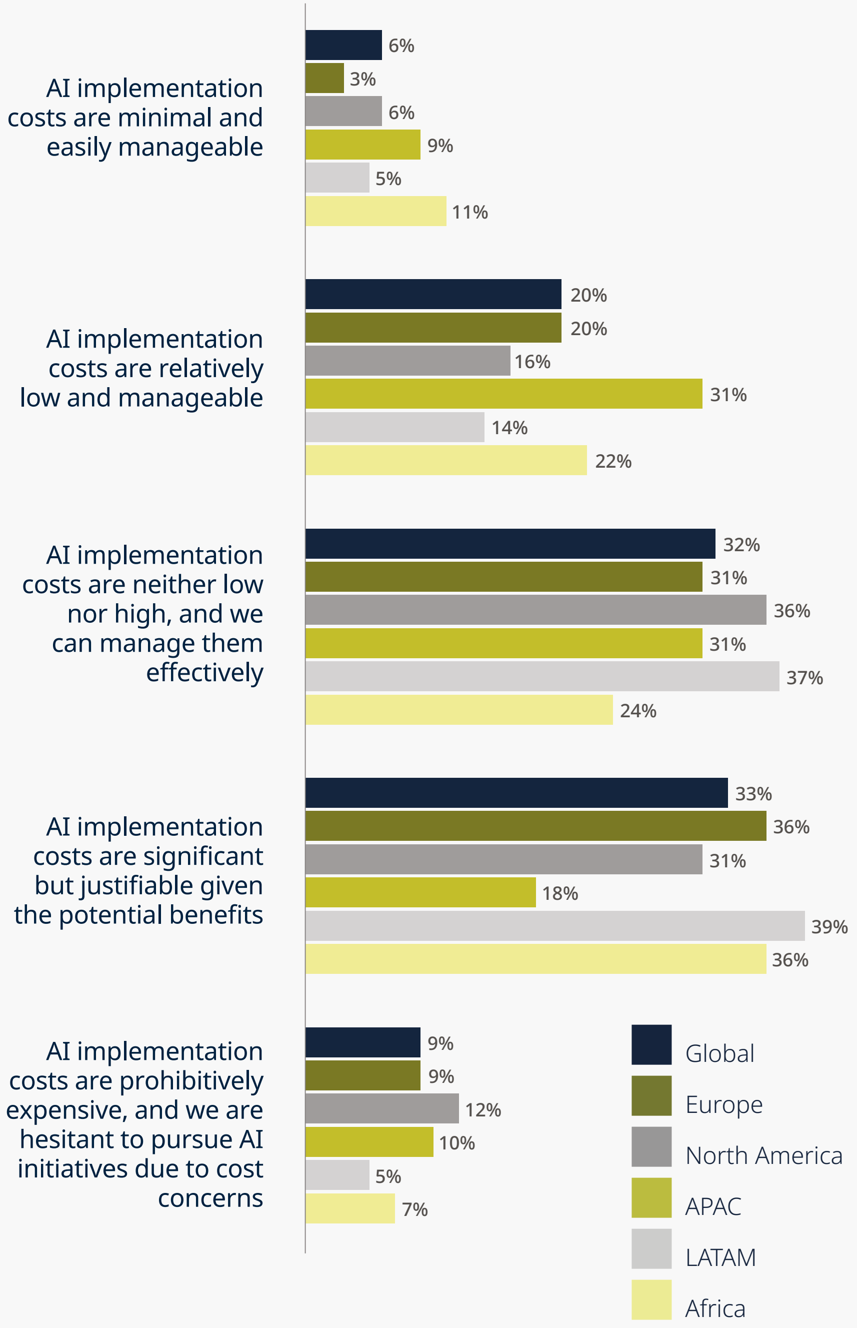
But perhaps more significantly, a third of respondents believe that the costs of implementation will be significant but justifiable given the huge potential of the technology and a similar proportion believe costs can be managed effectively.

> "AI is no longer just a technological advancement; it's a strategic imperative. Organizations must move beyond the fear of missing out and focus on integrating AI holistically across all functions. The real value of AI lies not just in cost efficiencies but in its ability to drive innovation, enhance decision-making, and fundamentally transform how businesses operate."
>
> **Niresh Rajah, Chief Data and AI Officer, DLA Piper**

Benefits of AI



Perception of the costs associated with AI implementation



Costs first, augmentation later

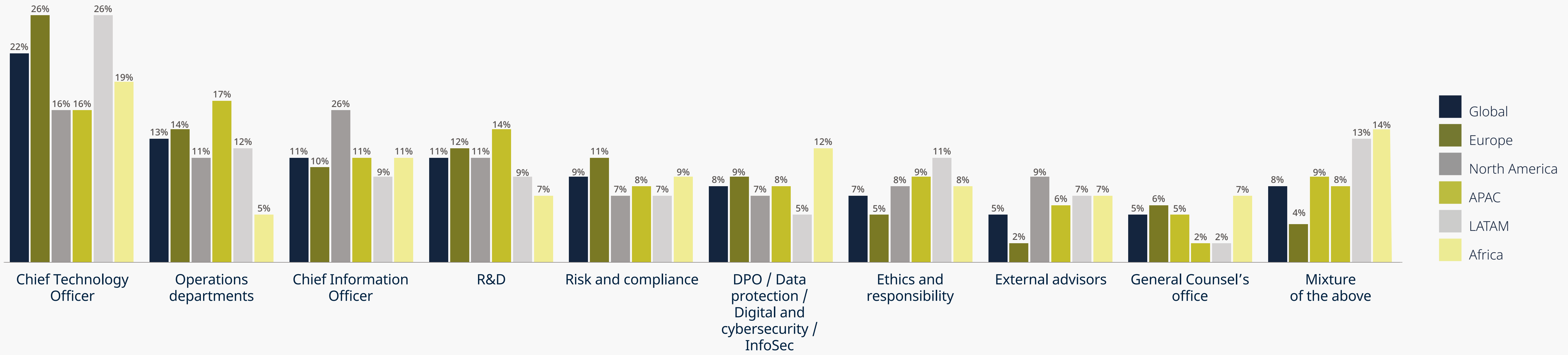
Another clear trend we are seeing in our practice is an early focus by businesses on how AI can help them to achieve cost efficiencies. But relatively few have moved to the next stage where AI is implemented to deliver truly transformative value.

It's a case of tackling the costs first and leaving augmentation to later.

Not surprising then that increased efficiency and productivity top the list of benefits that can be derived from deploying AI, with the ability to streamline operations and supply chains, reduce costs, improving workplace safety and reducing risks all featuring in the list of perceived benefits.

> Regulation always struggles to keep pace with technology, and that has certainly created significant uncertainty for organizations looking to invest in AI and other digital solutions.

Primary driving force behind the adoption of AI



Adoption of AI is most likely to be driven by the CTO, followed by the CIO and operations or R&D departments.

Leadership

Respondents appear certain that the main driving force behind AI adoption will come from the Chief Technology Officer, with operating departments and the Chief Information Officer also providing impetus.

But, oddly, the CEO does not feature as a driver of change at all. That's strange since the opportunities for deep business transformation and the significant financial, regulatory and ethical risks around deploying the technology certainly call for leadership from the very top as well as expert oversight from the boardroom.

A view shared by some financial services' regulators who have placed responsibility for governing the use of AI on management, including the UK where certain senior managers are considered personally accountable for the safe and responsible use of AI in their business.

In addition, for AI to achieve its true transformational potential it will require buy-in from right across the organization. It has to be a team effort, led by a strong captain.

> Without a doubt Artificial Intelligence is the one issue that dominates the agenda of most organizations, whether private or public sector.

Regulatory dissonance

Regulation always struggles to keep pace with technology, and that has certainly created significant uncertainty for organizations looking to invest in AI and other digital solutions.

Regulatory and legal compliance features quite high on the list of drawbacks our respondents see with implementing AI, as do ethical and social implications, while data privacy and security top the list.

In reality we are beginning to see extensive new regulation being enacted around AI, led by the EU with its AI Act, which came into force in the summer of 2024 and will be applied progressively over the next year or so.

As the world's first comprehensive regulatory framework for AI, the new law is already influencing thinking on AI regulation way beyond Europe's borders as authorities look how to govern this inherently complex area.

Despite that, it seems inevitable that the world will see a patchwork of regulation emerging in the coming years. It's notable, for example, that China has also passed hard law on AI regulation which in some instances goes further than the EU's AI Act.

There is a strong likelihood that regulation will go in different directions across different jurisdictions. For instance, since much AI development is still happening in the US, there is a danger that a West Coast American approach to managing issues such as bias and discrimination will be exported to other markets where entirely different ethical approaches are the norm.

Organizations will need to navigate a period of regulatory dissonance with great care. Even at this early stage, it is no longer enough to ask if AI systems are responsible and accountable, without also asking are they legal as well. But what's keeping people awake at night is a fundamental question: how do we innovate when there is so much regulatory uncertainty?

This is a particular issue in the US where state law is moving ahead much faster than federal law, especially in the febrile pre-election period where proposed federal legislation requiring bipartisan support is likely to get held up.

For example, Colorado has just passed a law on AI regulation, closely modeled on the EU AI Act. California is now promising to do the same. Businesses wanting to operate in these markets will need to make sure their processes comply.

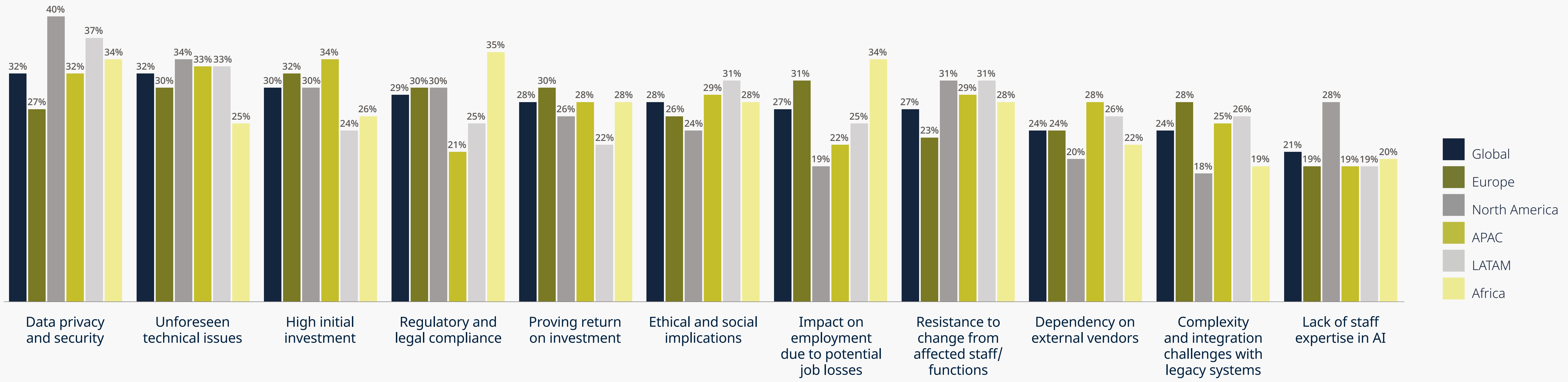
The same applies to US business looking to operate in the EU. Given that, some multinational businesses are looking to make sure they comply with the toughest regulatory regime and then apply that to their operations worldwide.

International efforts to co-ordinate the regulatory response to AI are helpful, not least the symposium first held at the UK's Bletchley Park, with follow up meetings in Korea and France, bringing together governments, regulators and leading developers.

It's important that major AI providers agreed at these meetings that their systems can be tested and certified, and it's notable that the UK's Department of Science, Innovation and Technology has set up an AI Safety Institute to do so.

But it remains to be seen to what extent such initiatives will create the degree of certainty that organizations crave.

Drawbacks of AI



Litigation risk

We're also seeing a growing risk of litigation around the deployment of AI. The first wave of litigation has been around copyright claims brought against the big AI developers.

But we are just beginning to see class action cases being brought against companies deploying AI models.

These are often around accusations of discriminatory hiring practices informed by an algorithm with an in-built bias, but we also expect to see more accuracy-based cases coming to the courts soon.

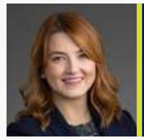
Keeping compliance up to date

Given the sheer level of legal complexity developing in this area, organizations are looking to test their AI systems not only for their technological capabilities but also for legal compliance.

Red teaming has been regularly used to test technologies, but we are now using legal red teaming to help clients put their systems through rigorous trials to identify areas where they might fall foul of the law.

It's work that has gained great traction with some of our major clients giving them greater confidence that their compliance and wider governance processes are up to date in a complex and fast-changing environment.

Key contributors



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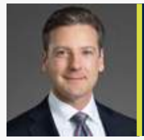

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Cybersecurity – the threat intensifies

Action to counter the threat of cybercrime has had little impact on the levels of attacks we are seeing across the globe. Cybercrime is becoming both bolder and more brash, with attacks continuing to rise in frequency and sophistication, in part increasingly powered by AI technology which bad actors have harnessed to their advantage. The threat remains asymmetrical, with criminals able to launch ever more devastating attacks much faster than private and public sector organizations, constrained by budgets and bureaucracy, can build defenses against them.

Cybercrime is becoming both bolder and more brash, with attacks continuing to rise in frequency and sophistication.

The use of AI by criminals is a major new trend.

It means that the tell-tale signs that once would hint at an email phishing attack – spelling mistakes, frequent typos, for instance – have been eradicated using powerful generative AI models, criminal versions of systems like ChatGPT. We're also seeing the use of deepfakes in the context of verification calls made in the wake of a phishing attack.

And AI has opened new threat inroads such as data poisoning where bad actors attack the data training the AI being used by organizations, so that whole systems reliant on it can become infected.

With the regulatory environment becoming tighter, bad actors are even turning this to their advantage.

As an added lever to extort a ransom, we've seen attackers actively approaching regulators (or threatening to do so) to report companies trying to keep quiet about suffering an attack when it should be promptly declared to the authorities.

In addition, the barriers of entry into this criminal activity have been lowered as 'ransomware as a service' becomes more prevalent. You don't need a team of expert coders to staff an attack. The software can be bought off-the-shelf from other criminals.

In an increasingly tense geopolitical environment, we are also seeing state-sponsored cyberattacks on the increase, whether for the purposes of industrial or military espionage, sanctions busting, or, increasingly, to attack important infrastructure including energy and water utilities, financial institutions and healthcare systems.

Even important regulators are not immune from attack. Canada's anti-money laundering authority, the Financial Transactions and Reports Analysis Centre (Fintrac) was hacked in March 2024, shutting down its web reporting system.

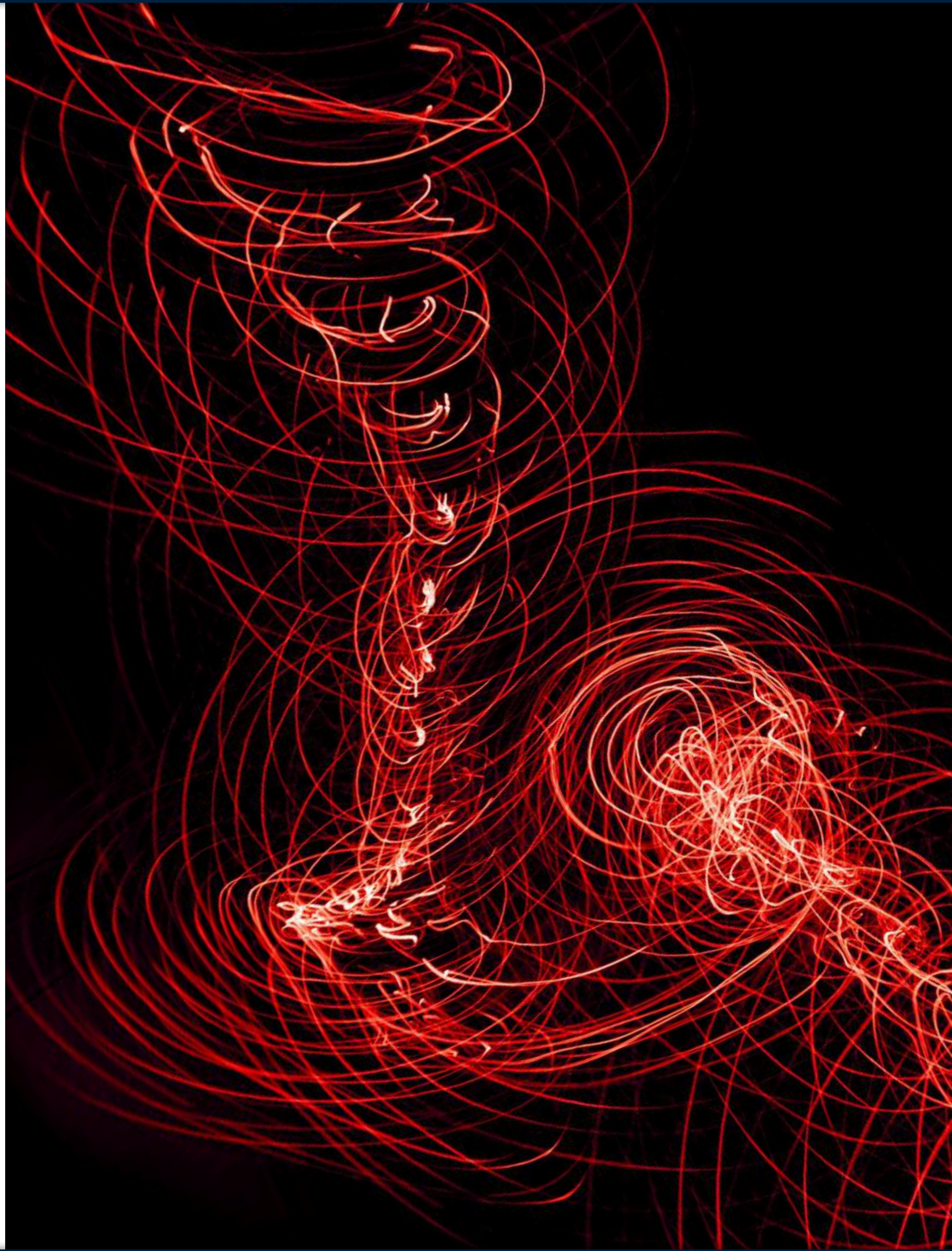
This collects data on millions of suspicious or large transactions providing key intelligence used in blocking money laundering and the financing of terrorists and organized crime.

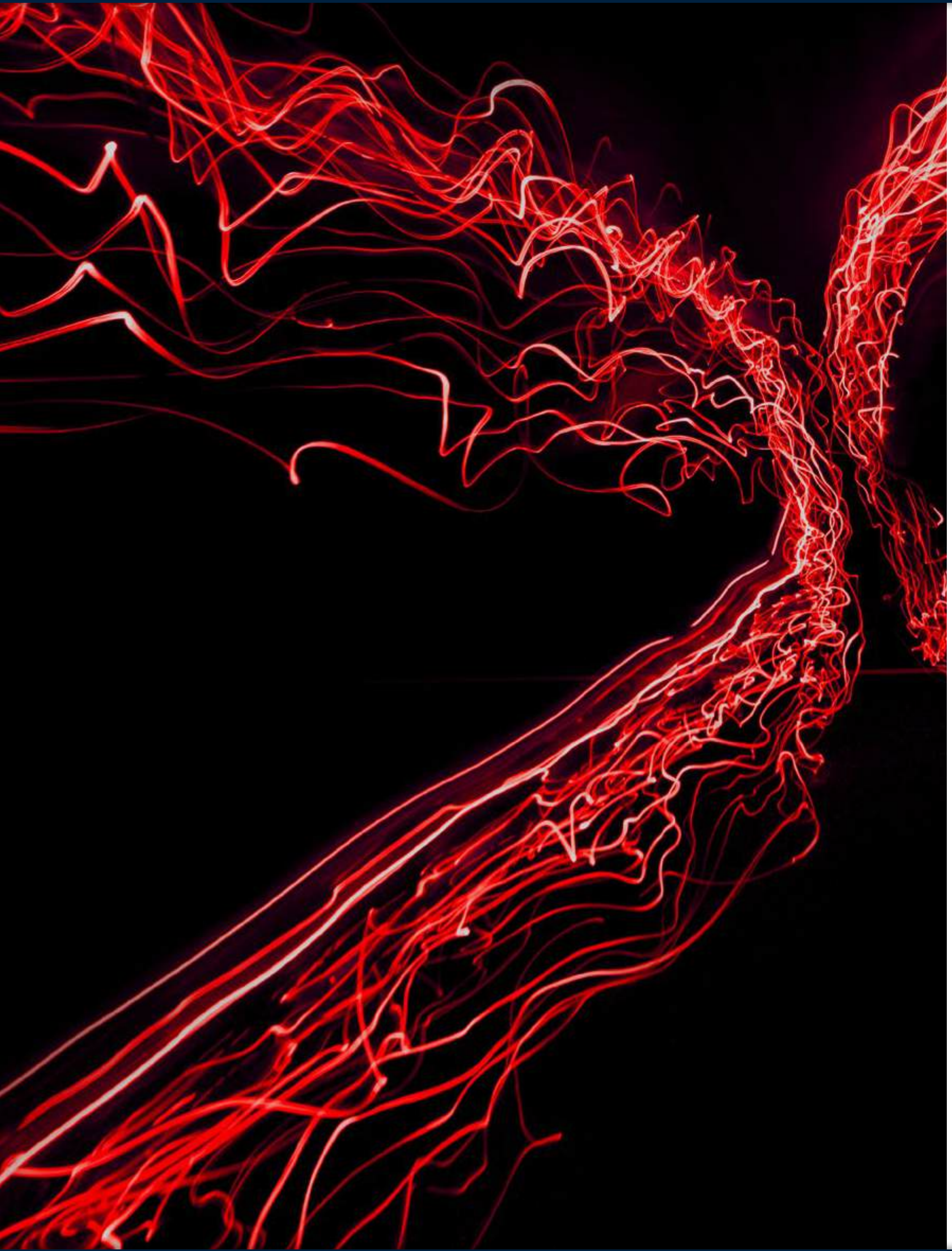
And the problem is a global concern.

Until two years ago, large-scale cyber incidents were rare in Australia. But that has all changed with high-profile attacks on Optus, the telecoms giant, and Medibank, the private health insurer, both impacting around nine million individuals. In 2023, Latitude, a financial services provider, also suffered an attack affecting an estimated 14 million current and former customers across Australia and New Zealand, despite some individuals not having dealt with the company for many years previously.

Elsewhere in Asia, we have seen a sudden and massive spike in attacks in recent months with personal data exfiltrated from systems and posted on the dark web to extort a ransom.

AI has opened new threat inroads such as data poisoning where bad actors attack the data training the AI.





The regulators fight back

That's left the target organizations in an uncomfortable pincer movement between growing amounts of criminal activity on the one hand, and ever more stringent cyber regulation on the other, often accompanied by increasingly punitive fines.

In the US, EU and UK, the requirements to report cyber breaches have become more stringent, with fines imposed on organizations succumbing to an attack escalating rapidly.

IBM, for instance, has estimated that the average cost of a cyber breach has now risen to nearly USD4.5 million, with ransomware attacks remaining the main threat vector.

Governments and regulators in these jurisdictions are also taking action to make vital national infrastructure more secure with tough new rules.

These include:

- The US Cyber Incident Reporting for Critical Infrastructure Act (CIRCIA) which imposes a 72-hour deadline to report an incident and a 24-hour window to report a ransom payment having been made.
- The EU's NIS2 directive which came into force in 2023 and aims to boost security across the bloc. It updates earlier regulation, placing obligations not just on infrastructure operators but also on companies providing services to them.
- The Digital Operational Resilience Act (DORA), which comes into force in January 2025 aimed at EU financial institutions but which also applies to UK institutions serving customers within the EU.

Following the fallout from recent large-scale incidents, such as the Optus breach, regulation is now being developed at pace in Australia alongside tougher privacy laws, with the government pledging to make Australia the most cyber-secure nation by 2030.

Beyond changes to legislation, the Australian incidents also demonstrate a more pro-active approach to enforcement, with Optus currently the subject of investigations by the privacy and the telecoms regulators, as well as several class action challenges in the courts. Earlier in 2024 Optus lost an appeal to prevent its forensic incident reports from being disclosed. That means that further details of the breach, and Optus' response to it, will be revealed and the rest of corporate Australia is watching these cases carefully.

The picture across the rest of Asia remains more complex, where regulation is far less mature and varies significantly from jurisdiction to jurisdiction. Generally, fines, if imposed at all, are much lighter. In some regimes there is no obligation to report incidents at all. However, operational risks (such as websites being blocked by regulators, or apps taken down off app stores) and corresponding contractual risks are prevalent and must be managed carefully.

Multinationals operating in the region are finding it is dangerous to rely on being compliant with EU or US regulation in the hope this will see them through in other markets. They need expertise on the ground, but often struggle to find cyber experts in the region with the right skills to fulfil this role.

> As cybercriminals leverage AI to outpace defenses, organizations must not only innovate but also fortify basic security practices to withstand an ever-evolving threat landscape.

Complacency?

Given this ever-worsening threat scenario and the march of increasingly strict regulation, our survey throws up some surprising results.

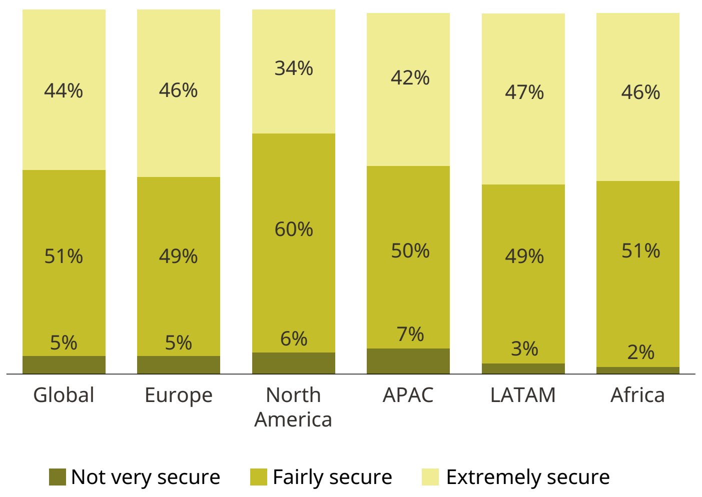
For instance, respondents have an extraordinarily high level of confidence in their security levels, with over half saying they feel at least fairly secure from cyber threats, a figure that rises to 60% in the US.

Overall, a quarter express real concerns that they may suffer an attack or breach, although that concern is most acute in Europe where a third of respondents say they have a high level of worry. But that indicates that the vast majority do not share the same concern.

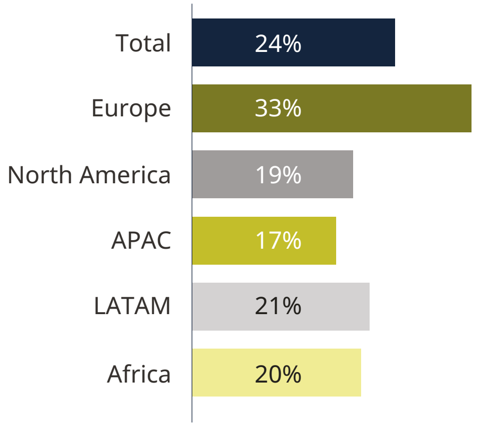
Organizations in our survey have a range of policies and procedures in place to meet basic cyber threats, most notably regular software updates, using cybersecurity software, banning the use of personal devices and strong password policies, the latter being particularly important in the US (72%).

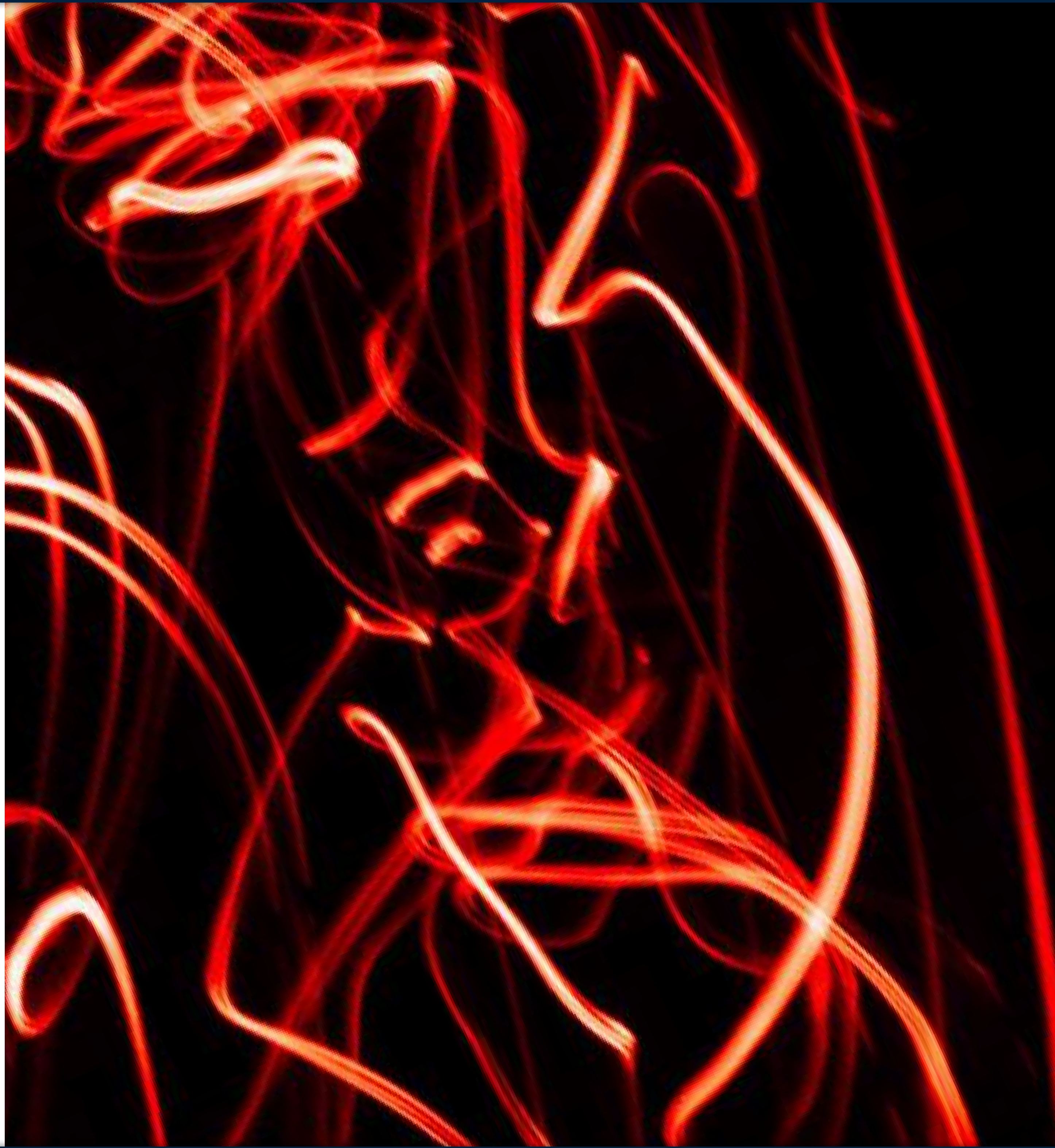
Yet there are some worrying findings here too. For instance, only 49% say they have strong digital security measures in place such as anti-virus software and firewalls and just 45% conduct regular risk assessments. That means over half, in both cases, don't – a stark finding.

Level of security against a cyberattack

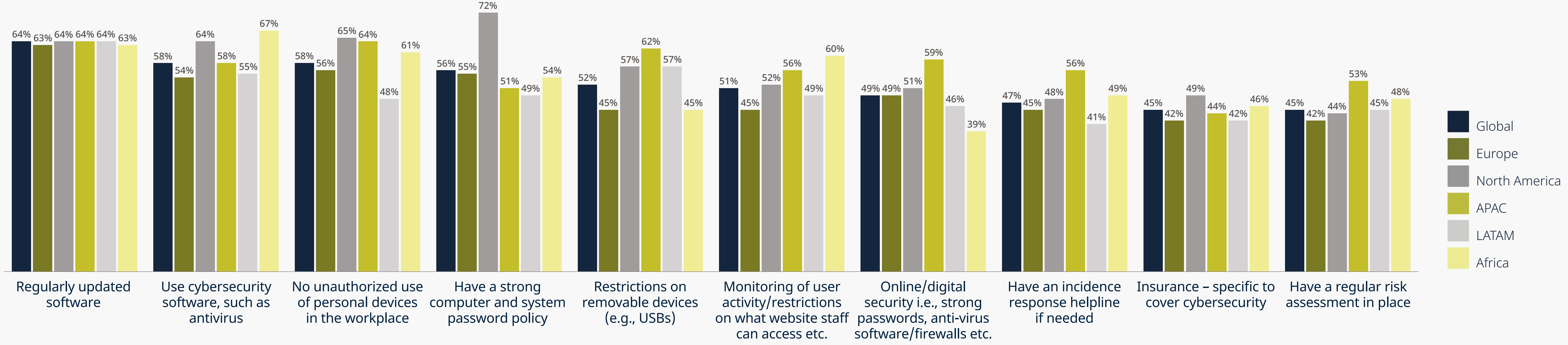


High level of worry about a cyberattack/breach





Policies in place to meet cybersecurity issues



Organizations have a range of policies in place to meet basic cybersecurity issues, most commonly software updates, use of cybersecurity software, prohibiting unauthorized use of personal devices and strong password policies, which is of particular importance in North America.

There are also some notable discrepancies in the data. Although 79% say they have a specific person in place to manage a cyber-attack, a slightly lower proportion (75%) say they have a response plan in place, suggesting there's a person in place with no plan.

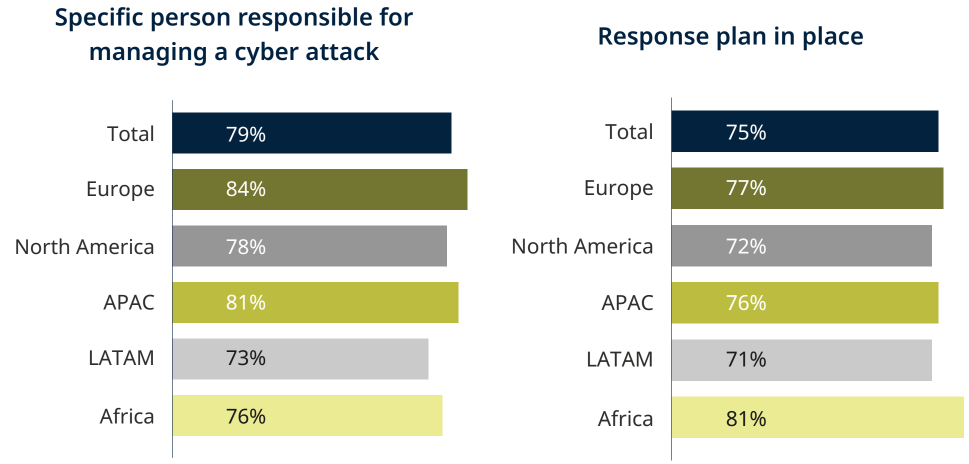
However, this might be misleading. In our experience organizations are more likely to have a response plan in place but have yet to put it into practice across the organization or test it in any meaningful way.

Whether these findings point to a sense of complacency is a moot point. Certainly, organizations that have experienced an attack and are therefore fully aware of the operational, reputational and boardroom liability risks and the escalating costs involved, have a much better understanding of the issue.

But some are in a different place, perhaps needing to go through an attack to learn those lessons. We are working hard to inform organizations with this mindset of all the actions they can take to protect themselves.

> Cybercrime is evolving at a pace that outstrips the ability of organizations to defend against it, leaving them vulnerable to increasingly sophisticated attacks.

Procedures in place for managing a cyber-attack



Around 3/4 of organizations across all regions have a specific person responsible for managing a cyber-attack and a similar proportion have a response plan in place.

Getting the basics right

The attitudes uncovered by our survey may just be an indication of a growing confidence among our respondents in the power of technology to guard against attacks. That is to some extent justifiable.

The worst-case scenario would be for organizations to react to the use of AI-enabled systems by threat actors by throwing up their hands in defeat, unable to see how they can fight these super-intelligent weapons.

But it’s worth noting that, although highly capable, these systems are not so intelligent that they cannot be beaten.

And there have been some positive law enforcement developments in combatting cybercrime, not least the joint operation led by the UK’s National Crime Agency in conjunction with Europol and the FBI to take down LockBit, one of the most active ransomware gangs with over USD120 million in ransom fees.

But, technology and enforcement aside, one truth continues to apply: the weakest link in the chain remains human beings – employees, managers, people out on the road who approve a phishing email without doing the proper checks. The AI being deployed in cyber-attacks will continue to exploit this human vulnerability.

That means it remains essential to get the basics right, through rigorous governance policies and procedures properly disseminated through the entire organization, continuous training, effective reporting and threat escalation lines and regular tabletop exercises to model and prepare for attacks.

There are some easy wins here. Such measures, carefully deployed and continuously updated, can make sure that people within an organization know their roles and are ready to take the right action when the inevitable happens.

> The weakest link in the chain remains human beings; employees, managers, people out on the road who approve a phishing email without doing the proper checks.

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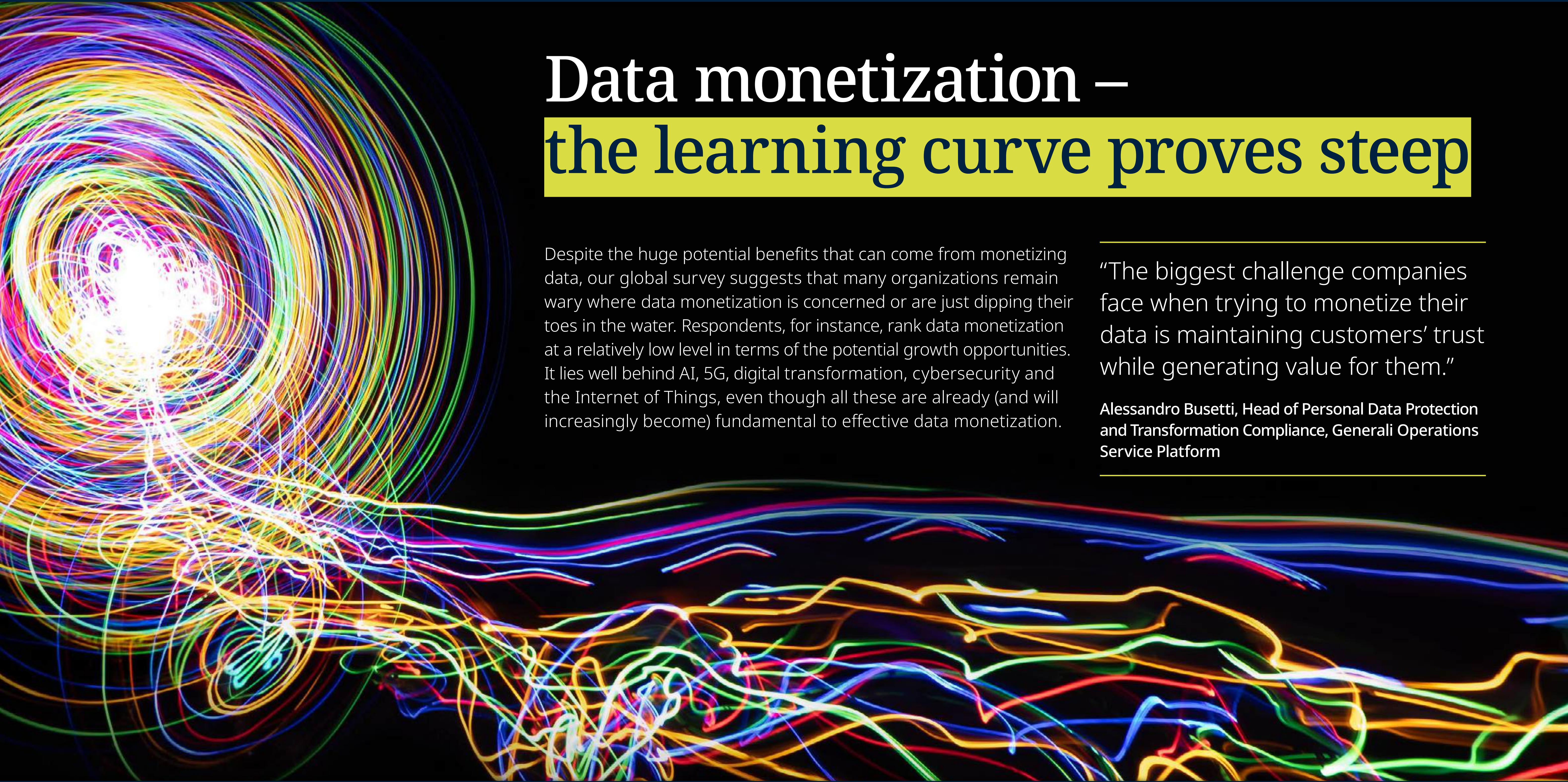
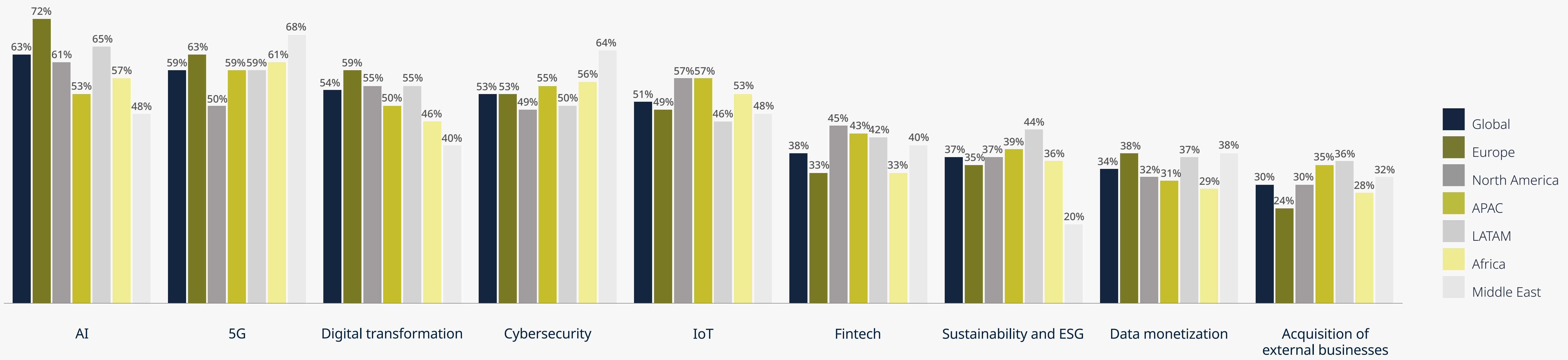


Data monetization – the learning curve proves steep

Despite the huge potential benefits that can come from monetizing data, our global survey suggests that many organizations remain wary where data monetization is concerned or are just dipping their toes in the water. Respondents, for instance, rank data monetization at a relatively low level in terms of the potential growth opportunities. It lies well behind AI, 5G, digital transformation, cybersecurity and the Internet of Things, even though all these are already (and will increasingly become) fundamental to effective data monetization.

> "The biggest challenge companies face when trying to monetize their data is maintaining customers' trust while generating value for them."
>
> **Alessandro Busetti, Head of Personal Data Protection and Transformation Compliance, Generali Operations Service Platform**

Potential opportunities for future business growth



Using a scale of 1-9, where 9 represents the highest potential and 1 the lowest, we asked respondents to rate the opportunities based on their potential for business growth. The percentages indicate the proportion of respondents who rated a particular opportunity as 7, 8, or 9. AI, 5G, digital transformation and cybersecurity were ranked highest.

> "As far as I'm concerned, data monetization is not viewed by banks as a major commercial opportunity, except for sharing data within their group of subsidiaries. That being said, I believe the biggest challenge for a company looking to monetize its own data is ensuring its security, both in terms of cybersecurity and intellectual property protection."
>
> **Alessandra luliucci, Senior Director, Intesa Sanpaolo**

It is perhaps not surprising, then, that less than a third of respondents say they are currently making full use of data monetization in their operations.

That figure rises to 38% for respondents in Europe, where data regulation is most mature and increasingly pervasive, but falls as low as 19% in the US and 18% in the Asia Pacific region where regulation is currently less developed and much weaker.

However, the number of organizations making limited use of data monetization is considerably higher, rising to 57% in the US and 54% in Asia Pacific, with scores in Latin America and Africa at 51% and 43%, respectively.

This finding is much more in-line with what we are seeing in our own practice, where clients are increasingly looking for opportunities to process and repurpose data, both in highly regulated sectors like financial services and life sciences, as well as those where regulation is lighter.

Although adoption may be slow, we believe many organizations do in fact see significant untapped potential in monetizing their data.

Potential roadblocks

Part of what is slowing adoption is a fundamental misunderstanding of what data monetization actually is.

In the early days it was almost exclusively associated with gathering personal information which, once used, could then be sold on to others.

This approach has been significantly curtailed by the introduction of tough new regulation, particularly the EU's GDPR data protection law which imposes strict limits on what kind of data can be retained and curbs on the purposes it can be used for.

Organizations whose GDPR journey proved difficult may well have been scarred by the experience and are, therefore, cagey about exploring other ways to monetize data.

But, in our experience, that is by no means the general picture.

Many organizations we work with define monetization in a broader and more useful way. For them, monetization means putting data gathered for one purpose to work in another context often to make efficiency gains and cost savings, with the potential to have a direct impact on both top and bottom-line financial performance.

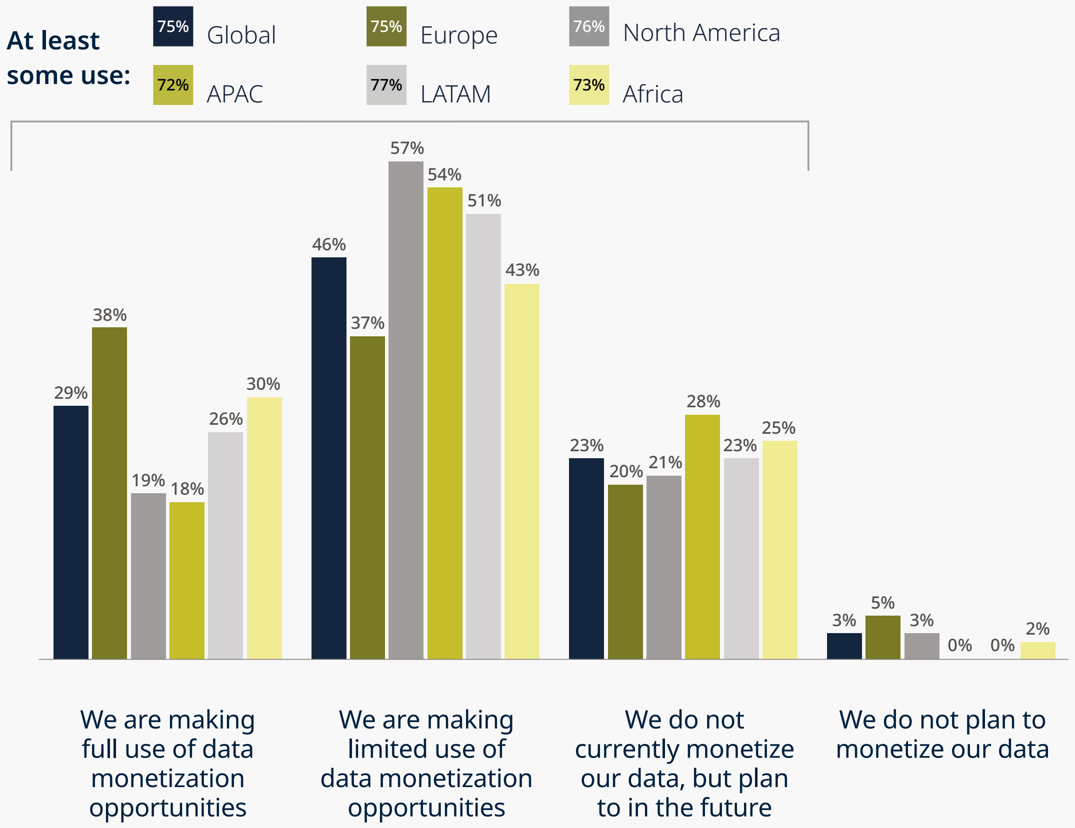
More often than not it is about improving customer relations and gaining a better understanding of customer preferences, to improve marketing and drive sales.

But equally it could be about improving industrial process to make production lines and maintenance processes more efficient, to improve supply chain management and procurement, or to fine-tune product development.

Our survey illustrates the variety of ways organizations are looking to monetize data. Asked which areas of the business benefit most from data monetization, our respondents put customer service and customer relationship management at the top, closely followed by improved IT systems, sales and marketing.

Over half of respondents in Africa highlighted the improvements in production processes as a significant benefit of data monetization, significantly higher than in any other region.

Use of data monetization



3/4 of those who see growth opportunities from data monetization are already making at least limited use of these opportunities. However, less than 1/3 globally are currently exploiting these opportunities to the full.

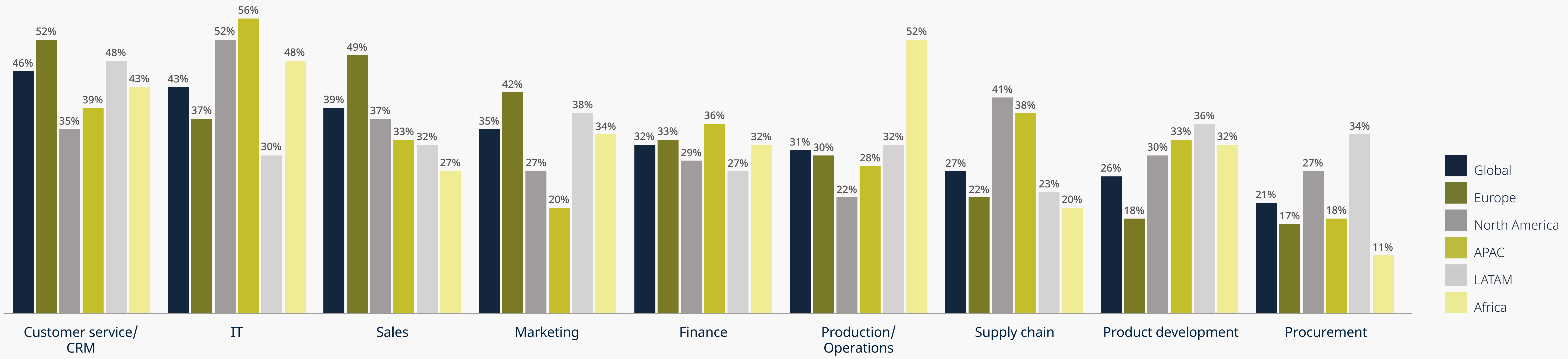
Good data, bad data

Many of the challenges of monetizing data highlighted by our survey relate to the quality of data. This is a critical issue since gaining optimum results from processing data depends on the input data being robust. If not, it is a case of garbage in, garbage out.

Many organizations are put off by the sheer scale of unstructured data they have at their disposal, often dispersed across different systems and different business functions.

They need to hire data scientists skilled not only in handling huge data sets but doing so with a commercial mindset to consolidate data into a searchable pool.

Departments to benefit most from data monetization



Customer service, IT and sales are identified as the departments to benefit most from data monetization. Particular benefits for production/operations are highlighted in Africa.

It is notable that the quality and consistency of data, collection and mining of data, data analytic skills, and commercial exploitation all feature among the main challenges of data monetization identified by our respondents. Knowing the provenance of ownership of data to make sure it can be used for the purposes intended is also identified as a challenge.

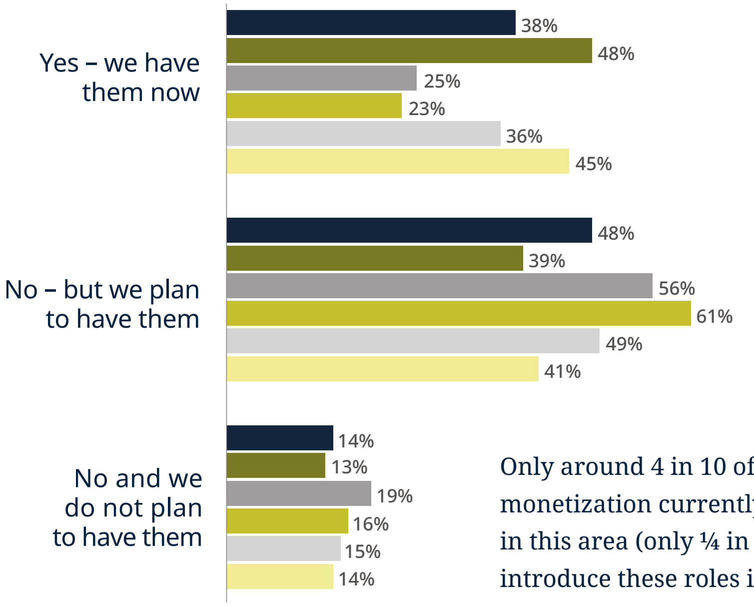
Data quality is often a particular challenge for public sector organizations, where data sets can be vast and can stretch back many years, even decades. Breaking down that data so that they are accessible to the power of data science is extremely challenging.

By contrast, start-up businesses that design their systems from scratch with the intention of making the best use of data are in a much more favorable position.

Given the central role of data scientists, it is interesting to note that just 38% of our respondents who see opportunities from data monetization currently employ data scientists, and that figure falls to just 25% in North America. However, half say they plan to do so in future.

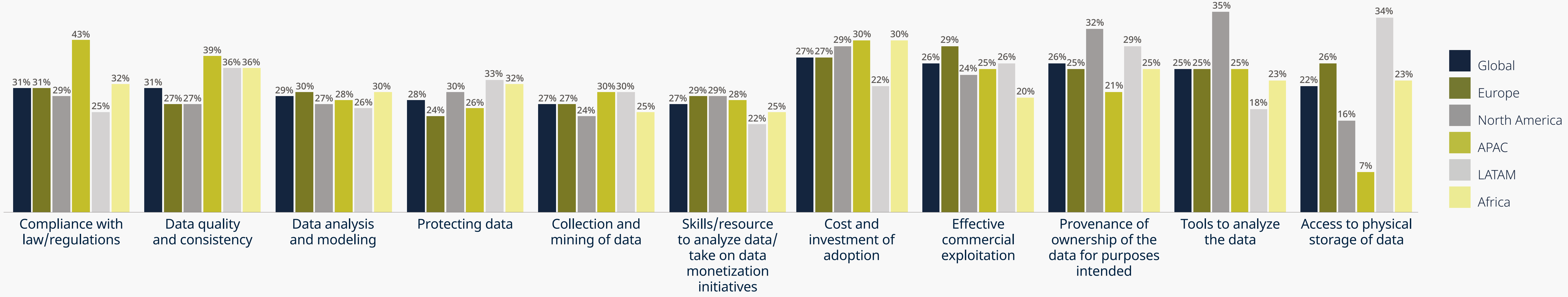
This reflects both an inherent misunderstanding about the role data scientists can play but also, perhaps, points to a wider issue around education and training. In some countries we are seeing a big push to design data science degree course which have a strong commercial as well as technological focus. Poland, perhaps because of the proliferation of multi-national companies setting up shared service centers there, is a case in point.

Presence of data scientists



Only around 4 in 10 of those who see opportunities from data monetization currently have specialist data scientists working in this area (only ¼ in North America and APAC) but ½ plan to introduce these roles in future.

Challenges in maximizing the benefits of data monetization



Regulation – a help or hindrance?

Given the volume of EU regulation that is already in force or in the pipeline directly impacting on data use or in related areas such as AI development, privacy and cybersecurity, it is surprising that concerns about regulatory compliance rank relatively low in our survey.

Some businesses clearly worry that the market is in danger of becoming over-regulated, yet under a third of respondents in Europe cite this as a challenge to maximizing the benefits of data monetization, almost on a par with findings from North America where 29% see it as a challenge.

That may reflect the fact that many organizations are beginning to recognize that, despite its increasing complexity, regulation can act as a spur to innovation rather than putting a brake on it. It is notable that a much higher proportion of respondents in Europe say they are making full use of data monetization, than in other regions – and twice as many as in North America.

The EU's Data Governance Act, which applied from September 2023, and the Data Act, for example, explicitly set out to facilitate voluntary data sharing and to empower businesses and consumers to access, use and share data securely. These measures should eventually make more data available to delve into and monetize.

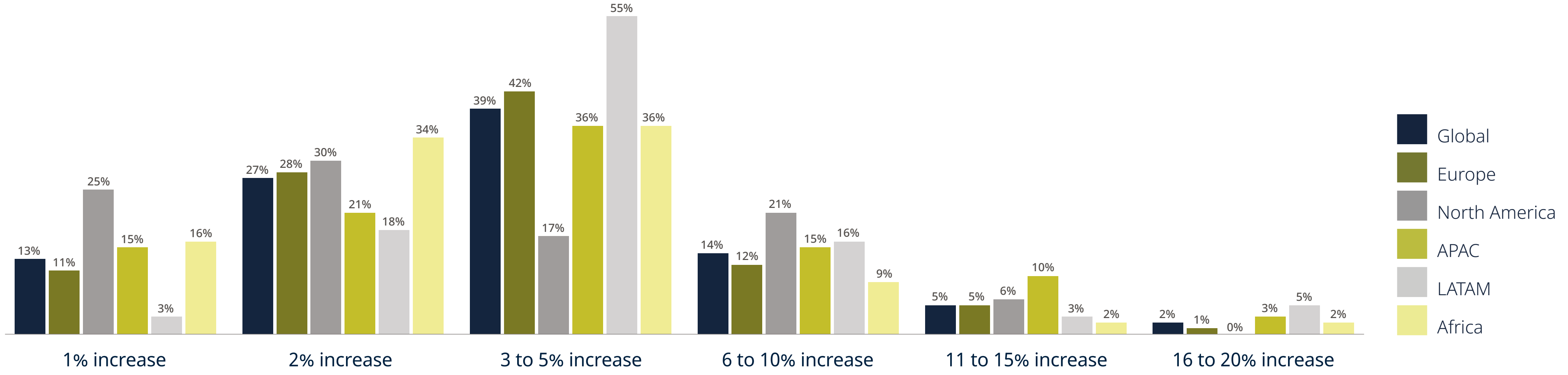
Financial benefits

Despite relatively low numbers currently making full use of data monetization, expectations around the financial benefits of monetizing data are strong and consistent across all regions.

And if organizations can find ways of monetizing data at relatively low cost, that will mean not only a healthy increase in turnover but also a significant boost to profit margins.

> "To turn data into currency, companies must navigate challenges like compliance with data protection regulations, maintaining data quality, and investing in technology and training. Success in a data-driven business model not only adds shareholder value but also enhances customer experience and generates social value."
>
> **Michela Pertino, General Counsel, Italtel**

Expected revenue increase from a focus on data monetization



Overall, 60% say that they expect their revenues to increase by 3% or more thanks to data monetization, and one in five put that figure at 6% or more.

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Digital transformation – the never-ending journey

Digital transformation has been a buzz word across almost all sectors for many years now, yet relatively few organizations have really grasped the scale of change that real transformation entails. In that sense the term digital transformation has become somewhat overused and misses the point – that organizations are on a continuous journey of deep digitalization, with the power to fundamentally change their business models, particularly with the introduction of advanced AI technologies.

After all, it's not long since the advent of blockchain technologies, which some believed would be equally transformative, but which are still finding their homes in business processes. Not all organizations have fully grasped the potential of cloud computing, and the Internet of Things is still in its relative infancy. Now we are beginning to glimpse the possibilities that AI can bring and next it will be quantum computing.

It's a case, therefore, of digital evolution rather than digital transformation and companies need to rethink their approach to becoming a technology and data-centric enterprise and be nimble enough to change and adapt to exploit the opportunities that new and improved technologies will present. That approach must embrace that true transformation is really an evolutionary journey.

Incremental change

For many companies, digital transformation tends to remain focused around discreet projects, with start and end dates and budgets allocated accordingly.

Generally, this approach results in particular systems and processes being optimized using digital technology rather than transforming the entirety of the enterprise to operate in a radically different way, becoming more digital-centric and, over time, more AI-centric.

Our Tech Index survey in many ways reflects this current state of play.

When asked about the greatest benefits of digital transformation our respondents highlight incremental improvements to the business consistent with a project-based approach to digital transformation.

For instance, achieving greater flexibility and agility to meet business needs is cited by a third of respondents. Having more streamlined processes ranks at the same level in most regions, although it is highest in North America.

Improved competitive edge, better information flows and speeding up the deployment of new technologies all come in at around the same level, although reducing IT costs were seen as a greater benefit in Africa than elsewhere.

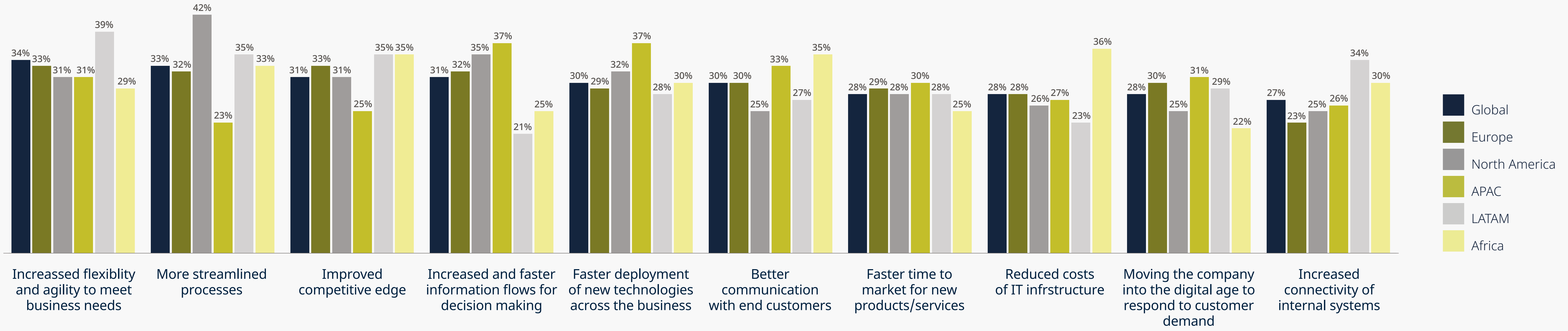
A similar picture emerges around what are perceived as the greatest challenges of digital transformation.

Here, exposure to greater security risks ranked highest, especially in Latin America. The cost of disruption to the existing business were cited by 40% of respondents globally, as did the integration of new and legacy systems – the latter highlighted by over half our respondents in Africa.

Nearly four in ten expressed concerns about the levels of investment required while finding the right technology partner was the biggest concern in the Asia Pacific region.

> It’s a case of digital evolution rather than digital transformation, and companies need to rethink their approach to becoming a technology and data-centric enterprise.

Greatest benefits of digital transformation



A range of benefits from digital transformation are recognized, including increased flexibility (particularly important in LATAM), more streamlined processes (particularly important in North America), improved competitive edge and improved information flows. Again, cost reductions are seen as an important benefit in Africa.

It's still early days

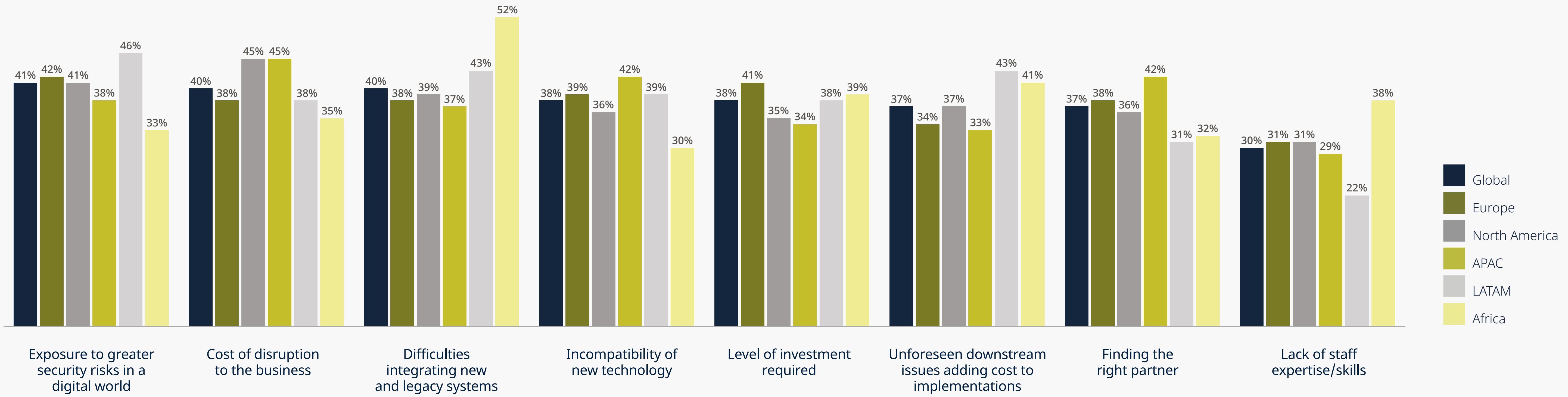
It's true that the power and potential of AI is on every organization's agenda right now.

However, as we note throughout this report, part of what's preventing organizations from understanding the scale of transformation they can achieve revolves around the fact that AI remains at a relatively early stage of development.

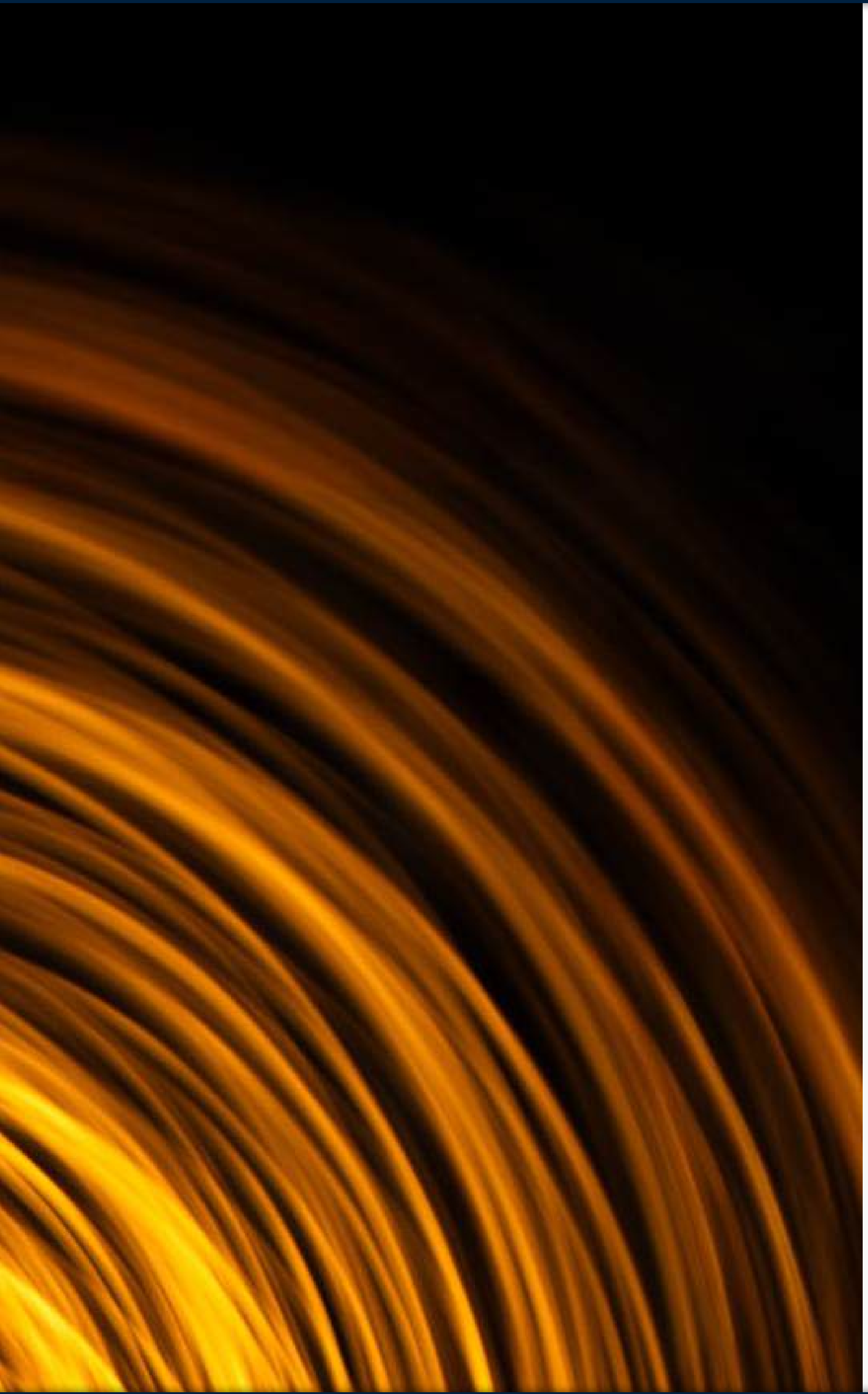
There's a tendency to call everything AI when a lot of what is being given that label refers to robotic processing automation, machine learning, more powerful search capabilities and prediction models that have been around for some time.

While organizations are starting to test the water when it comes to the adoption and implementation of generative AI solutions, the true transformative potential of generative AI is still several years away from being realized and put into operation, albeit we anticipate that the pace of change will increase rapidly and the regional variations are likely to influenced by regulation which, especially in the European Union, will put a significant compliance burden on organizations, akin to that of GDPR in 2018.

Key challenges to the implementation of digital transformation



There are a number of challenges associated with the implementation of digital transformation, including exposure to greater security risks, cost of disruption to the business and difficulties integrating new and legacy systems. The latter is particularly important in Africa.



Leadership and pressure for change

While that's the case, decision-making is becoming clogged and investment choices are being slowed down while organizations assess the risks of a multitude of AI offerings and work out where best to place their bets.

Our survey finds that the drive to embrace digital transformation is still coming mostly from internal IT teams followed by senior management (21%). However, almost as many (19%) say external suppliers are a main driving force. Customers, internal marketing and customer service teams rank much lower although that could be due to internal IT teams or senior management aptly anticipating customers' expectations, as many respondents cited responding to customer demand as one of the greatest benefits of digital transformation.

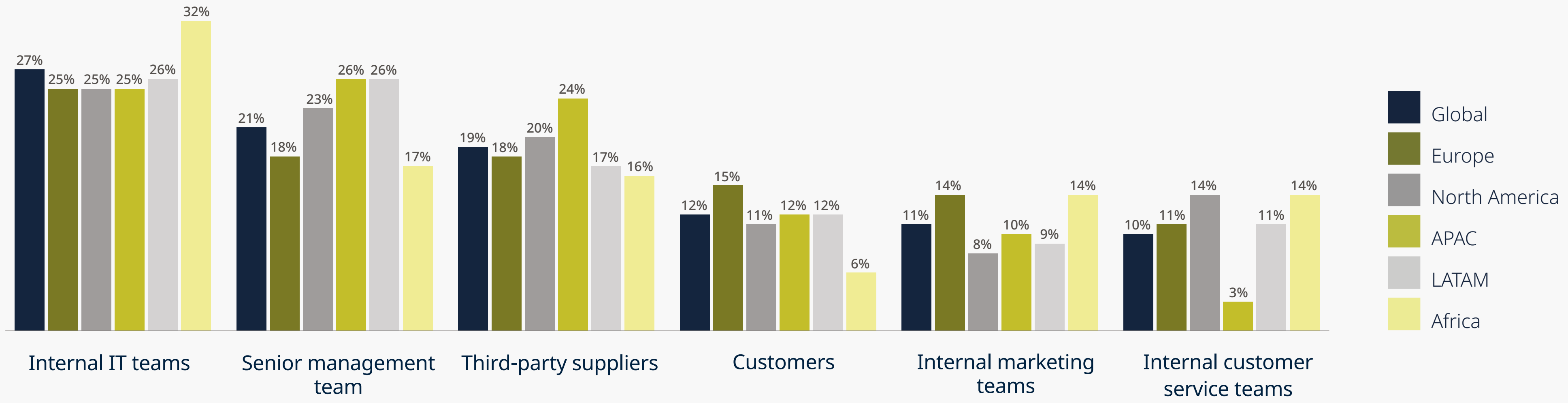
Given the scale of business model transformation that is likely to be enabled by AI, this is clearly an area requiring strong leadership, so its perhaps surprising that the role of senior management is cited by only a fifth of respondents.

Outsource providers are making their presence felt, coming to their clients with AI solutions in the hope of developing an eye-catching proposition that might have wider applications across different sectors. Often it's a case of AI looking for a problem to solve. This is only adding to the 'noise' that boards need to work through to make sensible, strategic decisions.

But the truth is that until the potential of AI is more visible, it's very difficult for boards to understand the strategic value of investing in it.

Companies that have people in the C-suite with a technology background are likely to make faster progress in understanding the transformative impact of AI and the real needs it can successfully address. Where that's not the case, the tendency for many leadership teams is to bide their time or to look for relatively easy wins.

Source of drive for digital transformation



The drive for digital transformation is most likely to be coming from internal IT teams, followed by the senior management team and third-party suppliers.

Questions of trust

Some organizations are, of course, further ahead. Some financial institutions have, for instance, moved beyond deploying chat bots to communicate with customers and are asking if AI could be used to analyze customer requirements and offer some measure of financial advice.

The potential benefits of a service like this are clear – the cost savings could be huge and the consistency of advice could be improved given the level of data being analysed at much higher speeds than humans could achieve.

But the ethical and liability challenges are equally huge. What if bias was introduced to the training data that discriminated against certain groups of customers, even on an unintentional basis? What if an automated version of mis-selling was created which the system repeats again and again, presenting the institution with huge and escalating liability?

Whether being deployed by a bank to improve customer advice on investment decisions, a life sciences company to make drug development more efficient and precise, or a retailer to offer personalized products or services, there is always a question of whether the AI can be trusted to come up with the right results.

AI can only be as good as the data it is trained with. Until we get comfortable that there are trustworthy safety checks that do not rely on human intervention, there will always be a reticence about applying it as widely as might otherwise be the case.

These are complex issues – technologically, ethically and operationally – that companies will have to navigate as their journey of digital evolution progresses, and in all likelihood, accelerates in the years ahead. The trust and implementation will, at least in part, be influenced by regulation which, as it develops, will put a significant compliance burden on organizations and is likely to vary regionally.

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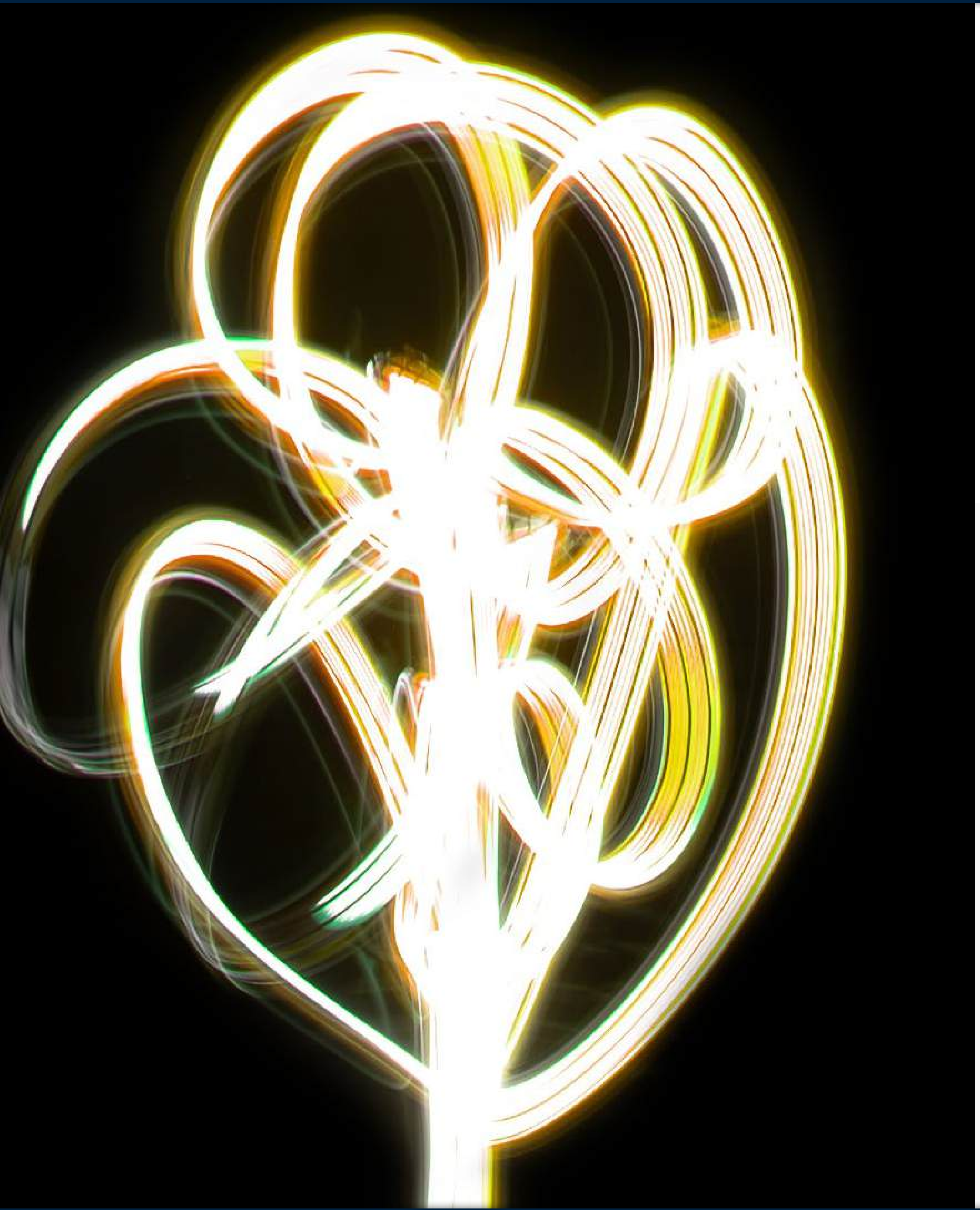
Sustainability and ESG – regulation accelerates change

Given the economic and political uncertainty of recent years, you'd be forgiven for thinking sustainability and ESG issues would have fallen down boardroom agendas.

That was a trend in past moments of economic turbulence, with companies retreating from commitments on the environment and social responsibility made in more prosperous times. One example is the immediate aftermath of the global financial crisis of 2008.

But our global survey of attitudes to ESG show a clear trend in the opposite direction. These issues have grown as priority areas for action right across the world and in most sectors.

ESG issues have grown as priority areas for action right across the world and in most sectors.



Just under half of organizations said ESG issues have grown in importance compared with other concerns, with 67% of respondents in Africa taking that position. That is significantly ahead of other regions, but unsurprising given the devastating impacts of climate change already felt across the continent.

Globally, a quarter reported no change in the importance they attach to this topic, and only 26% said ESG issues were now less of a priority.

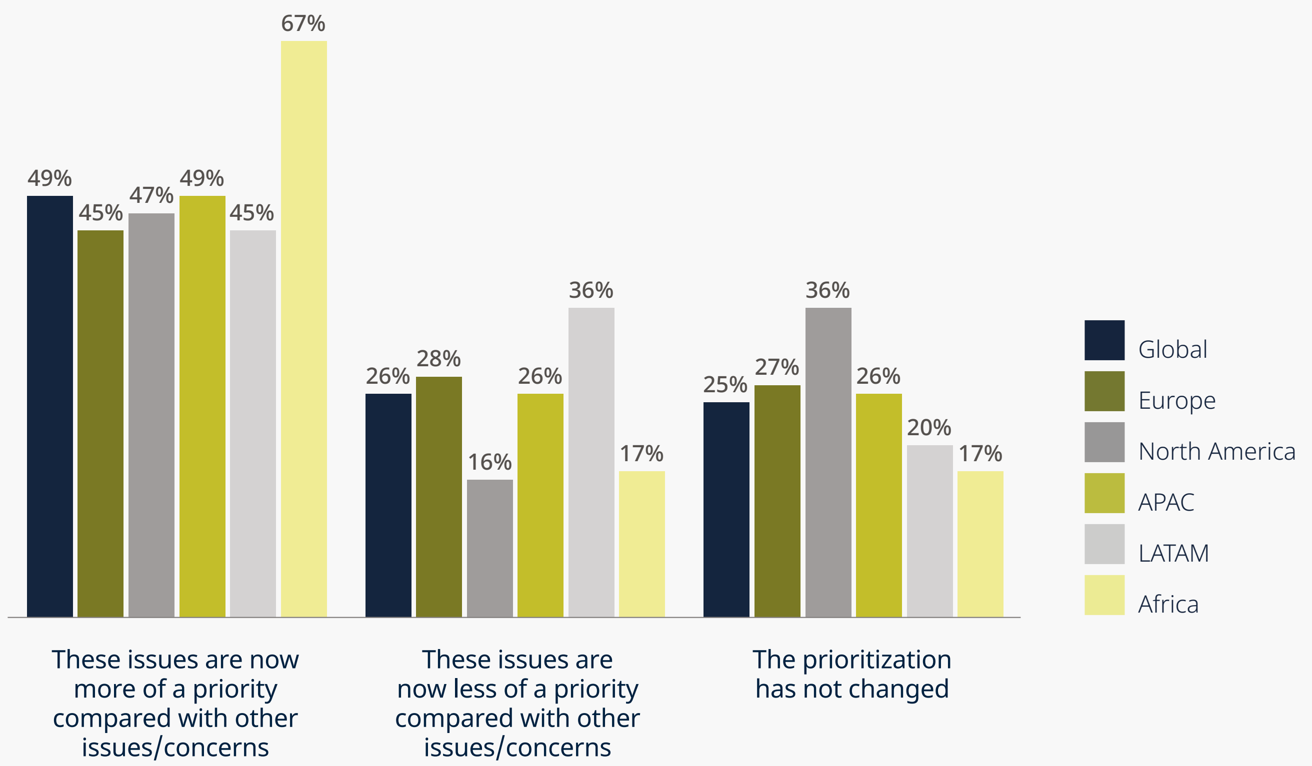
These findings match our client work across many jurisdictions. This is most often driven by recent or emerging regulation and growing stakeholder pressure as the world looks for solutions to tackle urgent environmental and social challenges, including climate change.

Companies most directly affected by regulation are increasingly engaged with sustainability. But many others in regions where regulation is less mature are still taking action, often driven by strong stakeholder pressure to prepare for potentially even stricter requirements by the time they are implemented.

But some sectors have deprioritized sustainability and ESG issues, particularly where the short-term focus has been on battling through a cost-of-living crisis and a high inflation environment.

> Sustainability reporting should be treated as the equivalent to financial reporting.

Changes in prioritization of ESG issues in the last two years



Europe in the vanguard

Europe is driving forward ESG issues the fastest, with a growing raft of EU regulation addressing key sectors.

Key EU legislation includes:

- **The Sustainable Finance Disclosure Regulation – particularly important for asset managers and funds looking to invest in tech companies**
- **The EU Taxonomy with its Green Asset Ratio – which pushes lenders to make sustainable investments**

These two pieces of legislation are testing asset managers and lenders on the sustainability of their portfolios. As a result, institutions leading the way with a credible strategy for managing ESG risks are finding it easier to raise capital.

More generally, the EU is moving to standardize sustainability reporting through the Corporate Sustainability Reporting Directive (CSRD).

This requires in-scope public and private companies, regardless of sector, to meet European Sustainability Reporting Standards, which demand much higher levels of disclosure backed with detailed data, allowing companies' performance to be clearly assessed.

Sustainability reporting should be treated as the equivalent to financial reporting, instead of being seen just as window dressing or a nice to have. It is the output of a broader, more disciplined approach to considering sustainability impacts, risks and opportunities for the reporter's business.

The EU's Corporate Sustainability Due Diligence Directive (CS3D) holds certain companies more accountable for what's going on in their supply chains and imposes a requirement for companies to publish clear transition plans for how they'll decarbonize their operations.

CS3D was introduced in May 2024 after a fairly tortuous process of consultation, during which the original ambition of the directive was greatly reduced. But it still has bite.

In the UK, two pieces of climate-related financial reporting legislation took effect from April 2022: one aimed at making reporting mandatory for certain public and private companies, banks and insurers, and the other targeting limited liability partnerships.

The regulations, which amend the Companies Act 2006, are driving in-scope organizations to develop ESG frameworks so they can report more clearly and concisely on climate risk.

An inflection point

Although regulatory regimes are far less well developed in the US, we've seen much greater engagement with sustainability issues across sectors in the last six to twelve months.

We're now at a turning point. Although big public companies are often well ahead of the curve, clients in mid-cap businesses and smaller enterprises are now striving to understand the likely future path of regulation and how ESG fits into their strategic business plans.

Much of the focus is on climate, not least because businesses are feeling the effects of global warming on their direct operations and their supply chains. But companies are increasingly examining a wider set of social and governance risks as well, such as human rights, modern slavery and forced labor.

This trend is gathering pace, even though the ESG debate has become highly politicized, with figures on the right associating it with the "woke" agenda and directly challenging proposed changes in regulation.

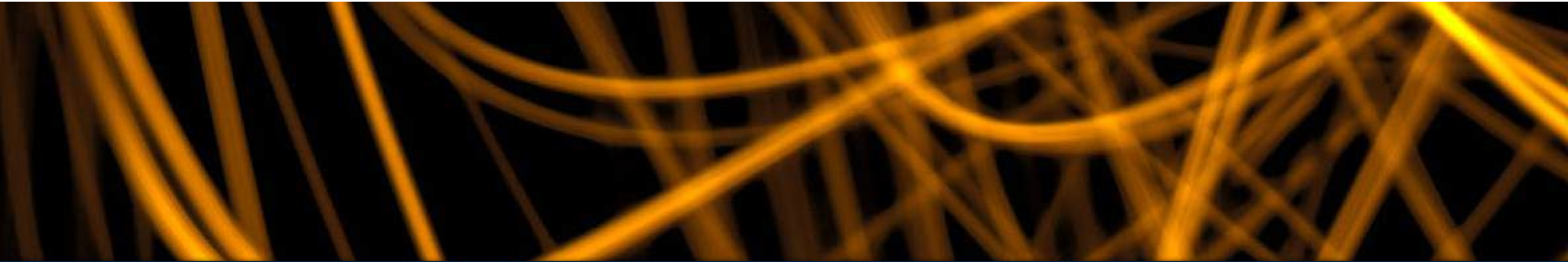
The Securities and Exchange Commission, for example, has voluntarily stayed new climate disclosure rules which were approved to come into force in March 2024.

The proposed rules immediately sparked legal challenges, mostly from "red" state attorneys general and certain companies who argued that the SEC did not have the authority to compel companies to make climate disclosures. Climate activists and NGOs also took to the courts, saying that the new rules did not go far enough.

Far from deterred by these polarized disputes, clients are looking for guidance on the complex new rules.

They want to know what they'll be required to do if the rules eventually come into force. There is a sense of when, not if, despite debate over the precise nature of the rules and the timing of implementation.

Those already pulled into scope of EU regulations – or pursuing advanced voluntary approaches – want to understand the differences between the respective regimes and where compliance gaps might lie.



Greenwashing or greenhushing?

The strongest regulatory action in the US is happening at state rather than federal level, especially in California, where bills have been passed on disclosing greenhouse gas emissions, covering Scope 1, 2, and 3, and financial risk disclosure.

The state has already taken action to stamp out greenwashing by any company, however big or small, selling products and services in California. The legislation, brought forward in October 2023 and initially meant to come into force in January 2024, is aimed at ensuring consumers have the right information to make informed purchasing decisions.

This has put huge pressure on companies to publicly substantiate any claims they're making on climate, net zero or carbon neutrality. Given the burden on businesses to comply in such a short space of time, the start date may be pushed back and the current proposal is that the legislation will take effect in January 2025.

But the direction of travel is clear, and is mirrored in the EU's Directive on Empowering Consumers for the Green Transition, which includes a ban on generic environmental claims and will come into force in 2026. A new Green Claims Directive is also now at the proposal stage, and is up for negotiation with the European Parliament, most probably in the next legislative cycle.

In the face of these ever-stricter regulations on greenwashing and the political kick back against them, there are concerns that, far from greenwashing their activities, companies will instead understate the action they are taking on sustainability. This behavior is called greenhushing.

We don't see much evidence of this. Rather than pull back from their commitments, our clients are taking far greater care to make sure they're measuring their activities effectively and reporting accurately, to avoid both litigation and regulatory risks.

We're seeing this need for greater rigor being reflected in the makeup of boards, with more companies looking to recruit directors who understand sustainability issues and can identify the risks and opportunities ESG presents. In the past, this governance aspect has been overlooked. Today it's likely to be viewed as critical.

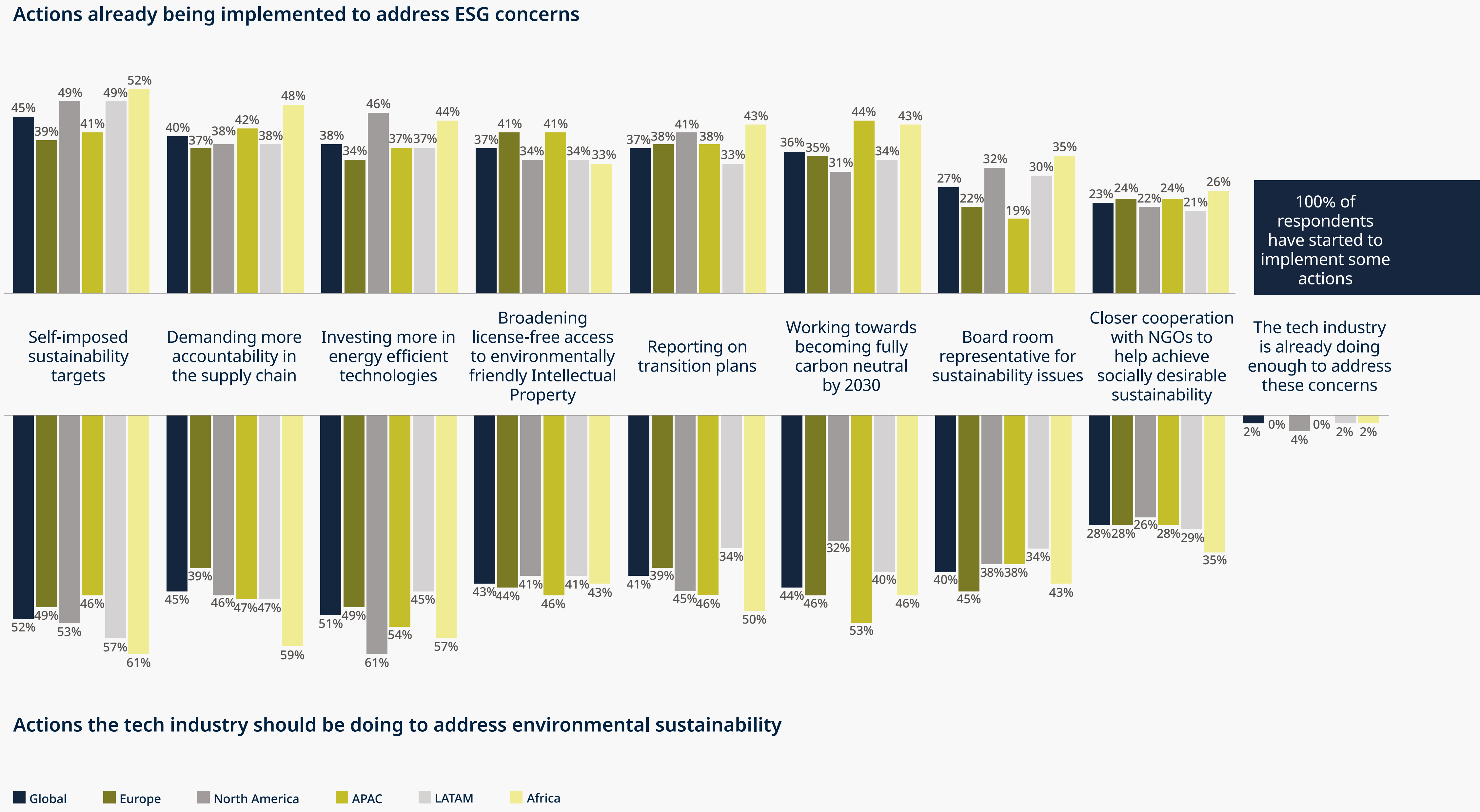
> There are concerns that companies will understate the action they are taking on sustainability. This behavior is called greenhushing.

Ambition

But it's not all down to regulation. Many companies are taking matters into their own hands, often in response to pressure from increasingly demanding stakeholders. These can be investors, employees, customers or the public, and, according to our survey, they have ambitions to increase their ESG-related activities and approaches.

Respondents unanimously say they've started taking action to address environmental sustainability. This includes using self-imposed sustainability targets (used by just under 50%), demanding greater accountability from suppliers (40%), investing in renewables, reporting on transition plans or working towards full carbon neutrality by 2030, each being pursued by over a third of respondents.

In terms of action being taken already, Africa, where the impact of climate change is felt so acutely, ranks most highly on almost all issues.



(Top) When asked what actions tech companies should be taking, 98% feel the industry could and should do more. But there are some clear gaps between what companies aspire to do and what they're actually doing, for example on boardroom responsibility for sustainability issues.

(Bottom) 80% of companies say they've at least put in place an emerging framework for managing ethical risks around the deployment of technologies, a figure that rises to around 90% in both Latin America and Africa. But only 40% say they have a comprehensive framework, and a good proportion still have no framework at all. That includes a quarter of respondents in the US, which is likely a reflection of the current political and regulatory uncertainty in the country.

The role of big tech

Our findings do not show the challenges and opportunities that technology presents. Technology can play a critical role in enhancing sustainability, improving performance across all aspects of ESG initiatives. For example, by optimizing the use of technology on a scalable basis (to reduce energy demands), and gathering and analyzing environmental data to build more robust risk management systems.

“Broadening license-free access to environmentally friendly intellectual property” scores highly among our respondents on what the tech industry is and should be doing.

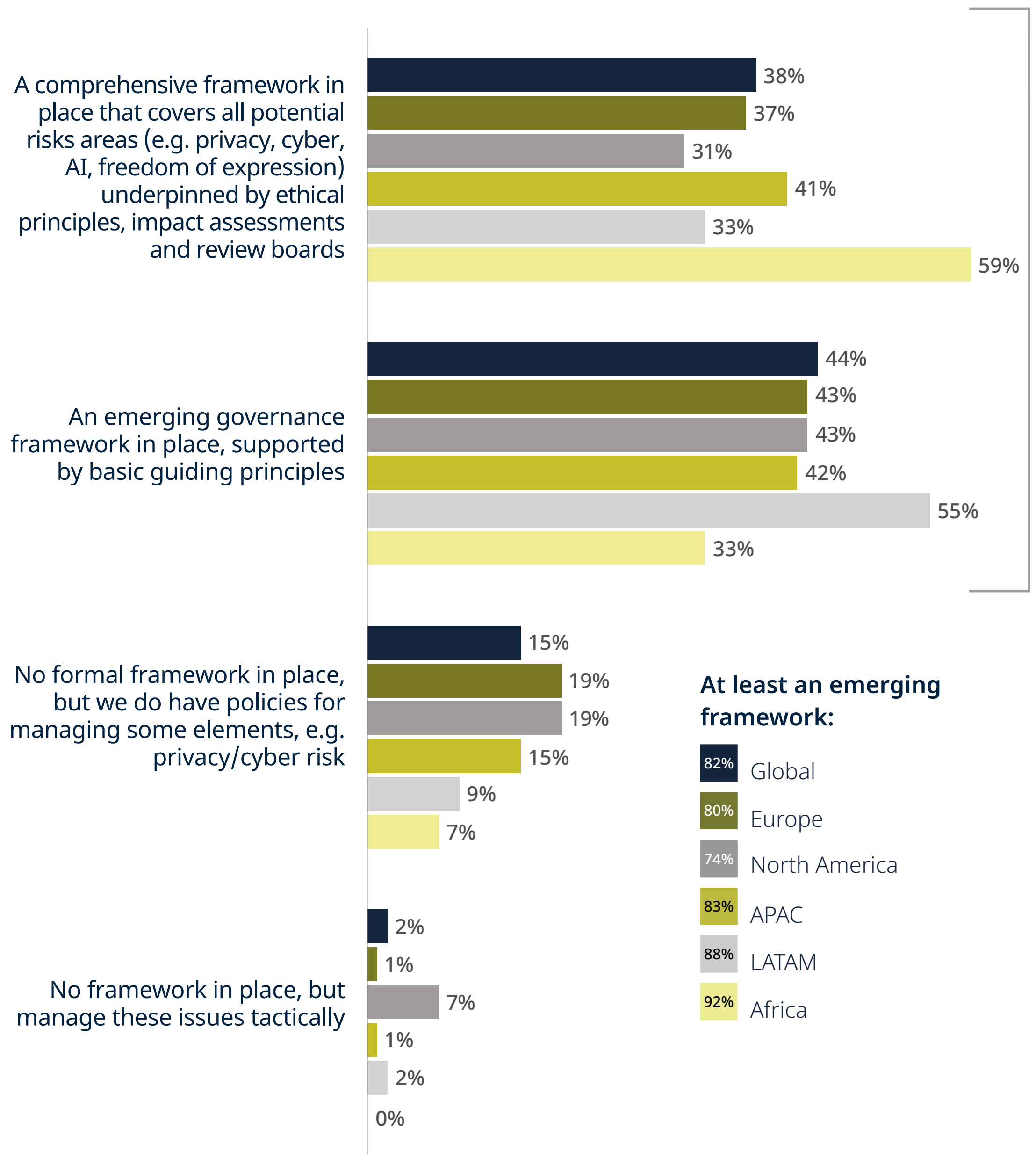
The tech industry is playing a proactive role in advancing the clean energy transition. For example, by investing in renewable energy to supply power-hungry data centers. By signing long-term power purchase agreements, they have given investors the confidence to finance a rapid expansion of wind and solar energy.

But as technology advances, there are downsides too.

With the emergence of increasingly powerful AI models, data center power consumption has shot up dramatically. This has made it challenging for some big tech operators to meet their own sustainability targets, with growing doubts they will meet 2030 net zero targets. The projected rise in quantum computing in the next decade will only add to this pressure.

Balancing this equation will be critical, not least because many companies in other industries are watching big tech closely to see what they can learn in managing their own transition to clean energy.

Framework for managing ethical risks associated with the deployment of technologies



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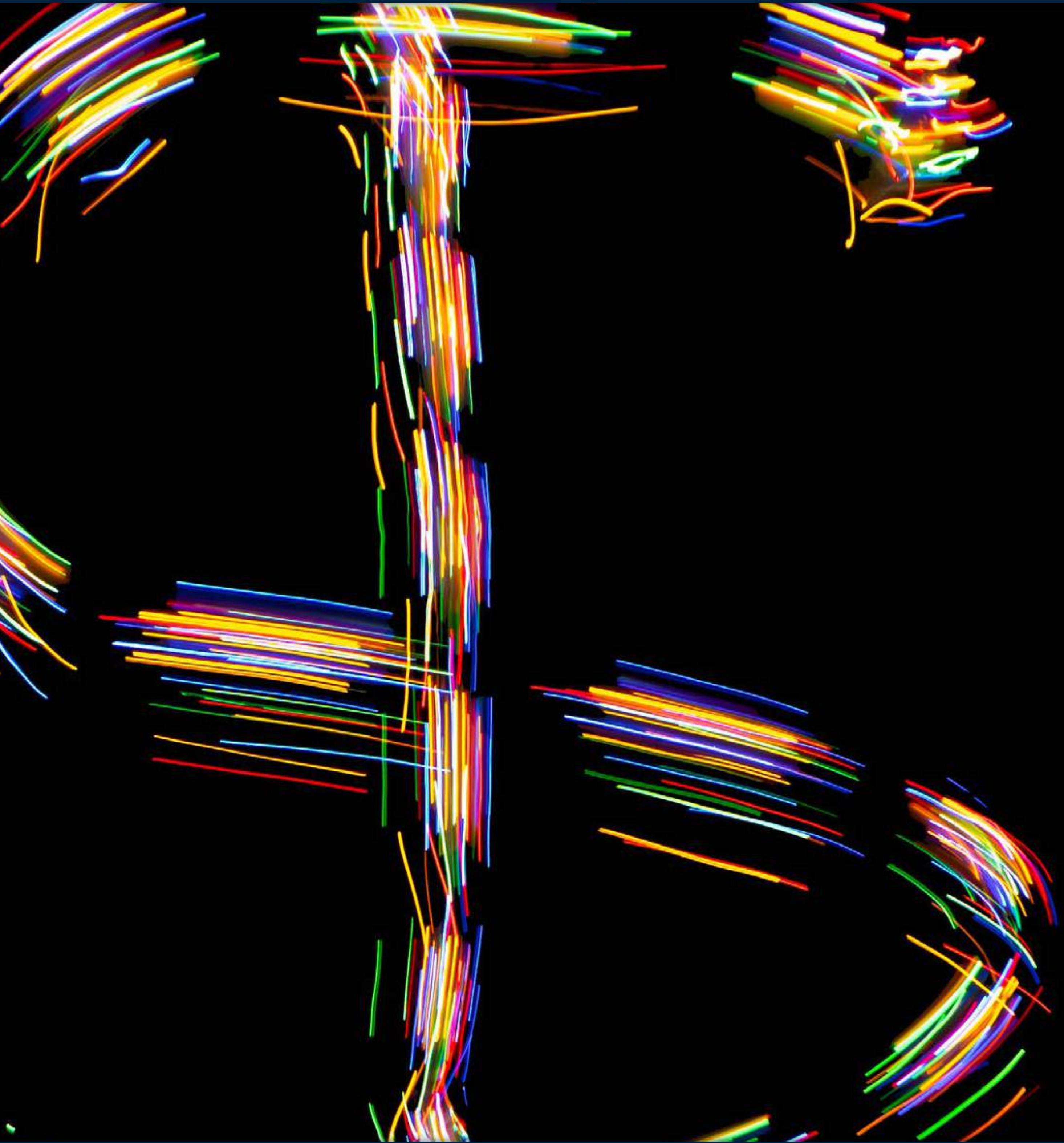


Fintech – squaring the circle

After two years of severe turbulence, a sense of calmness and confidence appears to be gradually returning to the fintech sector for innovators and investors alike.

Our survey finds that fintech features as one of the key areas for potential business growth although it ranks in sixth place, well behind AI, 5G, digital transformation, cybersecurity and the Internet of Things.

This suggests that uncertainty still lingers over the sector, although it's worth noting that respondents in North America, the Asia Pacific region and Latin America all see greater growth potential in the sector than in other regions.

A sense of calmness and confidence appears to be gradually returning to the fintech sector for innovators and investors alike.

Two factors in particular rocked the sector in 2022, with the impact persisting through most of 2023 – uncertainties over the future direction of regulation and valuations of early stage and even more mature fintech businesses which had sky-rocketed and become unsustainable.

Against that backdrop there was a sharp correction, with many of these businesses either extending funding runways or forced into down rounds, often having to accept a significant valuation haircut, and with backers pressurizing start-ups to cut costs and staff.

Funding remains constrained in 2024, but we are seeing valuations improving with multiples strengthening and settling at more sustainable levels.

There is also a continuing trend for established financial services companies to partner with tech innovators rather than acquire them, if they do not have the capability to develop systems in-house.

Given the turbulence of recent years, our expectation is that we will see a flight to safety, with customers increasingly looking to established financial services players who have successfully adopted digital banking technologies rather than relying on pure fintech operators.

Concerns about the financial strength of fintech providers continues to score almost as highly in our 2024 findings as in our last survey two years ago. But, this year, other issues rank higher as perceived drawbacks from fintech investment.

Disruption to the existing business, concerns around security, the difficult of integrating new technology with legacy systems and lack of skilled staff all rank higher.

Our expectation is that we will see a flight to safety.

The regulatory challenge

One other issue continues to concern our survey respondents – meeting regulatory requirements. This remains as much of a concern as it was at the time of the market correction.

However, we believe there has been a significant shift where regulation is concerned.

Aspects of the fintech world were initially only lightly regulated and some were hardly regulated at all, particularly around crypto and other virtual assets. Indeed, the lack of regulation initially proved a major attraction for some investors.

But, in a significant reversal, we've seen regulation in this area being tightened and, in many jurisdictions, this is now bringing a sense of predictability to the market which is, in turn, boosting investor confidence.

In the 20 or so years since financial services started to be digitized, we've seen different parts of the market go through familiar cycles of development.

Typically, a tech solution emerges and takes time to prove itself.

The businesses looking to deploy the technology also need time to assess not just its effectiveness, but also to gauge the likely return on the considerable investment involved and to test if there is real customer demand for it.

Invariably, law and regulation governing these new digital services lag the technology and are in a constant race to catch up when development is moving at such a rapid pace.

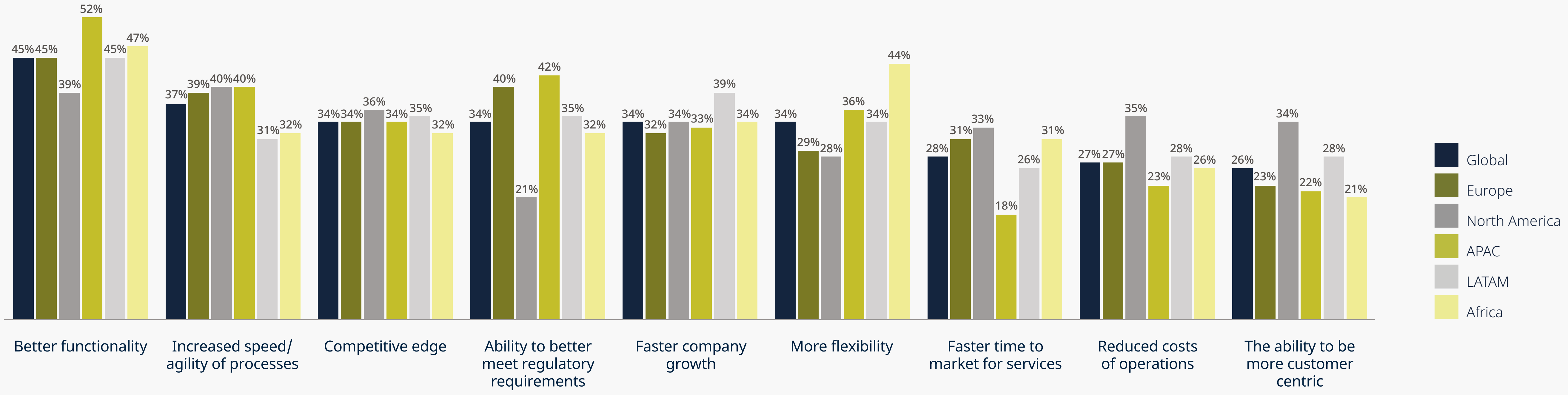
It's a constant effort to square this tech/investment/demand/ regulation circle and inevitably it is a lengthy process.

We saw that in the early days of online banking services, as new technologies like electronic signatures and electronic records were brought to market. It took time to prove the efficiency and accuracy of these systems and for customer demand to be established, something that was greatly accelerated during the pandemic.

Over the last decade or so the focus has been on cryptocurrencies, blockchain and the whole complex area of virtual/digital assets and we've seen this go through the same process. Now we are seeing the same journey being undertaken where AI is concerned, with all the thorny ethical issues, beyond technological capabilities, it throws up.

But in each case, as the law adapts to meet the advancement in technology, we see the market strengthening as financiers, lenders and the capital markets gain the confidence to invest at much higher levels.

Benefits of fintech



Key benefits of fintech are seen to be better functionality and increased speed/ agility. Flexibility is particularly important in Africa and the ability to better meet regulatory requirements is particularly important in Europe and APAC. Reduced operations costs and the ability to be more customer centric are of greater importance in North America compared with other regions.

Embracing regulation

This has been particularly evident around the regulation of crypto and other digital assets in the last two years across several jurisdictions.

Dubai has taken a strong lead in creating its Virtual Asset Regulatory Authority (VARA), a regime we played a central role in developing and implementing in 2023.

In the post-Covid world, and as the Middle East more widely attempts to diversify away from its economic dependence on fossil fuels, Dubai has recognized that virtual assets can be a powerful engine for economic growth. Similarly, it has recognized that stable and predictable regulation can promote investment rather than deter it.

Hong Kong, a jurisdiction where early crypto innovators were drawn because of relatively light regulation, is coming to the same conclusion and is actively tightening licensing regimes while also looking to stimulate fintech investment and the development of exchanges through direct government support. Singapore is on its own journey where regulation is concerned but the trend is also towards more regulation.

Demand for Fintech solutions in Africa is growing rapidly but a consistent legal regime to govern it will be needed to enable development and deployment.

In Europe, the Markets in Crypto-asset (MiCA) regulation also came into force in mid-2023 aiming to foster the development new technologies by creating a framework covering virtual assets, issuers and service providers.

As in other regimes, the aim of regulation is about striking a balance between protecting users and stimulating innovation but it's interesting that some major markets are taking longer to develop laws, namely the UK and the US.

The UK government is consulting widely with industry before establishing a regime, in the hope that it can align with developments in the EU while also being distinctive from it.

In other areas we've seen this approach reap benefits. Open banking, for instance, has grown rapidly in the UK, allowing users to access multiple accounts across different providers from one platform, and this is now playing out in the business world through open financing. It's a trend which is now being followed at pace in many other markets.

Development of regulation in the US has also moved at a slower pace, partly in recognition that there can be dangers in rushing to regulate. Initially the focus has been on anti-money laundering and illicit finance.

In the virtual asset space, however, the Financial Innovation and Technology for the 21st Century (FIT21) Act directing how virtual securities and commodities will be regulated, has recently passed through the House of Representatives and awaits scrutiny by the Senate. The journey continues.

With regimes developing at different speeds, the eventual goal must be to create some level of global standardization and interoperability in regulation for what is a global industry. If too many authorities go out of the gate too soon there is a real danger that regulation will cause friction that could stifle innovation and investment.

> Regulation is now bringing a sense of predictability to the market, boosting investor confidence.

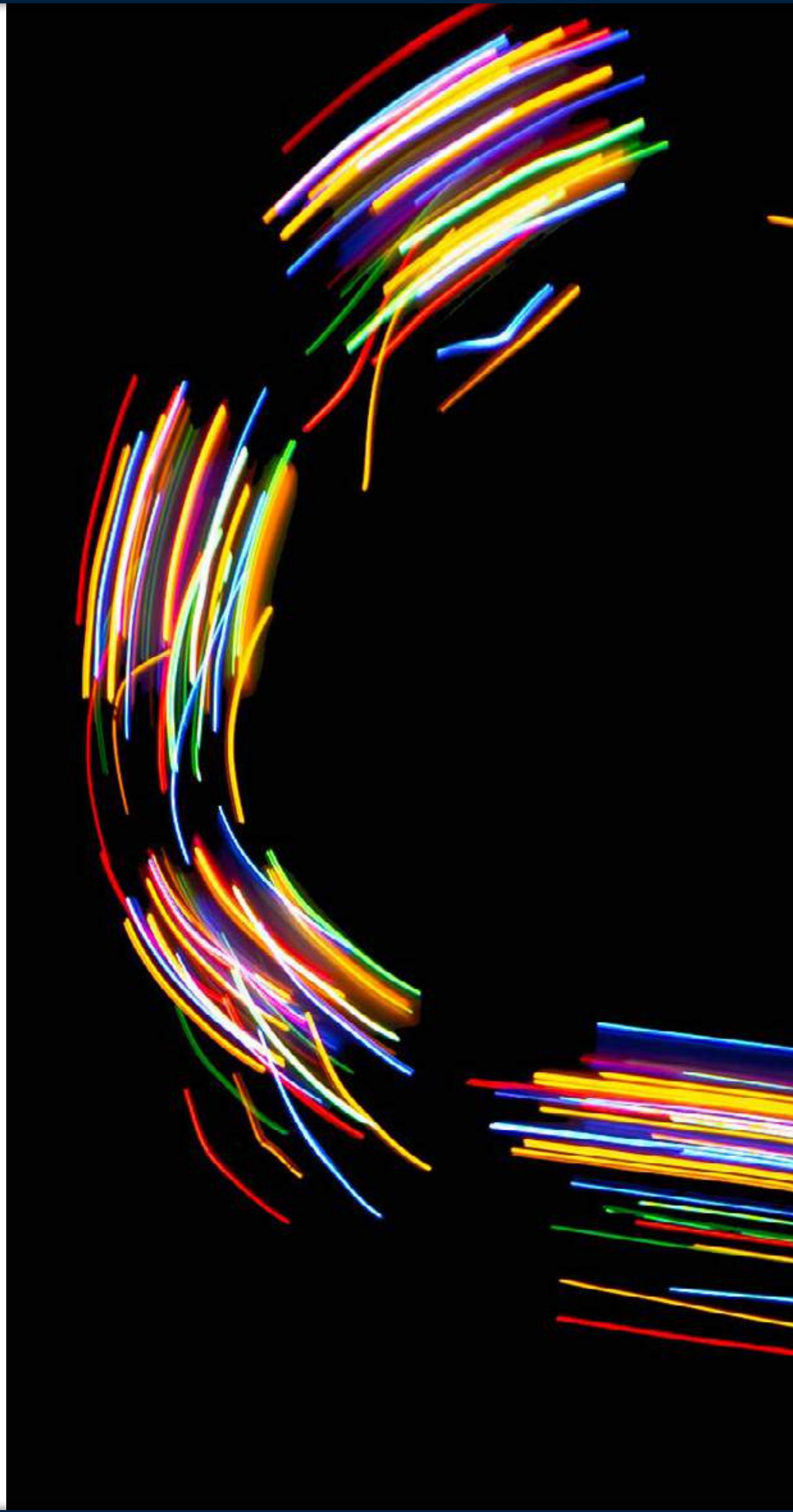
Resistance to change

While the pressure to digitally transform financial services continues to strengthen, a pressure that will only grow as AI technologies advance, development can meet points of resistance within institutions.

Internal resistance can come from a wide variety of sources, whether it's from frontline staff unwilling to reskill or fearful for their jobs, from legal departments anxious to see how regulation evolves, from CTOs concerned about increased security and data risks, to CEOs who remain unconvinced that the huge investment required is justified.

In working with clients on their digital transformation journeys we look to encourage them to prioritize two issues in particular. Firstly, to make sure that all the relevant people with the business come to the table to plan for that transformation. And secondly to take a step-by-step approach, picking key parts of the process to focus on.

Trying to do it all at once – boiling an ocean – is not only very expensive, but often means becoming bogged down and overwhelmed.



Greatest potential

Our survey finds that respondents still see the greatest potential for fintech development in areas where the technology has had time to evolve, notably payment services and digital banking.

The former scores particularly highly in the Asia Pacific region, where currency conversion issues continue to make money transfer difficult. Digital banking scores most highly in Africa.

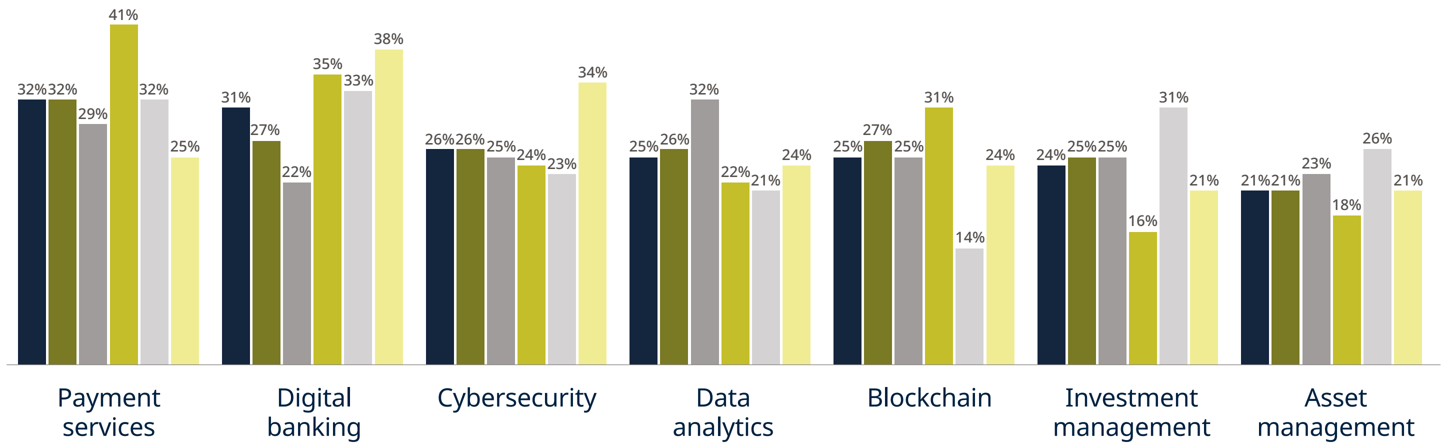
There are some interesting divergences in the findings around blockchain and cryptocurrencies where the greatest potential is seen in Asia, and alternative lending which ranks most highly in Africa.

Regtech – where dedicated risk management technology can help institutions to navigate a complex regulatory environment – is seen as a particular important growth area in North America.

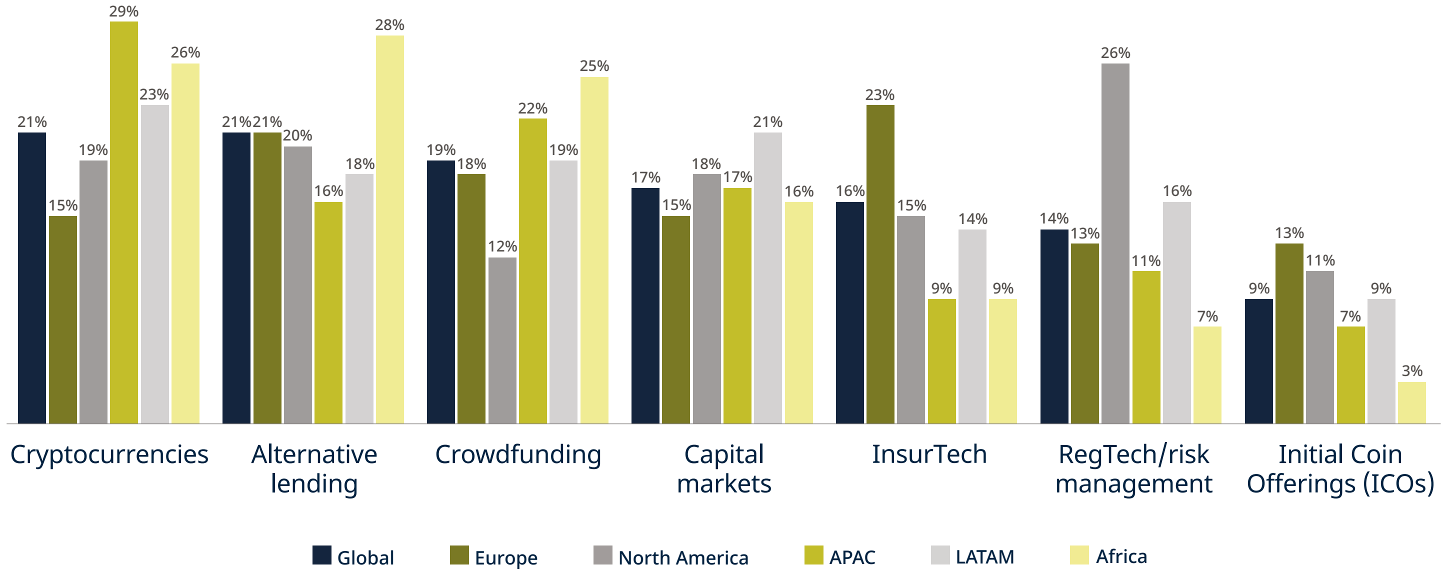
It's one we also expect to expand, particularly as supporting AI solutions come on stream, as institutions look for better and more efficient ways to remain compliant across different jurisdictions and regulatory regimes.

> Digital banking scores most highly in Africa, reflecting its potential to transform financial services accessibility.

Areas with most potential across the fintech ecosystem



Respondents see the greatest potential value from Fintech for customers coming from payment services and digital banking, followed by cybersecurity, which is particularly important in Africa.



Other areas are slightly less likely to be identified, but 1 in 5 respondents highlight applications such as cryptocurrencies, alternative lending or crowdfunding as having high potential. Some areas are of particular importance for certain regions, for example alternative lending in Africa and RegTech in North America.

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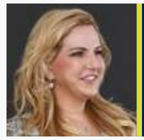

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Internet of Things – industrial and consumer use cases set to multiply

Private and public sector organizations across the world are beginning to find an increasingly wide range of ways to deploy Internet of Things (IoT) technology, powered by super-fast connectivity. But these are still relatively early days for IoT applications, and the variety and scale of use cases is only likely to multiply in the years ahead, particularly as they become enhanced by AI.

They may not catch the headlines, but industrial settings have already become a rich proving ground for IoT devices using and processing huge amounts of previously uncollectable data to fine-tune the efficiency of production lines and supply chain management. Systems in this setting, as in others, are becoming more accurate and reliable thanks to advances in technology, including error-correcting back propagation learning.

Manufacturers are also using IoT solutions to improve the performance of their products. The auto industry is a case in point, where a growing number of sensors are being built into car models to collect data, otherwise impossible for humans to understand and interpret, and process it in real-time to improve vehicle performance and increase safety.

> Private and public sector organizations across the world are finding an increasingly wide range of ways to deploy Internet of Things technology, powered by super-fast connectivity.

High-end luxury cars can, for instance, scan the road ahead to detect potholes, quickly adjusting the vehicle's suspension, with incredibly fast connectivity allowing real-time data processing that only a few years ago would have been rudimentary. Devices can also act as virtual personal assistants, for instance managing routine maintenance, even up to the point of booking a service when it becomes due.

Airlines are looking to deploy IoT in innovative ways, for instance enabling real-time maintenance reporting while planes are in flight so that engineering crews are ready to respond on landing. Some are also looking at creating intelligent labels linked to ticketing information to ensure luggage goes to the right destination.

Governments and local authorities are putting the technology to growing use to enable smart cities, processing much deeper data to plan traffic management systems and new roads and railway lines.

In regions increasingly prone to extreme natural events, such as floods and earthquakes, IoT technology is playing an increasingly powerful role in predicting and recovering from disasters. Energy and water utilities are using it to better manage grid and pipeline capacity.

Although some sectors, like financial services, may struggle to find many use cases for IoT devices, that's not true of all. Much more accurate health-tech devices, such as wearables, are able not only to send data to healthcare professionals but also to insurers with health, wellbeing and lifestyle readings affecting the premiums individuals pay.

High-end luxury cars scan the road ahead to detect potholes, quickly adjusting the vehicle's suspension, with incredibly fast connectivity allowing real-time data processing that only a few years ago would have been rudimentary.

Conquering connectivity blackspots

Connectivity is key to ensuring rapid processing of data in real-time and the development of new connectivity models, such as low earth orbit satellites, means that IoT services are now reality even in areas poorly served by conventional forms of connectivity, like fiber networks.

Take New Zealand, where agriculture is still such an important contributor to national economic output. Its cities enjoy world class connectivity through fiber networks and through investment by incumbent telcos in 5G, with 90% enjoying high-speed broadband services. Not so in rural areas.

But, thanks to satellite-connected IoT devices we are seeing groundbreaking new approaches to dairy and beef farming and forestry even in what would otherwise be connectivity blackspots.

Take a pasture management system invented by the agritech company, Halter. This allows beef and dairy farmers to manage grazing patterns without the need for fences, thanks to a remotely controlled collar attached to each animal which allows farmers to create virtual fences, directs animals to chosen parts of the pasture, brings them in for milking and monitors their health in real-time. It's a system which is now breaking through in export markets.

Realizing the benefits

Given the growing number of IoT use cases being developed, and the huge scope for further innovation, it's not surprising that respondents to our survey are realizing the wide-ranging benefits of this technology.

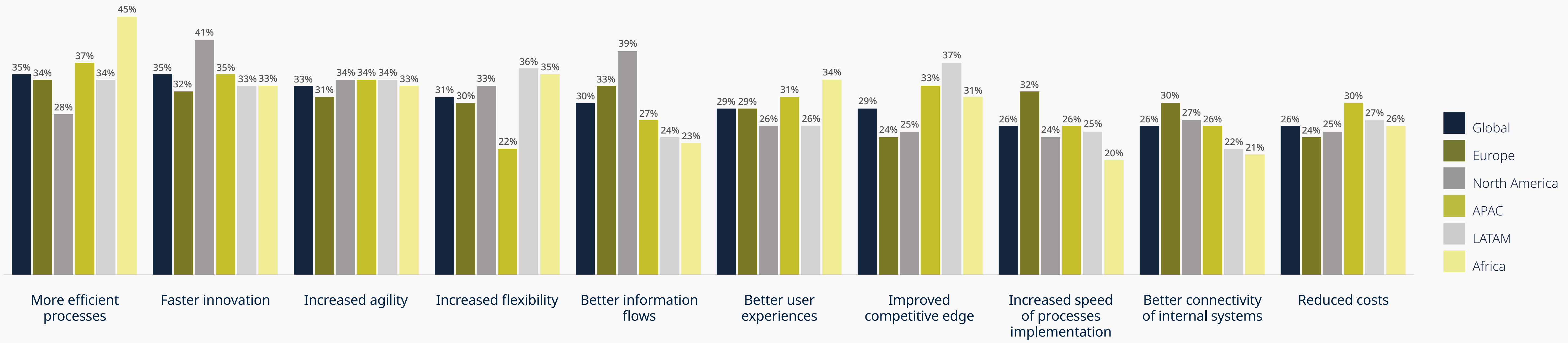
Top amongst these are the ability to make processes more efficient (seen as a particular benefit in Africa), faster innovation and increased agility, better information flows and improved user experiences.

Perhaps surprisingly, reduced costs comes well down the list of benefits since many use cases already being deployed are boosting efficiency and producing significant savings.

The perceived drawbacks of IoT/connectivity are, however, more predictable and certainly mirror what we are seeing in our client work, with increased regulation and privacy risks riding highest.

> In regions increasingly prone to extreme natural events, such as floods and earthquakes, IoT technology is playing an increasingly powerful role in predicting disasters.

Benefits of Internet of Things



A range of benefits are associated with IoT/ connectivity, including more efficient processes (particularly important in Africa), faster innovation (particularly important in North America) and increased agility and flexibility.

It's interesting to note that, at 52%, increased regulatory requirements are seen as more problematic in Europe, where regulatory reform continues to be most intense, than in other regions. Meanwhile privacy worries rank most highly in North America (53%).

That's not surprising given that some consumer facing IoT devices rely on processing huge amounts of personal data. Organizations we work alongside often also cite this as a major concern, particularly where there is a danger of data being transferred across borders.

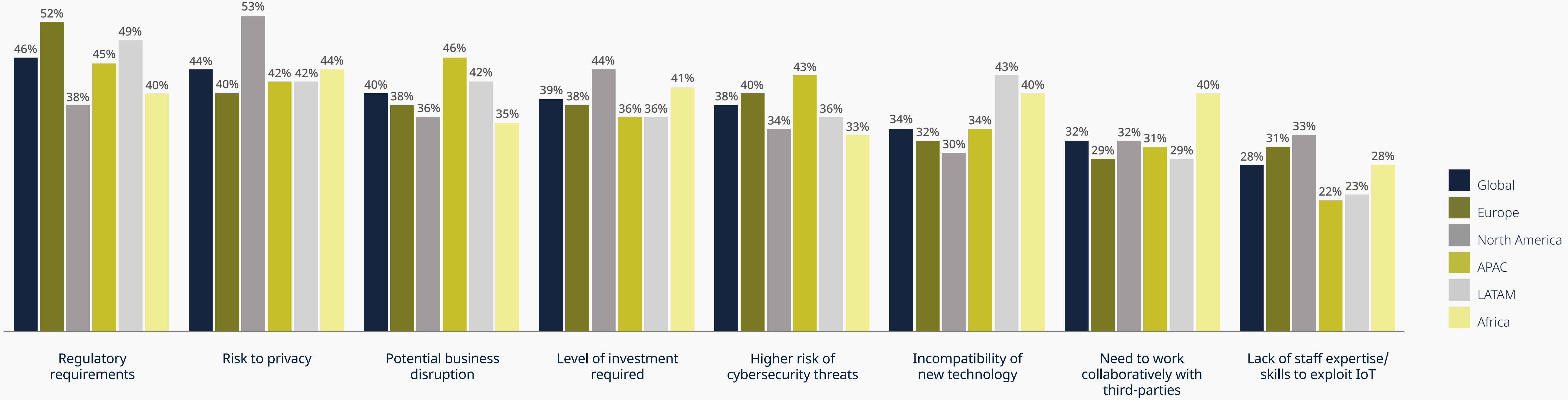
We're also seeing issues around product safety liability and in regulatory efforts to encourage a circular economy approach to manufacturing.

A good example is in the production of computer printers, where the OEM business models are built around the sale of ink cartridges rather than the printers themselves. They are keen to prevent to stamp out fraudulent copies of their cartridges but also want to frustrate remanufacturers repurposing them for re-use. To do so, they are installing chips in printers to reduce functionality when re-filled cartridges are detected, in the process running up against regulations encouraging closed-loop manufacturing.

Among other drawbacks identified in our survey, respondents in North America express greatest concern about the levels of investment required in deploying IoT technologies. Potential for business disruption and increased cybersecurity risk are ranked highest in the Asia Pacific Region.

> The rapid convergence of all the technologies that will continue to make IoT devices increasingly capable means that we are only seeing the tip of the iceberg in terms of the variety and scale of IoT development.

Drawbacks of Internet of Things



The key drawbacks to the implementation of IoT are seen to be regulatory requirement and the risk to privacy, which is particularly important in North America.

Emerging threats

A clear majority of respondents (56%) say that they are most likely to rely on a mix of external and internal resources to meet their IoT/connectivity needs, with that number rising to 64% in North America and 58% in Africa. Over a quarter say they will rely on external suppliers only, a figure that rises to 36% in the Asia Pacific region.

Given this high reliance on external suppliers, it's perhaps surprising that some emerging threats do not appear among drawbacks cited in our survey.

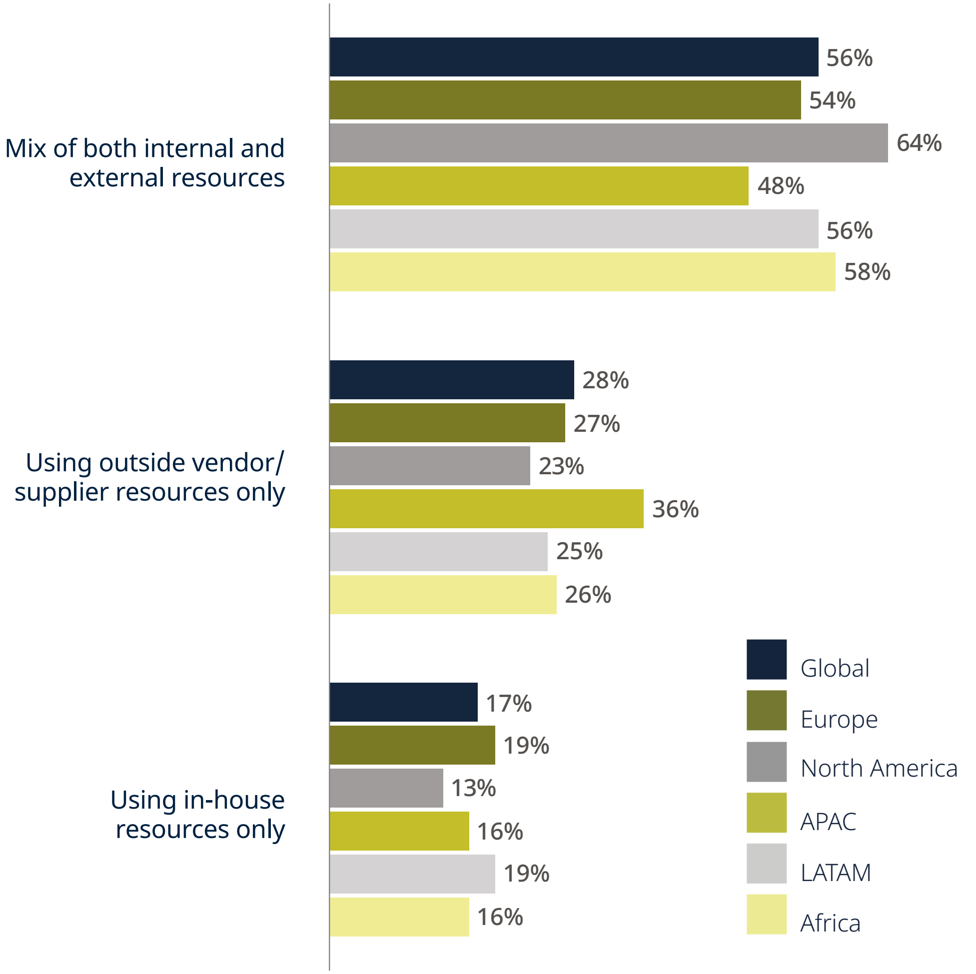
We're seeing growing evidence that companies whose data is being used to power externally supplied IoT devices are increasingly worried about their data being shared with competitors in the process exposing important trade secrets. They are also concerned about 'web scraping' by suppliers hungry for data to train their devices.

It will be interesting to see how regulation and law develop around these new risks and whether they act as something of a brake on the future development of IoT devices.

We think it is unlikely to have a significant impact, however.

The rapid convergence of all the technologies that will continue to make IoT devices increasingly capable – sensor development, powerful new connectivity solutions, the cloud and AI, for example – means that we are currently only seeing the tip of the iceberg in terms of the variety and scale of IoT development and deployment in years to come.

Meeting IoT needs



Organizations are most likely to be using a mix of internal and external resources to meet IoT/ connectivity needs. Those in APAC are more likely to be relying solely on third parties than those in other regions.

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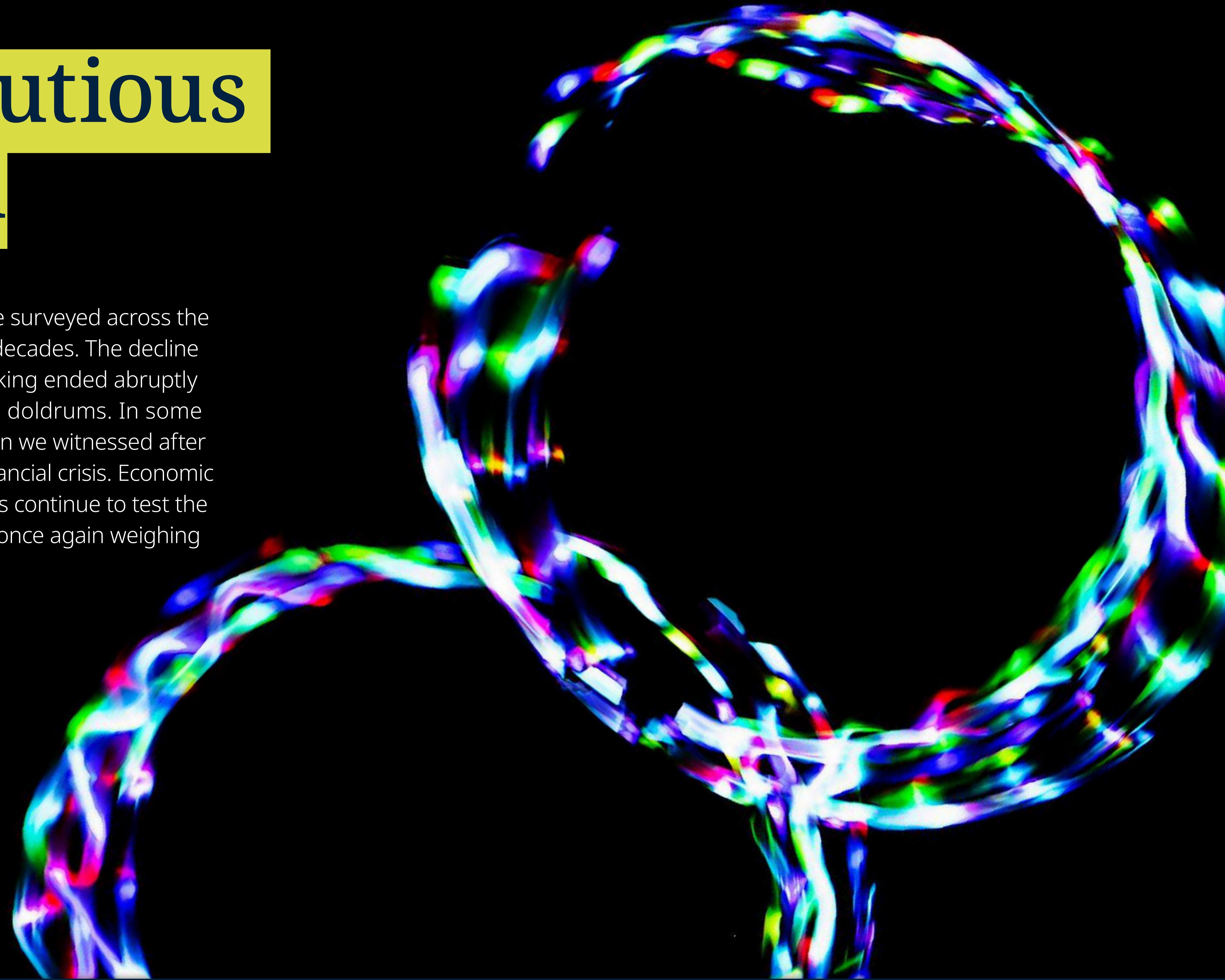
M&A – makes a cautious but uneven return

Acquisitions appear to be back on the agenda for many of the companies we surveyed across the world after one of the steepest downturns in M&A activity we have seen in decades. The decline in the volume and value of deals since the extraordinary bull run of dealmaking ended abruptly in 2022, has seen an 18-month period where the market was idling in the doldrums. In some markets the decline in activity has been even more severe than the downturn we witnessed after the dot-com bubble burst in the early 2000s and in the wake of the 2008 financial crisis. Economic conditions remain challenging in 2024 and growing geopolitical uncertainties continue to test the nerve of investors. But our survey shows that many of our respondents are once again weighing up the varied advantages of acquiring external businesses.

Economic conditions remain challenging in 2024, testing the nerve of investors considering acquisitions.



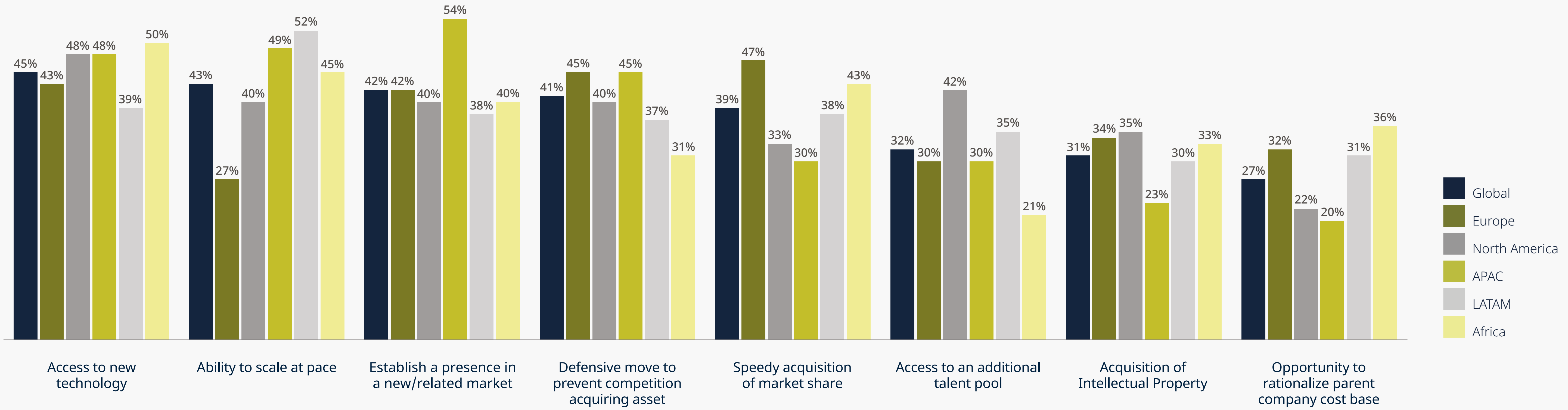
Consistently, across the world, respondents highlight the benefits they expect to enjoy once back on the acquisitions trail. Accessing new technology, the ability to scale at pace and establish a presence in new markets, increasing market share rapidly, and accessing skilled talent, all rank highly as reasons to do transactions.

However, key priorities emerged in different regions.

For instance, in Europe accelerating market share growth ranked higher than in other markets. Establishing a presence in other markets was a priority in the Asia Pacific region, while in North America widening the talent pool ranked higher than elsewhere.

> The US is leading the return to M&A activity, with a noticeable uptick in sell-side transactions

Most beneficial features from the acquisition of an external company



The key areas of benefit from the acquisition of an external company include access to new technology, ability to scale at pace and being able to establish a presence in other markets (particularly important in APAC).

Green shoots?

If these responses indicate a growing confidence among companies to contemplate external acquisitions – as they appear to – that certainly matches what we are seeing in our own practice.

Key deal markets are beginning to see a return to growth, but it is by no means a consistent picture across the globe and there is still a high degree of caution on display.

The US appears to be leading the return to activity. After a protracted period of very limited activity (dominated by buy-side transactions and some cross-border deals), we've seen a noticeable uptick since the beginning of March with sell-side activity reigniting.

Currently the market is dominated by strategic buyers, helped by the fact they are completing deals using cash or equity.

Financial sponsors, however, appear to be biding their time, adjusting to the prospect of interest rates remaining relatively high and unlikely to come down this year as quickly as some had hoped.

Rate volatility remains a major concern for these investors who had grown used to having ready access to cheap debt during the M&A boom when, for a protracted period, rates stayed at historic lows, at or close to zero. With 'free money' no longer on offer, sponsors will need to adjust their models to the new reality – something that will happen but will become much easier if rate levels stabilize and become more predictable.

By contrast in Australia, where we are also beginning to see an uptick in transactions, activity is being dominated by private equity houses or their portfolio companies, with North American buyers in the lead.

They are looking to build out their platforms by bolting on businesses that give them access to sought after software, other technologies or talented teams. In addition, these transactions can allow them to establish a bridgehead into chosen markets across the Asia Pacific region, whether that's in Australia, Southeast Asia, China or Japan.

Software as a Service (SaaS) companies remain a key target here as in many other markets and transactions giving access to important AI solutions remain the most sought after.

That's particularly true for more sophisticated tech investors who understand how AI is likely to enhance their business models and where it might prove to be a threat. For others the learning curve is steep.

We are mostly seeing smaller ticket M&A in the range of USD20 million to USD100 million in this region but completing a series of bolt-on deals in this value bracket can quickly allow investors to expand and enhance their platforms.

Occasionally we are seeing much larger deals by strategic players, but transformational megadeals, which were common at the height of the bull market, remain few and far between.

European M&A markets continue to lag the return to activity we are seeing in North America. Investors are kicking the tyres on prospective deals rather than launching full tilt into transactions. Deal term sheets may be doing the rounds, but few transactions are yet completing, unless there is an AI or SaaS element to the deals. These tend to proceed faster.

> Accessing new technology, scaling quickly, and entering new markets are key reasons for pursuing acquisitions.

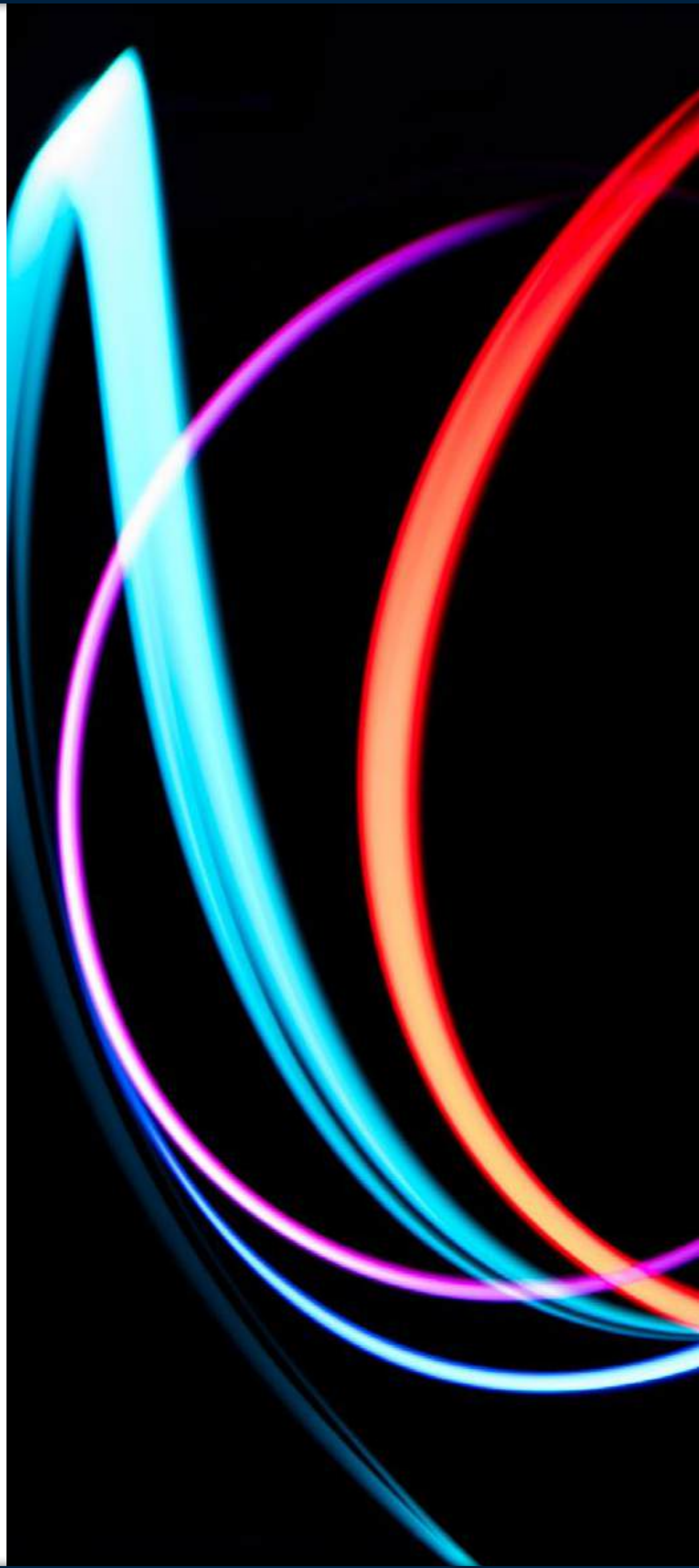
Market shifts

As the market slowly recovers, some significant differences in market dynamics have emerged strongly contrasting with the peak of the bull market in 2020 and 2021.

For instance, we are seeing far greater discipline in due diligence processes, with buyers very reluctant to find themselves launching into what then turns out to be a duff deal. Extra caution is being taken around valuations and in verifying whether tech solutions being accessed or talent being acquired (so-called acquihiring) can really enhance capability and value.

Deal multiples – which reached exorbitant and often unsustainable levels in the boom – have certainly tempered. Fast growing software companies that would then have achieved valuations at 10x to 12x ARR, are more likely to be nearer to 6x to 7x ARR now, apart from deals involving AI.

Against that backdrop we've seen a reversal of a trend that became prevalent a few years ago, where investors did so-called roll-up deals in the knowledge that a company would see its value immediately and significantly enhanced just by being brought into a portfolio of similar businesses, under the umbrella of an active and successful PE fund.



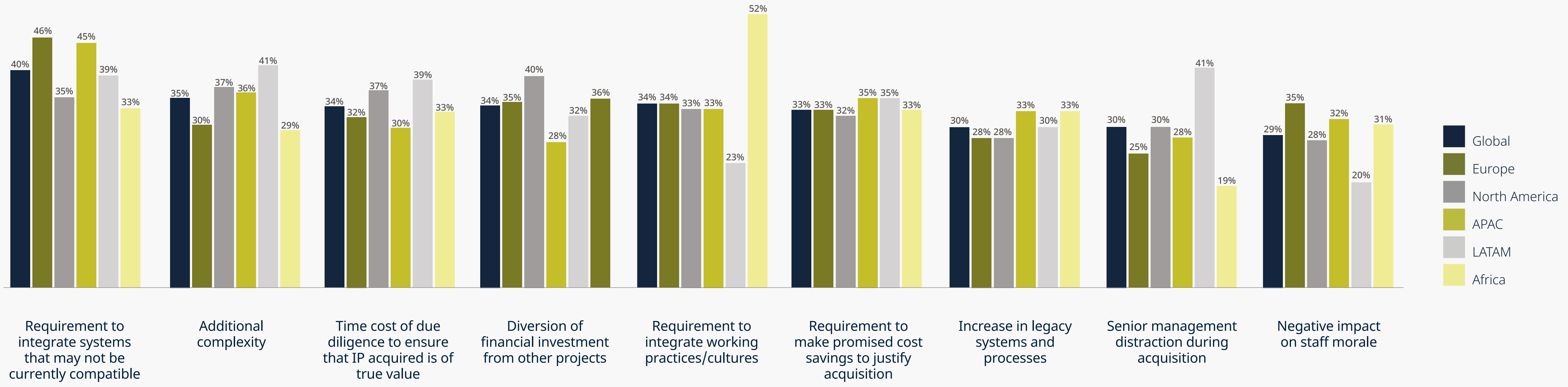
Confronting challenges

Asked what are the major drawbacks from acquiring an external business, our respondents are again fairly consistent across different markets. Integrating systems across the enlarged business, restructuring complexities, the cost of detailed due diligence and fear that this will divert funding from other crucial projects, all ride high in their concerns.

Again, there are some interesting regional variations. Integrating working practices and cultures is a particular concern for African organizations, while due diligence costs and the threat that an acquisition will act as a distraction for senior management score highly in Latin America. In Europe and APAC more respondents worry about integrating systems that may not be compatible.

Economic conditions remain challenging in 2024, testing the nerve of investors considering acquisitions.

Biggest drawbacks of acquiring an external company



The biggest drawbacks of acquiring an external company are the requirement to integrate systems and the additional complexity involved. The requirement to integrate working practices/ cultures is seen as a particular drawback in Africa

Heightened uncertainties

A number of significant potential challenges do not feature in the responses from our survey, however, even though we do know from clients that these are seen as real.

With some 1.5 billion people in more than 50 countries going to the polls in 2024, the uncertain outcome of key elections does weigh heavy as a risk factor for companies considering making acquisitions. Unsurprisingly, given the importance of the US to the global economy, November's presidential election will be watched carefully although, curiously, it is not always seen as a major risk factor by US investors.

The deteriorating state of geopolitics, not least Russia's continuing war in Ukraine, is also seen as a major concern. It ranks particularly highly in central and eastern Europe, where investors are staying their hands despite there being many attractive corporate targets in the region. That might, in turn, explain why US investors are looking for opportunities elsewhere in the world, notably in Latin America and Australia.

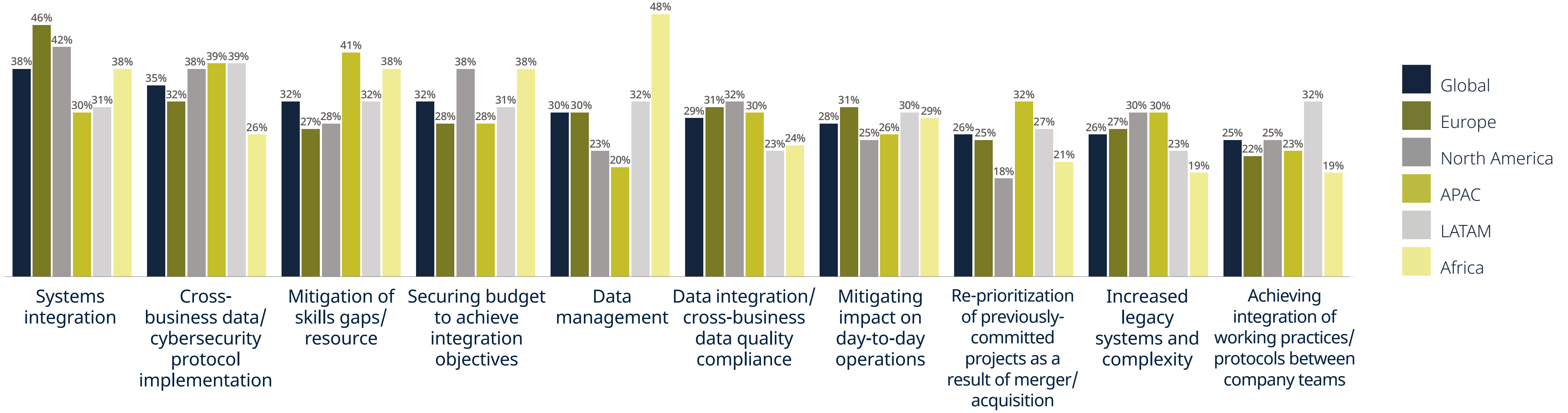
Regulation is also a factor influencing when and where investors do deals, not least the continuing focus on measures to protect national security.

The CFIUS regime in the US, whose activities were greatly increased under the Trump presidency, has continued just as strongly under President Biden, adding complexity and cost to dealmaking. Under US pressure, similar approaches have been pursued across much although not all of Europe and in parts of the Asia Pacific region.

Antitrust action by competition authorities has perhaps hit the tech sector disproportionately and continues to be a worry for many investors. In some instances, we are seeing investors opting to pursue joint ventures and other partnership arrangements rather than transactions, for fear of triggering investigations by the regulators.

Given all that, the fact that our respondents seem genuinely optimistic about the economic outlook and generally comfortable about the regulatory environment, should bode well for continued steady, if more measured, growth in transaction activity.

Key technological challenges when acquiring an external company



Key technological challenges related to acquiring an external company include systems integration, implementing cross-business security protocols, migration of skills gaps/ resource and securing budget for integration objectives. Data management is a particular challenge in Africa.

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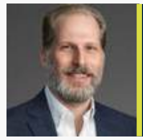
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In some instances, we are seeing investors opting to pursue joint ventures and other partnership arrangements rather than transactions, for fear of triggering investigations by the regulators.

5G – yet to prove its full potential

Despite a number of challenges that have impacted the development and deployment of 5G, it continues to be seen as a technology with huge potential, not least because of the advantages it offers in terms of data processing speeds, increased connectivity and lower latency.

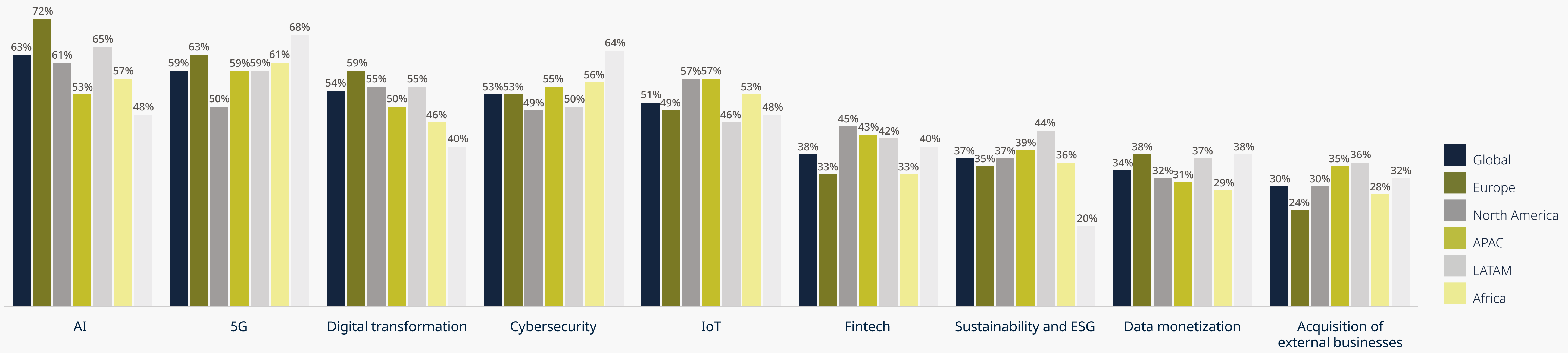
Respondents to our survey placed 5G second only to AI in terms of technologies that offer greatest potential in the coming years, ahead of digital transformation, cybersecurity, the Internet of Things and Fintech.

Overall, 5G scores 59% in our global survey – compared with AI at 63% – and it scores considerably higher than that in Europe (63%), Africa (61%) and the Middle East (68%). In some ways that is surprising, since 5G has faced a series of significant hurdles since it began to be rolled out across networks.

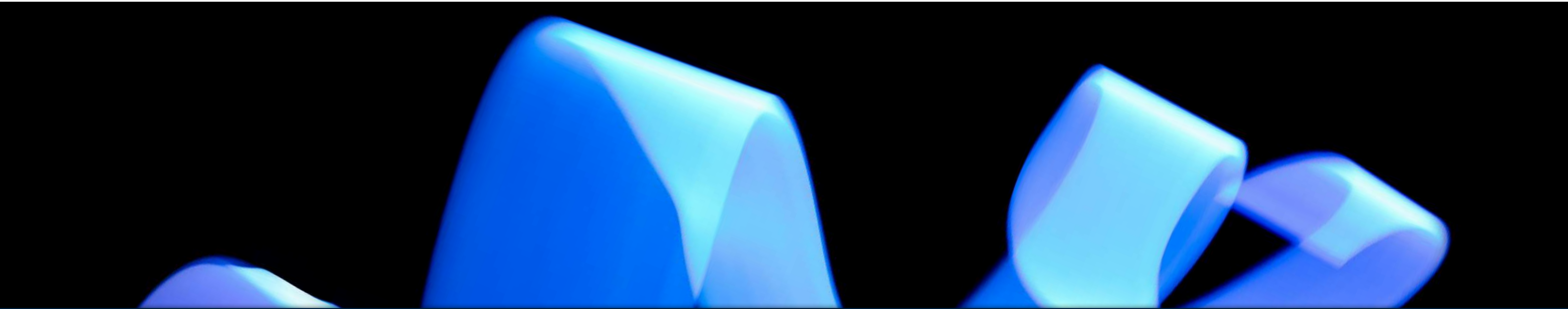
These include the imposition of tough national security controls to restrict Chinese involvement in key markets, a struggle to prove at scale many of the anticipated use cases for the technology, and concerns over spectrum allocation. Most of these issues continue to bite.

> Despite significant hurdles, 5G is still seen as a technology with huge potential due to its data processing speeds, increased connectivity, and lower latency.

Potential opportunities of 5G



AI, 5G, digital transformation and cybersecurity are all seen as key opportunities for future business growth.



Impact of national security controls

Take national security controls. Measures introduced under the Trump administration to bar Chinese vendors, most notably Huawei, from supplying equipment have not, as some hoped, been watered down by President Biden. Irrespective of the outcome of this year's presidential election, we can expect such restrictions to continue.

In reality, the restrictions have had less impact on the US market than elsewhere, particularly Europe.

The big US telecom operators were able to clear their networks of Chinese equipment relatively quickly.

Smaller providers faced a tougher challenge despite initially being supported by a funded rip-and-replace scheme. However, there is now some concern that further funding will not be available to complete the job of removing switching devices.

Under strong US pressure many European governments imposed similar controls, often forcing operators to remove capable components from their networks and replace them with equipment that was often less effective and more expensive.

This has certainly slowed the roll-out of 5G networks and has undoubtedly angered operators.

It may also have deterred investment in network development to some extent, although it is interesting that one of the main arguments Vodafone UK and 3UK are using to justify their proposed merger to competition authorities is the proposition that it will unleash huge resources to build out 5G networks.

By contrast, many African countries have continued to use Chinese vendors. Given the levels of infrastructure investment by China across the Continent, Governments have been careful not to jeopardize that inflow of funding. Operators here will only be worried about using Chinese equipment if they are trying to attract US funding or fear they will fall foul of sanctions.

Given that picture, it's not surprising that across the world 83% of respondents to our survey believe national security controls around 5G have gone far enough, a score which is consistent across regions.

Of the small number that believe additional measures are needed, the highest number believe stricter regulatory controls should be imposed for infrastructure projects. Some, clearly believing that government intervention can give them a competitive edge, call for increased state funding of domestic manufacturers and the imposition of market share caps on foreign providers.

Security implications of 5G

Organizations who believe there are enough provisions in place to protect national security in their country for the introduction of 5G.

Region	%
Global	83%
Europe	84%
North America	82%
APAC	83%
LATAM	83%
Africa	83%
Middle East	82%

Of those organizations who were concerned that measures were not sufficient, they advocated the following measures should be implemented

Measure	%
Tighter regulatory controls in infrastructure developments / implementation	63%
Tighter monitoring of supplier risk profiles	59%
Increased state investment in home country infrastructure manufacturers and software providers	58%
Impose market share cap on non-domestic providers	53%

83% of survey respondents believe national security controls around 5G have gone far enough, a consistent sentiment across regions.

Testing use cases

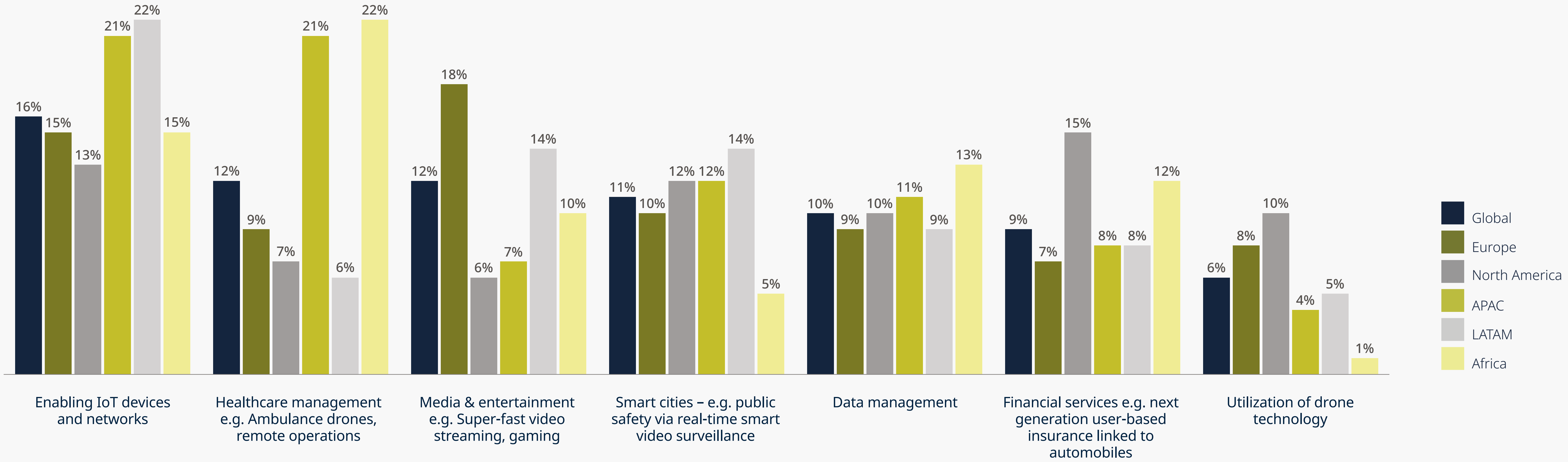
A wide range of apparently compelling use cases were anticipated as 5G networks began rolling out, and many of these are highlighted in our survey as having the most growth potential.

Enabling Internet of Things devices and networks comes out on top, followed by media and entertainment, such as super-fast video and streaming. Managing healthcare services and the development of smart cities, using real-time video surveillance of roads and other transport services, score relatively highly too while financial services is seen to have greater potential in North America than elsewhere.

Interestingly one area where 5G is expected to have a growing impact – autonomous vehicles and smart cars – does not feature directly.We are seeing progress in some of these areas. But the area where 5G is achieving greatest traction is in industrial settings, where network slicing, something that is not available on earlier generation technology, is allowing businesses to set up private networks, for instance in factories and ports, to automate production and maintenance processes at high speed and in real-time.

It still remains to be seen if 5G will deliver some of the more exotic use cases envisaged for it, such as remote surgical and other medical procedures. There's little evidence of that happening yet.

Cases where 5G offers most growth potential



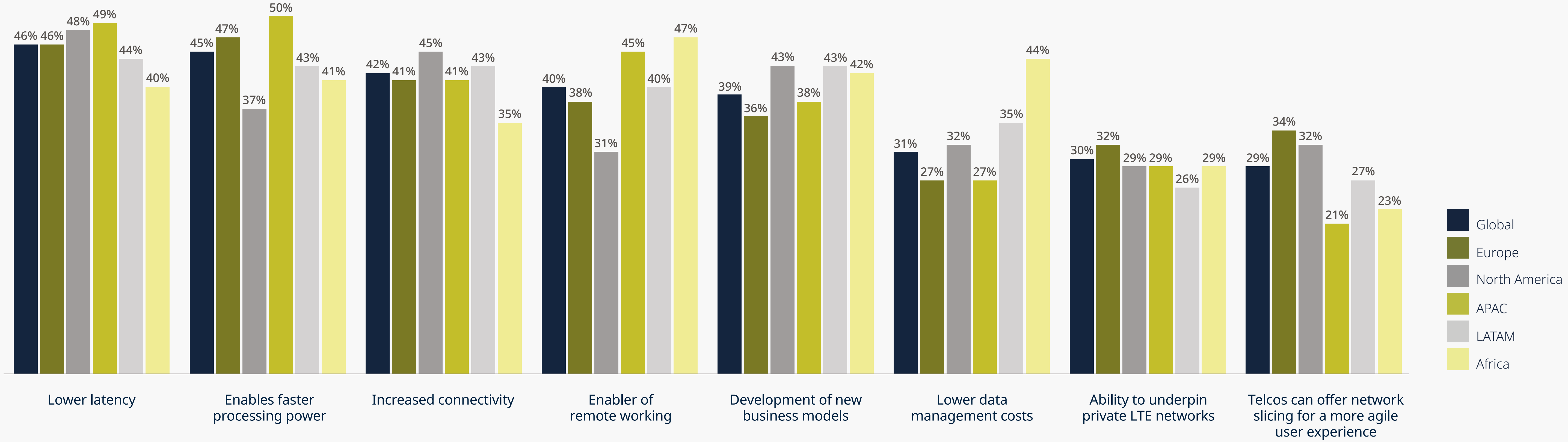
5G is seen to offer the most growth potential in relation to enabling IoT devices/ networks, followed by media/entertainment (particularly important in APAC and Africa), healthcare management (particularly important in Europe) and smart cities. In North America, use cases related to financial services are most commonly identified.

Getting the consumer proposition right

There’s a clear consensus among our respondents about the benefits of 5G technology, with lower latency, faster processing power, increased connectivity and the ability to re-engineer business models widely cited.

And there is broad agreement on the challenges of deploying 5G more widely where increased costs of connectivity, worries about the underlying technology and the difficulty of developing credible use cases rank highly.

Benefits of 5G



Top benefits of 5G are felt to be lower latency, faster processing power, increased connectivity and enabling remote working. Lower data management costs are particularly important in Africa.

Interestingly, the fact that the solutions delivered by operators may not meet customer expectations also features amongst leading challenges and ranks most highly in Africa.

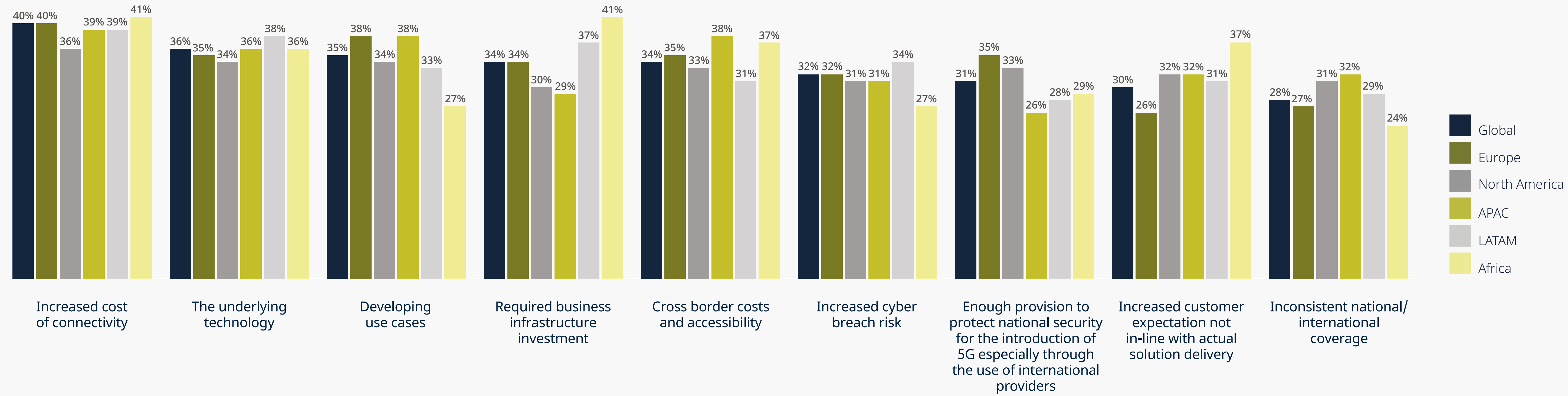
We've certainly seen evidence that operators are struggling to turn 5G into a convincing customer proposition worth spending more money on, particularly when they are already being served with decent broadband internet connections via 4G networks.

That's a significant contrast to what we saw with the transition from 3G to 4G technology, where operators were able in their marketing to point to a clear set of advantages from the new generation of mobile, advantages that were worth paying more for.

In the first phase of the 5G roll-out, US operators who focused on low band delivery were unable to demonstrate the full capabilities of the technology. Though the technology is pretty ubiquitous both in dense urban settings and in connecting remote rural areas, there is a sense that, until networks are further developed, the real benefits of 5G are still around the corner.

Operators are struggling to turn 5G into a convincing customer proposition, especially when many consumers are satisfied with 4G services.

Challenges of 5G connectivity



The spectrum challenge

Whether or not these advantages will be realized depends in part on making more spectrum available.

Although 5G offers greater flexibility in spectrum use than 4G, there is clearly a need to replenish the pipeline, not least since there is already talk of the next generation of mobile technology, 6G, becoming a reality by 2030.

In the US it’s an issue that is made all the more complex since the Federal Communications Commission’s authority to auction spectrum was allowed to expire in 2023 and has not yet been re-activated.

But it’s an issue we are seeing in other regions too. In South Africa the challenge has to some extent been resolved after additional spectrum was released two years ago.

However, the digital migration programme has yet to be fully completed meaning that not all the promised additional spectrum is yet available. And it is common for spectrum auctions to result in litigation as rival operators mount challenges to ensure they secure their slice of the action.

Inevitably that leads to delays. The latest auction process, for example, was held up for almost 10 years because of numerous legal challenges.

> 5G offers the most growth potential in enabling IoT networks, media and entertainment, healthcare management, and smart cities.

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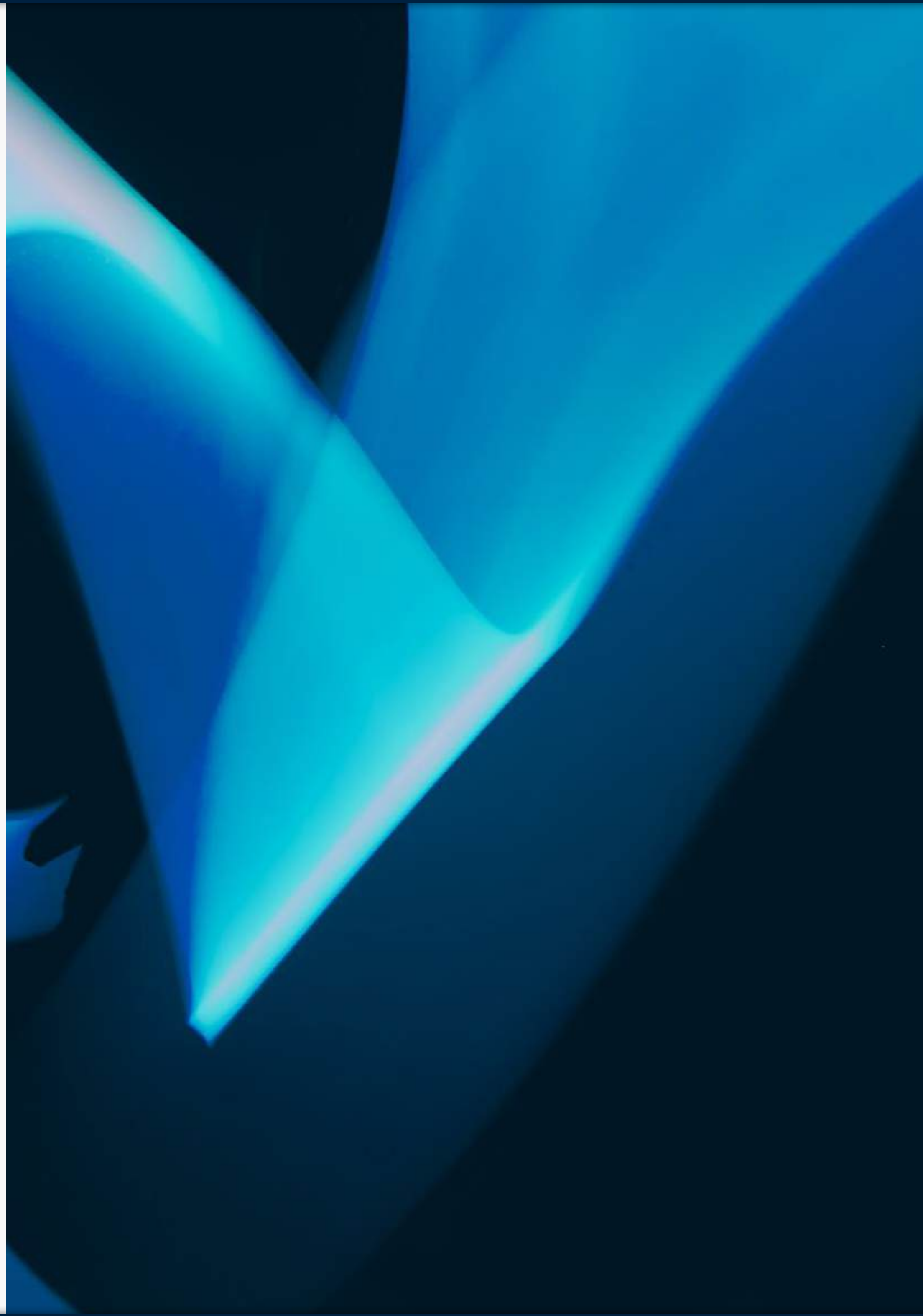


Riding the next big wave – some final thoughts

Some glowing reflections

Two things stand out from DLA Piper's seventh biennial Tech Index surveying the mood of organizations and key individuals in the technology sector – an overwhelming sense of optimism and an absolute focus on the promises, and potential threats, of Artificial Intelligence.

The first is, perhaps, more surprising than the second.

The global economy is in much better shape than when we last took soundings in 2022. Then inflation was soaring, and interest rates had climbed to a 40-year high, abruptly ending a protracted period of readily available debt financing and a transactions boom in which the tech sector played a starring role.

Today the outlook is brighter but not without continuing uncertainties.

Inflation may have come down swiftly to more sustainable levels, but interest rates have not fallen as fast as many had hoped, and all the forecasts project a prolonged period of relatively sluggish growth in many major economies, most notably Europe and the UK.

It's, therefore, surprising that our latest survey finds the tech sector in such exuberant mood, with a record overall tech score of 71, three points higher than in 2022, which was itself the highest score we had ever seen in our surveys.

And importantly we have, this year, taken a truly global approach to canvassing opinion so have a picture of consistent confidence across the world, with only slight variations from region to region.

Past editions of the Tech Index have focused exclusively on Europe, relying on a much smaller sample size.

But this year we have widened the scope, questioning a total of 1,200 respondents and testing the opinions of businesses with revenues ranging from USD10 million to USD10 billion to give us a deeper insight into the mood of tech businesses and the people shaping tech policy in government and regulatory authorities.

Respondents this time not only express confidence in the current economic climate but are equally upbeat on the current direction of regulation, availability of skills, the state of venture capital markets and the current IP environment. All are seen overwhelmingly to be conducive to growth, although there are bigger doubts expressed about tax regimes.

Against that backdrop, respondents expect revenues to increase in the year ahead, with more than a third expecting growth of more than 6%.

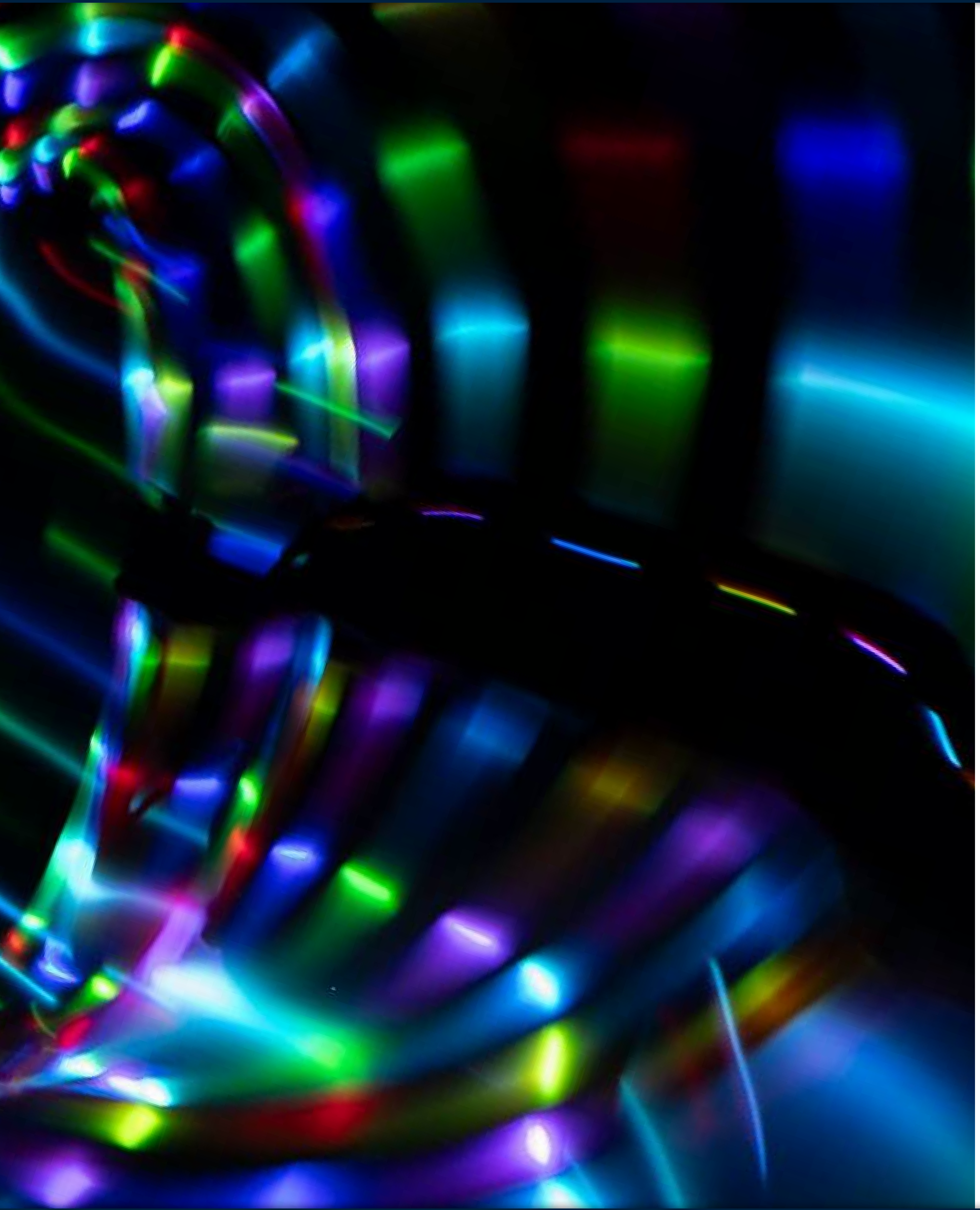
It's only when asked for their reflections on the current febrile geopolitical environment that their confidence dips and by a considerable margin.

That's only to be expected against a backdrop of war in Ukraine, conflict in Gaza, continuing trade tensions between the US and China, and a raft of elections that have proved highly unpredictable, with growing concerns over the rise of protectionist and divisive populist movements. The outcome of November's US election is a particular cause for concern, given the global economic and political impact it will undoubtedly have.

How, then, to explain the confidence that tech businesses are expressing in a world which, at this level, remains troubled?

We believe it's a reflection of the resilience businesses have had to show after coming through what has been a very unsettling few years. They've coped with the shock of the pandemic and the sharp economic correction that came in its wake, including soaring energy prices, supply chain disruptions, the need to retrench and restructure and conflict.

It's as if they have emerged with a new mindset – ready to see uncertainty as the new norm, better able to navigate unpredictable challenges and always eager to look for new opportunities for growth.



Preparing for the AI revolution

The accelerating pace of tech development is playing a major part in boosting confidence, and particularly advances in AI.

Since our last survey, we've seen this technology take a big leap forward not least with the rapid development of large language models and generative AI, which were only on the horizon for most organizations in 2022.

Now almost every conversation around technology turns to the subject of AI almost immediately as organizations try to get a handle on the transformative potential of this technology.

That's not always an easy task. As we note in the report, AI is, despite recent advances, still in its infancy. It has, what we call, a "last mile problem," yet to demonstrate in any really tangible ways what its potential impact, for good or bad, will be – not only on businesses but deeply throughout society.

Organizations, therefore, face a dilemma, caught between the fear that they will be left behind if they don't start investing in new solutions coming to market but, equally, very concerned that they may end up backing the wrong horse at great cost, financially, ethically, and reputationally.

A clear sense emerges from our survey that organizations are ready to invest, and indeed are already doing so. Even those who think that investment will be significant express a willingness to invest because they believe the benefits AI will bring will be truly transformational.

And it's worth remembering that other solutions that only relatively recently were also touted as transformative – Internet of Things, cloud computing, 5G, and data monetization, for example – are still at an early stage in proving their potential, as our findings clearly show. Why would AI be any different?

All these solutions will be enhanced by AI, of course. It touches everything. Even in the murky world of cybercrime, AI is already being harnessed by the bad actors and faster than it is being used to build cyber defenses.

Caution

Given the difficulty of gauging the likely impact of AI it's not surprising that many organizations are taking a cautious approach.

We see the same level of caution being exercised by dealmakers as M&A markets venture capital raising start to recover after an almost total shutdown through most of 2022 and 2023.

A long and record-breaking boom in transactions was brought to a standstill as inflation and interest rates soared. With valuations climbing to unsustainable levels in the bull market, many investments were carried out in haste and, in many cases, led to buyer's regret. Some of those investments have turned sour and we saw a spate of massive redundancies in the sector as chickens came home to roost.

The recovery which began in the US in the Spring of 2024 is beginning to pick up pace and to spread to other markets, with multiples now settling at much more reasonable levels. Many of the talented people let go by start-ups as they were forced to make cutbacks are now launching new businesses to bring their ideas to market.

But we see dealmakers and financial backers taking a much more disciplined approach to due diligence this time to assess the businesses or the talent they are acquiring – which can only be a good thing.

Despite economic uncertainties and a turbulent geopolitical landscape, the tech sector is showing unprecedented optimism, with a record overall tech score of 71.

The regulatory conundrum

Regulation provides another reason why organizations are taking time to get the lay of the land before investing in technology or moving to acquire external businesses.

The last two years have seen an avalanche of new regulation, whether that's around data protection and privacy, cybersecurity and vital infrastructure, national security, sustainability and ESG reporting, or virtual assets, much of it driven by laws coming out of the EU, although by no means exclusively.

It's notable that China has gone further in introducing hard law around AI than any other country, going further even than the EU's AI Act, the world's first comprehensive framework for regulating this technology. In the US, federal law is moving far slower than regulations coming out of individual states, such as California and Colorado.

Regulation never moves as fast as technological advances. It's always in a race to catch up. And different priorities always emerge in different jurisdictions leading to the creation of an ever more complex patchwork of overlapping and sometimes contradictory laws that organizations, wanting to do business in any particular market must comply with.

Inevitably, that has re-ignited the debate over whether regulation threatens to stifle innovation or if, when carefully devised and applied, it can actually promote the development of innovative new systems and business models.

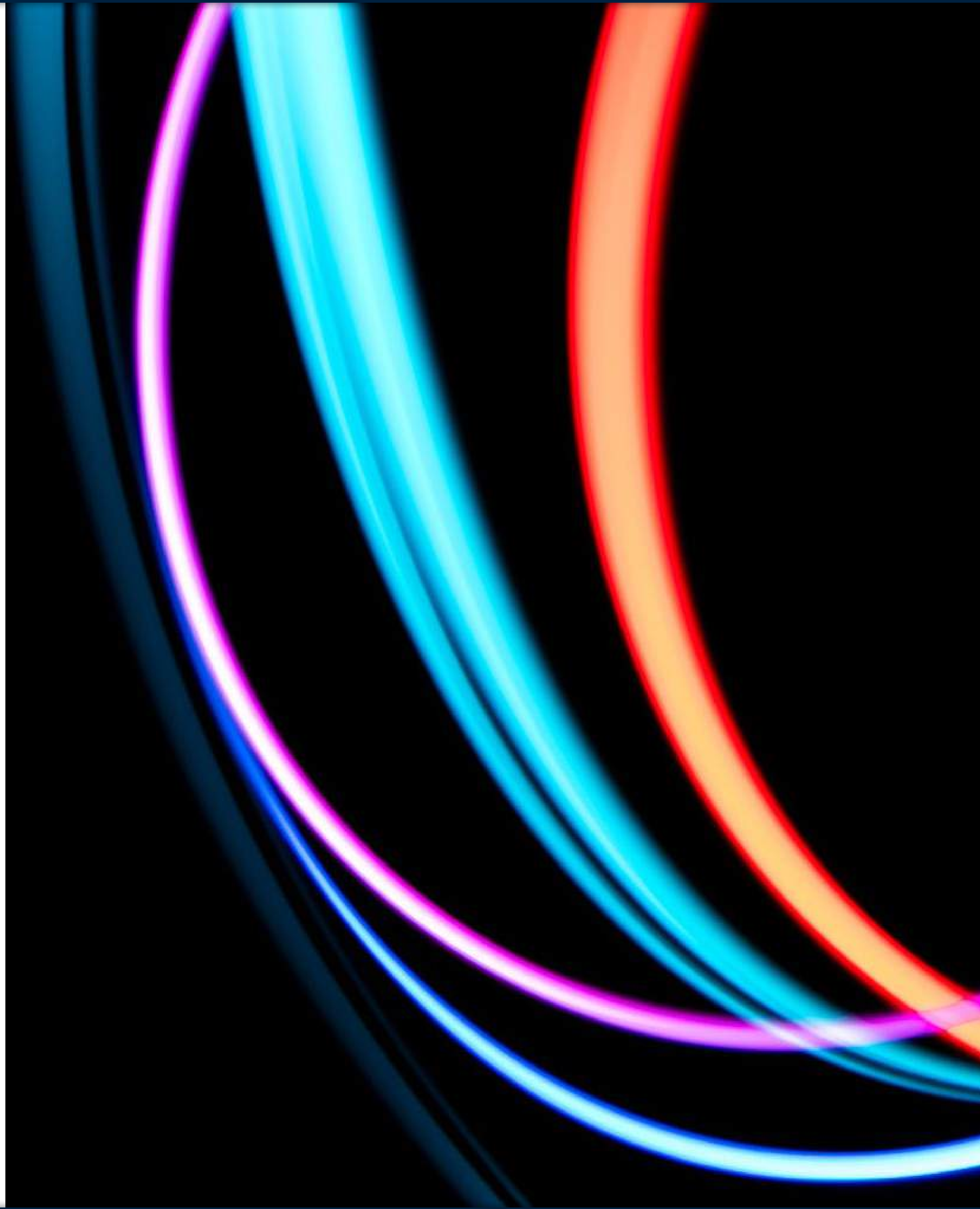
It's easy to find people in the former camp who believe we are in an era of regulatory overreach. But, increasingly, there are those taking the opposite view.

It's interesting, for instance, that in one area that initially attracted investment because it was subject to very little regulation, if any – crypto and other virtual assets – there's been a sea change in attitudes.

Dubai, Hong Kong, Singapore and the EU have, in different ways, moved to implement clear and transparent regulatory or licensing regimes to promote growth in this area, often in direct response to investor demands for greater certainty. Key African countries looking to build regulatory regimes around data, cyber and AI are increasingly using developments in the EU as a benchmark, if not a gold standard.

The hope is that, over time, we will see greater harmonization between regulatory regimes, though that remains a fairly distant dream. For now, organizations are having to get used to a period of regulatory dissonance and all the complexities that come with it.

Organizations are not just ready to invest in AI; they are already doing so, driven by the belief that AI will bring truly transformational benefits.



From transformation to evolution

Taken together these trends in the tech sector suggest that we are at a very important inflection point.

In the end, that is perhaps what most inspires the optimism that is clearly on display in this latest edition of the Tech Index – a sense that we are on the brink of an even more profound change in the way that commerce and society works powered by the AI industrial revolution and further out by the advent of quantum computing.

But as organizations prepare to ride the next wave, they need to rethink their approach to digital transformation.

Most companies have grasped the value and necessity of embracing transformation. But too many, to date, have used it as an opportunity to optimize their existing systems rather than as a chance to fundamentally rethink their business models.

As technology continues to advance at breakneck speed, it's clear that transformation is not a one-off event. Rather it is a continuous process of adaptation – a case of digital evolution rather than digital transformation.

This mammoth task of process change requires buy-in right across organizations because it will inevitably touch every corner of the enterprise.

Above all it requires strong and far-sighted leadership from the very top, guided by the CEO, supported by key colleagues in the C-suite and overseen by boards with the right expertise to understand the challenges and the huge opportunities that lie in wait.

Transformation is no longer a one-off event but a continuous process of digital evolution, requiring strong leadership and a fundamental rethink of business models.

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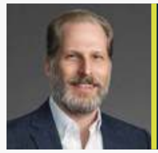

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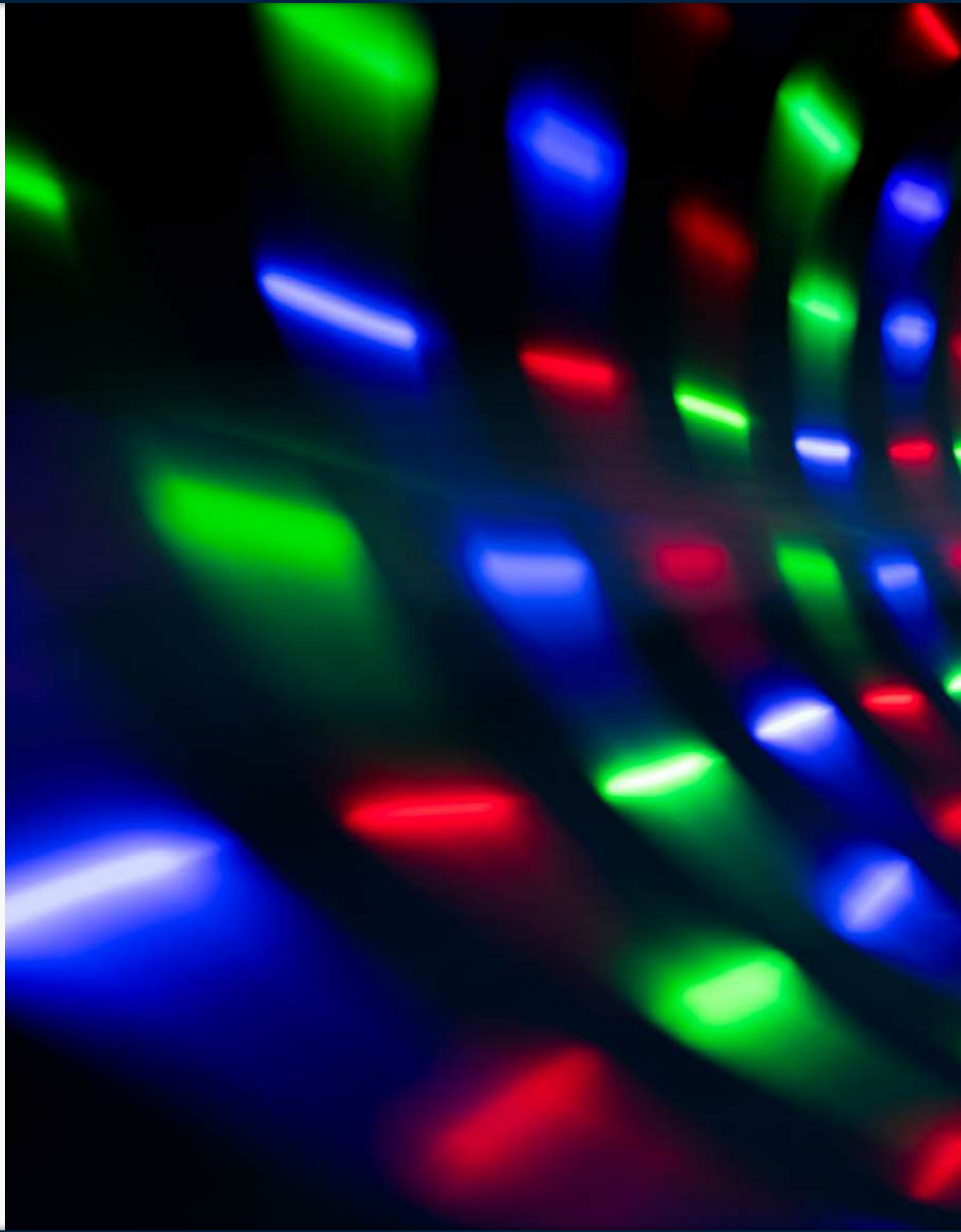


Methodology

Methodology and respondent profile

DLA Piper commissioned Coleman Parkes Research to gather insights from global companies to identify growth opportunities and the challenges they face.

In April and May 2024, 1,200 online interviews were conducted globally, with supplementary telephone research where necessary. These interviews were with executives from key technology businesses with revenues ranging from USD10 million to over USD10 billion. Additional perspectives were also obtained from members of the investment community and policymakers involved with the tech sector. All interviews were conducted in the respondents' local languages.

Previous editions of the Tech Index focused exclusively on Europe and were based on a smaller sample size. This year, we expanded our research globally, providing us with deeper insights into the perspectives of tech businesses and the individuals shaping tech policy within government and regulatory authorities.

Technology index methodology

The results have been compiled and weighted to generate DLA Piper's Technology Index Score. This score is derived from a diffusion index that considers the proportion of positive, negative, and neutral responses. The results are presented as a scorecard for each monitored business area, as well as an overall index.

How do we get the score?

Please note that due to rounding, percentages may not always appear to add up to 100%.

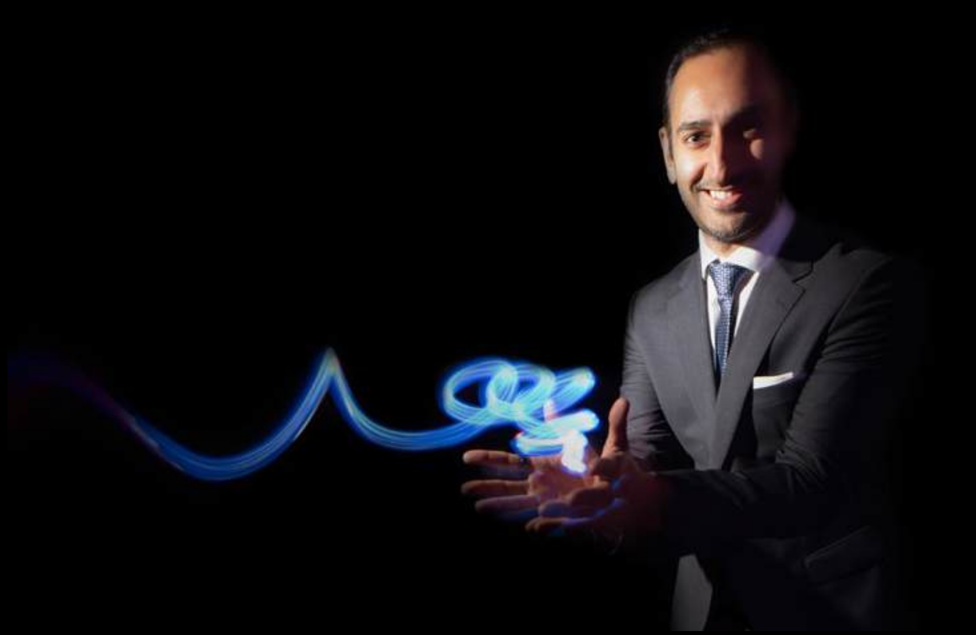
A glimpse behind the scenes – colleagues became artists for a day

Colleagues became artists for a day, capturing the magic of light painting to keep the human touch alive in our digital world.

The images featured in this report were all taken by DLA Piper colleagues specifically for this project. In an increasingly digital world, we believe it’s essential to keep the human touch alive!

We used light photography to capture a number of striking images. During an engaging afternoon workshop at DLA Piper’s London office, light painting photographer Tim Simpson inspired our colleagues to tap into their inner creativity.

Light painting captures the art of drawing in the air with a light source while the camera’s shutter remains open. The results, featured in this report, showcase a collection of unique images that trace the light’s path through darkness, brought to life through long exposure photography.



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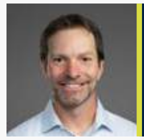
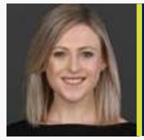


Ben Goodall
Senior Marketing & BD Manager,
US Technology Sector, Palo Alto

Ed Griffiths
Partner, London

Jennifer Holyoake
Legal Director, London



Lauren Hurcombe
Partner, Hong Kong


Ewa Kurowska-Tober
Partner, Warsaw


Sophie Lessar
Partner, London


Natasha Luther-Jones
Partner, Leeds and London


John Magee
Partner, Dublin


Morgan McDonald
Partner, Vancouver


Siobhan Mullen
Legal Director, London


Edel O'Kelly
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Sandra Oyewole
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Nick Rock
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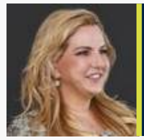
Tomáš Ščerba
Partner, Prague

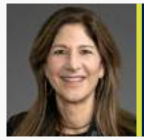

Sarah Jayne Scott
Marketing & BD Manager, Edinburgh

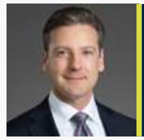

Hannah Smith
Senior Marketing & BD Manager, London


Michael Stead
Senior Associate, Birmingham


Gareth Stokes
Partner, Birmingham

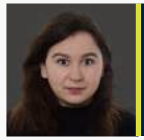

Kristi Swartz
Partner, Hong Kong and Dubai


Margo Tank
Partner, Washington, DC


Danny Tobey
Partner, Dallas


Nick Valentine
Partner, Auckland


Tzung-bor (Chung) Wei
Partner, Palo Alto


Magdalena Zmorka
Marketing & BD Manager, Warsaw

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